

**MACHINE LEARNING-BASED THREE-YEAR PREDICTION OF
CARDIOVASCULAR DISEASE AND MAJOR EVENTS IN
PATIENTS WITH NO PRIOR HISTORY**

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MACHINE LEARNING-BASED THREE-YEAR PREDICTION OF CARDIOVASCULAR DISEASE AND MAJOR EVENTS IN PATIENTS WITH NO PRIOR HISTORY

ABSTRACT

Cardiovascular disease (CVD) has been the leading cause of death in Malaysia since the 1980s, with myocardial infarction (MI) and stroke being significant contributors. Conventional risk scores like the Framingham Risk Score (FRS) are widely used for predicting CVD, but their 10-year risk assessment often lacks population specificity, particularly for the Malaysian demographic, and fails to capture short- to medium-term risks. This limitation may lead to missed opportunities for timely interventions, especially given the rapid progression of CVD in some patients. In response to this gap, this study developed a 3-year CVD risk prediction and event classification model tailored to Malaysia's diverse demographic landscape, addressing the need for a more accurate and timely risk assessment tool. Utilizing data from the REDISCOVER registry, which included 12,617 participants from eight Malaysian states, two distinct models were constructed. The first model predicted the occurrence of CVD events (CVD vs. No CVD), while the second classified specific CVD events, particularly MI and stroke. Data preprocessing involved median imputation for missing values and resampling to balance the classes. Feature selection was performed using Recursive Feature Elimination (RFE), SHAP-based selection, backward elimination as well as correlation analysis for noise reduction. The models trained included Random Forest (RF), XGBoost, Decision Trees (DT), Gradient Boosting Machine (GBM), Support Vector Classifier (SVC), and K-Nearest Neighbors (KNN). Ensemble models were also created using a stacking approach. Model performance was evaluated based on accuracy, area under the curve (AUC), classification reports, and confusion matrices. In the first model, the RF model

using the top 26 selected features achieved an AUC of 0.8587, outperforming the Framingham Risk Score (FRS), which had an AUC of 0.8343. The RF model also outperformed FRS in terms of the Net Reclassification Index (NRI), showing a 19% improvement at a 5% cut-off point. While the calibration of the RF model was similar to FRS, the Brier score was slightly higher for the RF model (0.01176) compared to FRS (0.01173), and the log loss was marginally greater (0.05720 for RF vs. 0.0568 for FRS). The second model, aimed at classifying specific CVD events, utilized an RF ensemble approach with the top 14 selected features. The model demonstrated strong predictive performance, achieving an AUC of 0.944, indicating excellent discrimination between the different CVD events. Despite the limited sample size, the confusion matrix demonstrated good classification performance, correctly classifying most cases of MI and stroke. This outcome suggests that even with a relatively small number of features, the model effectively captured the critical patterns associated with these specific CVD events. This study demonstrates that the developed models offer robust and accurate 3-year predictions specifically tailored to the Malaysian population. The first model provides a reliable approach to predicting CVD risk within a 3-year timeframe, while the second model effectively classifies specific cardiovascular events over the same period. The results indicate significant improvements over traditional methods, underscoring the potential for these models to enhance CVD risk assessment and intervention in Malaysia.

ABSTRAK

Penyakit kardiovaskular (CVD) telah menjadi punca utama kematian di Malaysia sejak 1980-an, dengan infarksi miokardium (MI) dan strok menjadi penyumbang penting. Skor risiko konvensional seperti Skor Risiko Framingham (FRS) digunakan secara meluas untuk meramalkan CVD, tetapi penilaian risiko 10 tahun mereka sering kekurangan kekhususan populasi, terutamanya untuk demografi Malaysia, dan gagal menangkap risiko jangka pendek hingga sederhana. Had ini boleh menyebabkan terlepas peluang untuk campur tangan tepat pada masanya, terutamanya memandangkan perkembangan pesat CVD dalam sesetengah pesakit. Sebagai tindak balas kepada jurang ini, kajian ini membangunkan model ramalan risiko CVD dan klasifikasi peristiwa 3 tahun yang disesuaikan dengan landskap demografi Malaysia yang pelbagai, menangani keperluan untuk alat penilaian risiko yang lebih tepat dan tepat pada masanya. Menggunakan data daripada pendaftaran REDISCOVER, yang merangkumi 12,617 peserta dari lapan negeri di Malaysia, dua model berbeza telah dibina. Model pertama meramalkan kejadian CVD (CVD vs. No CVD), manakala yang kedua mengelaskan kejadian CVD khusus, terutamanya MI dan strok. Prapemprosesan data melibatkan imputasi median untuk nilai yang hilang dan pensampelan semula untuk mengimbangi kelas. Pemilihan ciri dilakukan menggunakan Penghapusan Ciri Rekursif(RFE), pemilihan berasaskan SHAP, penghapusan ke belakang serta analisis korelasi untuk pengurangan hingar. Model yang dilatih termasuk Random Forest (RF), XGBoost, Decision Trees (DT), Gradient Boosting Machine (GBM), Support Vector Classifier (SVC) dan K-Nearest Neighbors (KNN). Model ensemble juga dicipta menggunakan pendekatan susun. Prestasi model dinilai berdasarkan ketepatan, kawasan di bawah lengkung (AUC), laporan pengelasan dan matriks kekeliruan. Dalam model pertama, model RF menggunakan 26 ciri terpilih teratas mencapai AUC sebanyak 0.8587, mengatasi prestasi Framingham Risk Score (FRS), yang mempunyai AUC sebanyak 0.8343. Model RF juga mengatasi prestasi FRS dari

segi Indeks Pengelasan Semula Bersih (NRI), menunjukkan peningkatan 19% pada titik pemotongan 5%. Walaupun penentuan model RF adalah serupa dengan FRS, skor Brier adalah lebih tinggi sedikit untuk model RF (0.01176) berbanding FRS (0.01173), dan kehilangan log adalah lebih besar sedikit (0.05720 untuk RF berbanding 0.0568 untuk FRS). Model kedua, bertujuan untuk mengklasifikasikan acara CVD tertentu, menggunakan pendekatan ensemble RF dengan 14 ciri terpilih teratas. Model ini menunjukkan prestasi ramalan yang kuat, mencapai AUC sebanyak 0.944, menunjukkan diskriminasi yang sangat baik antara peristiwa CVD yang berbeza. Walaupun saiz sampel terhad, matriks kekeliruan menunjukkan prestasi pengelasan yang baik, dengan betul mengklasifikasikan kebanyakan kes MI dan strok. Hasil ini menunjukkan bahawa walaupun dengan bilangan ciri yang agak kecil, model ini secara berkesan menangkap corak kritikal yang dikaitkan dengan peristiwa CVD khusus ini. Kajian ini menunjukkan bahawa model yang dibangunkan menawarkan ramalan 3 tahun yang mantap dan tepat yang khusus disesuaikan dengan penduduk Malaysia. Model pertama menyediakan pendekatan yang boleh dipercayai untuk meramalkan risiko CVD dalam jangka masa 3 tahun, manakala model kedua secara berkesan mengklasifikasikan kejadian kardiovaskular tertentu dalam tempoh yang sama. Keputusan menunjukkan peningkatan yang ketara berbanding kaedah tradisional, menggariskan potensi model ini untuk meningkatkan penilaian risiko CVD dan campur tangan di Malaysia.

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LIST OF SYMBOLS AND ABBREVIATIONS

AI	Artificial Intelligence
AHA	American Heart Association
ACC	American College of Cardiology
AUC	Area Under The ROC
ACS	acute coronary syndromes
BE	Backward Elimination
BMI	body mass index
CAD	Coronary artery disease
CCS	Chronic Coronary Syndrome
CCS-class	Canadian Cardiovascular Society classification
CHD	coronary heart disease
CT	Computed tomography
DALYs	disability-adjusted life years
DL	Deep Learning
DM	diabetes mellitus
DOSM	Department of Statistics, Malaysia
DT	Decision Tree
ECG	Electrocardiogram
EL	Ensemble learning
FH	familial hypercholesterolemia
FHS	Framingham Heart Study
FN	False Negative
FP	False Positive
FRS	Framingham Risk Score
GBM	Gradient Boosting Machine

HDL	High-density lipoprotein
HF	Heart failure
ICH	intracerebral hemorrhage
IHD	Ischemic heart disease
IHME	Institute for Health Metrics and Evaluation
IPH	Institute for Public Health
KNN	K-Nearest Neighbors
LDL	Low-density lipoprotein
LDL-C	low-density lipoprotein cholesterol
LV	left ventricular
MAR	missing at random
MCAR	missing completely at random
MI	Myocardial Infarction
ML	Machine learning
MNAR	missing not at random
MOH	Ministry of Health
MRI	Magnetic Resonance Imaging
NCD	Non-communicable diseases
NCVD	National Cardiovascular Disease
NHAM	National Heart Association of Malaysia
NHMS	National Health and Morbidity Survey
NIH	National Institutes of Health
NRI	Net reclassification improvement
NSTEMI	non-ST-segment elevation myocardial infarction
PAD	peripheral artery disease
REDISCOVER	Responding to Increasing Cardiovascular Disease Prevalence

RF	Random Forest
RFE	Recursive Feature Elimination
ROC	Receiver Operating Characteristic
ROS	Random Over-Sampling
RPCE	Revised Pooled Cohort Equations
RUS	Random Under-Sampling
SAH	subarachnoid hemorrhage
SCORE2	Systematic Coronary Risk Evaluation 2
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SVC	Suport Vector Classifier
SVM	Support Vector Machine
STEMI	ST-segment elevation myocardial infarction
TIA	transient ischemic attacks
TN	True Negative
TP	True Positive
UA	unstable angina
WHO	World Health Organization
XGBoost	eXtreme Gradient Boosting

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Chapter 1: INTRODUCTION

1.1 Study Background

Cardiovascular disease (CVD) remains the leading cause of death worldwide, responsible for an estimated 18 million deaths annually, which accounts for nearly one-third of all global deaths (WHO, 2021). This broad category includes a number of conditions affecting the heart and blood vessels, such as coronary artery disease, cerebrovascular disease, and peripheral arterial disease. The global burden of CVD has seen a marked shift over recent decades, with low and middle-income countries experiencing a significant rise in incidence due to rapid urbanization, lifestyle changes, and increased life expectancy (Mocumbi, 2024; Gaziano et al., 2010). Albeit advancements in medical technology and healthcare, the growing prevalence of CVD in these regions underscores the need for effective prevention and management strategies.

In Malaysia, CVD has been the leading cause of death since the 1980s, and its impact has only grown over the years (National Health Association of Malaysia (NHAM), 2017). The Department of Statistics Malaysia (DOSM) reports that ischemic heart disease (IHD) has consistently been the top cause of mortality, contributing to 16.1% of total deaths in 2023 (DOSM, 2023). Stroke follows closely behind, accounting for 7.2% of all deaths, making these two conditions significant contributors to the overall burden of disease in Malaysia.

By 2019, it was reported that a significant portion of the Malaysian adult population faced major cardiovascular risk factors: approximately 4-in-10 of all adults had elevated hypercholesterolemia, around 3-in-10 were diagnosed with hypertension, and 1-in-5 were living with diabetes mellitus. (NHMS, 2019). These risk factors are compounded by lifestyle choices such as poor diet, physical inactivity, and smoking, which further exacerbate the risk of developing CVD.

Despite efforts to control these risk factors, the incidence of CVD continues to rise, placing a significant strain on the Malaysian healthcare system. A study carried out by the Institute for Health Metrics and Evaluation (IHME) suggests that by 2025, the number of premature CVD deaths will rise by approximately 18%, reaching around 31,000 cases compared to the 23,000 cases reported in 2015. The increasing prevalence of CVD and its associated risk factors highlights the need for effective prediction strategies, particularly those tailored to the specific needs of the Malaysian population.

The burden of cardiovascular disease extends far beyond its impact on mortality and morbidity. CVD is also associated with significant economic costs, both in terms of direct healthcare expenses and indirect costs related to lost productivity and disability. In Malaysia, the Ministry of Health (MOH) reported that in 2017, CVD accounted for 1.4 million disability-adjusted life years (DALYs), translating into an estimated economic loss of RM60 billion (MOH, 2020). This economic burden is expected to increase as the prevalence of CVD continues to rise, placing additional strain on the healthcare system and the broader economy.

The economic impact of CVD is not limited to healthcare costs. It also includes the loss of productivity due to illness, disability, and premature death (Institute for Public Health (IPH), 2020). In many cases, individuals with CVD are unable to work, either temporarily or permanently, thereby leading to a loss of income and reduced economic output. This is particularly concerning in low and middle-income countries, where the economic burden of CVD can heavily impact families and communities (Gheorghe et al., 2018; Bloom et al., 2011). The societal impact of CVD is also significant. CVD often leads to a reduced quality of life for patients and their families, as well as increased healthcare needs and dependence on caregivers.

Given the significant health and economic burden of cardiovascular disease, there is a clear need for effective screening and prediction tools that can identify individuals at risk of developing CVD. Historically, the Framingham Risk Score (FRS) has been one of the most widely used tools for predicting cardiovascular disease risk. Developed in the United States of America based on data from the Framingham Heart Study, the FRS estimates an individual's 10-year risk of developing CVD based on multiple factors including age, gender, cholesterol levels, blood pressure, smoking status, and diabetes (D'Agostino et al., 2008). The FRS has been widely adopted in clinical practice and is recommended by many guidelines for assessing CVD risk.

However, despite its widespread use, the FRS and similar risk prediction tools have significant limitations, particularly when applied to populations outside of the United States. These tools were developed based on data from predominantly Caucasian populations and may not accurately reflect the risk factors and disease patterns in other ethnic groups (Selvarajah et al., 2014). Although previously articles like "Primary & Secondary Prevention of Cardiovascular Disease 2017" recommends the use of the FRS to assess the 10-year risk of developing CVD in the Malaysian population, many validation studies in Malaysia have shown that the FRS has moderate discrimination (AUC ranging from 0.63 to 0.76) and poor calibration, indicating that it may not be well-suited for predicting CVD risk in the Malaysian population (Chia et al., 2015; Kasim et al., 2023).

A significant limitation of traditional CVD risk prediction tools like the FRS is their focus on a 10-year prediction horizon, which may not be sufficient for identifying individuals at risk in the short to medium term. The increasing recognition of the need for short-term risk models is driven by the dynamic nature of CVD risk factors, such as hypertension and dyslipidemia, which can fluctuate over short periods due to various

interventions (Bays et al., 2021). Shorter prediction horizons offer a more immediate assessment of risk, enabling more timely and effective interventions. Recent studies have demonstrated that the prediction accuracy for myocardial infarction (MI) is often higher over moderate follow-up periods, particularly around 1500 days, compared to longer timeframes (Simon et al., 2022). This highlights the potential advantage of developing models with shorter prediction horizons.

Machine Learning (ML), a subset of artificial intelligence, offers a powerful alternative for CVD risk prediction by allowing for the modelling of complex relationships among variables. Unlike traditional statistical models, ML algorithms can analyse large volumes of data, identify patterns, and make predictions without relying on predefined assumptions. This flexibility makes ML particularly well-suited to CVD risk prediction, where multiple factors ranging from genetic predispositions to lifestyle choices interact in complex ways to influence an individual's risk of disease. Machine learning models can also incorporate a broader range of variables, including biomarkers, imaging data, and even genetic information, alongside traditional risk factors like age, blood pressure and cholesterol levels. This ability to integrate diverse data sources allows ML models to provide more personalized and accurate risk assessments.

In recent years, several studies have demonstrated the potential of ML in improving CVD risk prediction. For example, Weng et al. (2017) showed that ML algorithms such as Random Forests and Gradient Boosting machines can significantly improve the accuracy of predicting CVD events. Similarly, Cho et al. (2021) found that ML models provided better discrimination and calibration than conventional models in a multi-ethnic Asian population, highlighting the potential of these approaches to improve risk prediction in diverse populations like Malaysia's. Studies have increasingly shown that ML methods outperform traditional CVD risk prediction models, particularly in their

ability to handle complex and high-dimensional datasets (Cho et al., 2021; Sajeew et al., 2021; Weng et al., 2017). Unlike conventional models, which often rely on predefined assumptions about linear relationships between risk factors and outcomes, ML algorithms can adaptively learn from data. This adaptability allows them to uncover intricate patterns and nonlinear relationships, leading to more precise risk predictions, especially when feature selection methods are employed to enhance the model's performance (Kuhn & Johnson, 2013). Despite these advancements, there are limited studies regarding the application of ML algorithms for predicting general CVD risk and mortality in specific populations, such as in Malaysia.

Existing studies that have applied CVD risk prediction models to the Malaysian population predominantly focus on long-term (10-year) CVD risk prediction (Putri et al., 2023; Yew et al., 2019). However, these studies do not address the short-term prediction of CVD and CVD events like myocardial infarction (MI) and stroke, which are crucial for timely clinical interventions. Traditional models tend to oversimplify the complexity of risk variables, assuming linear relationships with clinical outcomes (Pal et al., 2022), whereas ML offers a more nuanced, data-driven approach. This highlights the need for developing and validating ML-based prediction models that can assess short-term (3-year) risk, particularly for MI and stroke patients, within the Malaysian context. This represents a significant research gap, as no studies to date have focused on ML-based 3-year prediction models for these specific CVD events in the Malaysian population.

1.2 Overview of the Study

The title of the study is “Machine Learning-Based Three-Year Prediction of Cardiovascular Disease and Major Events in Patients with No Prior History.” This study focuses on developing and validating two distinct machine learning models tailored to predict cardiovascular disease (CVD) risk within a 3-year timeframe, specifically

targeting the diverse Malaysian demographic. The first model predicts the occurrence of CVD events (CVD vs. No CVD), while the second model classifies specific CVD events, particularly myocardial infarction (MI) and stroke, over the same period.

The motivation behind this study stems from the limitations of conventional risk scores like the Framingham Risk Score (FRS), which, despite being widely used, often lack population specificity and fail to accurately predict short to medium-term CVD risks in the Malaysian population. Given the rapid progression of CVD in some patients, a shorter prediction horizon and a model tailored to the Malaysian demographic are crucial for timely and effective intervention.

1.3 Problem Statement

Cardiovascular disease has been the leading cause of death in Malaysia since the 1980s, with myocardial infarction and stroke being significant contributors. The conventional risk prediction tools, particularly the Framingham Risk Score (FRS), are based on 10-year risk assessments that often lack specificity for the Malaysian population. These tools tend to either overestimate or underestimate CVD risk in this demographic, leading to missed opportunities for timely interventions. Additionally, these conventional tools do not focus on short to medium-term risks, which are critical for preventing the rapid onset of CVD in high-risk individuals.

The critical issue prompting this study is the need for a more accurate, population-specific CVD risk prediction model tailored to the Malaysian demographic. Specifically, there is a need for models that can predict both the general risk of developing CVD and the likelihood of experiencing specific CVD events, such as myocardial infarction and stroke, within a 3-year timeframe.

1.4 Research Questions

RQ1: What are the key factors affecting the 3-year risk prediction of cardiovascular disease in the Malaysian population?

RQ2: Which machine learning model is most suitable for predicting the 3-year risk of CVD events in the Malaysian population?

RQ3: How does the performance of the newly developed machine learning model compare with the conventional Framingham Risk Score in predicting 3-year CVD risk?

RQ4: Which features are most significant in classifying specific cardiovascular events (myocardial infarction and stroke) within a 3-year period?

1.5 Research Objectives

The study aims to address the limitations of conventional CVD risk prediction tools by developing and validating machine learning models tailored to the Malaysian population. The specific objectives are as follows:

RO1: To identify the key predictors of developing cardiovascular disease in the Malaysian population.

RO2: To develop a general 3-year CVD risk model for the Malaysian population.

RO3: To develop a model for classifying myocardial infarction and stroke events within a 3-year timeframe.

RO4: To compare the performance of the general CVD risk model with conventional FRS risk scores.

1.6 Scope of Research

This study exclusively relies on the : Responding to Increasing Cardiovascular Disease Prevalence (REDISCOVER) dataset, which includes comprehensive data on 12,617 participants from across Malaysia. The dataset includes demographic details, medical history, lifestyle factors and clinical measurements. The study focuses on developing machine learning models that are specifically tailored to predict both the general risk of CVD and the risk of specific events, such as myocardial infarction and stroke, within a 3-year period. By using the REDISCOVER dataset, the study aims to create models that are highly relevant to the Malaysian population, addressing the limitations of existing prediction tools and offering a more accurate and timely risk assessment framework.

For model development, the study will employ various machine learning algorithms, including Random Forest (RF), XGBoost, Decision Trees (DT), Gradient Boosting Machine (GBM), Support Vector Classifier (SVC) and K-Nearest Neighbors (KNN). The study also includes ensemble models created using a stacking approach to improve predictive performance.

1.7 Thesis Outline

Chapter 1: Introduction: This chapter provides an overview of the study, highlighting the significance of cardiovascular disease in Malaysia, the limitations of existing risk prediction tools, and the necessity for a more tailored, population-specific approach. The chapter also outlines the research objectives, questions and scope.

Chapter 2: Literature Review: This chapter reviews the existing literature on cardiovascular disease, focusing on the global and Malaysian context. It examines traditional risk prediction tools, their limitations in diverse populations, and recent advancements in machine learning models for CVD risk prediction.

Chapter 3: Methodology: This chapter details the methodologies used for data preprocessing, model development, and evaluation. It explains the selection of machine learning algorithms, the process of feature selection, and the metrics used to assess model performance.

Chapter 4: Results: This chapter presents the results of the study, including the performance of the machine learning models in predicting both general CVD risk and specific cardiovascular events. Comparative analyses with the Framingham Risk Score are also discussed.

Chapter 5: Discussion: This chapter discusses the implications of the study's findings, comparing the effectiveness of the developed models with existing tools. It also explores the significance of the identified risk factors and their implications for public health interventions in Malaysia.

Chapter 6: Conclusion: The final chapter summarizes the study's key findings, highlighting the contributions to the field of CVD risk prediction and the potential for these models to improve clinical and public health outcomes in Malaysia.

Chapter 2: LITERATURE REVIEW

2.1 Cardiovascular Disease

Cardiovascular disease is a broad term that refers to a group of disorders affecting the heart and blood vessels. These disorders can range from conditions like coronary artery disease (CAD) and cerebrovascular disease to peripheral artery disease (PAD) (Olvera Lopez et al., 2023). These conditions are primarily caused by atherosclerosis, a process in which plaques made of fat, cholesterol and other substances build up in the arteries (Shao et al., 2020). Over time, this plaque can harden and narrow the arteries, limiting the flow of oxygen-rich blood to vital organs, including the heart and brain (National Institute of Health (NIH), 2022). In addition to atherosclerosis, other factors such as inflammation, oxidative stress and genetic predispositions play significant roles in the progression of CVD (de Almeida et al., 2020). These factors can influence how quickly atherosclerosis develops and how severe it becomes. Among the various forms of CVD, IHD and stroke are two of the most critical, responsible for a significant 85% of all cardiovascular-related deaths globally in 2019 (Di Cesare et al., 2024). MI, commonly known as a heart attack, is a critical and severe manifestation of IHD. This high percentage underscores the deadly impact of these conditions, with IHD causing 9.1 million deaths and stroke contributing to 6.6 million deaths in 2019 (Di Cesare et al., 2024).

In Malaysia, cardiovascular disease presents a severe and growing challenge to public health, impacting not only patients and their families but also the nation's economy. As of 2020, CVD was the leading cause of death in Malaysia and ranked third for hospitalizations in MOH hospitals (Khaw & Yong, 2024; Health Facts, 2021). A significant portion of hospitalizations was due to acute coronary syndrome (ACS), with 12% to 42% of hospitalized patients having previously experienced a coronary event

(Machta et al., 2022). In 2017, the treatment costs for CVD were estimated to be RM 3.93 billion, accounting for over 40% of the total direct healthcare costs associated with non-communicable diseases (NCDs) in the country (The Direct Healthcare Cost of Non-Communicable Diseases in Malaysia, 2022). The hospitalization costs alone amounted to RM 1.01 billion, with the average cost per hospital episode of CVD being RM 5,374. In addition to these direct medical costs, nearly RM 900 million was spent on testing for CVD risk, and medication expenses for CVD patients reached almost half of the RM 1.7 billion spent on drugs for NCDs (The Direct Healthcare Cost of Non-Communicable Diseases in Malaysia, 2022). The economic impact extends beyond healthcare costs, as CVD also results in significant productivity losses. These losses were estimated to range between RM 8.91 billion and RM 12.88 billion in 2017, depending on the method used to calculate productivity loss due to absenteeism, presenteeism, and premature death (MOH, 2020). Moreover, the total DALYs lost due to CVD in 2017 were valued at RM 59.85 billion, with a higher burden seen in males compared to females (The Direct Healthcare Cost of Non-Communicable Diseases in Malaysia, 2022). Without effective interventions, the economic and health burdens of CVD are projected to worsen, with CVD-related deaths potentially reaching around 31,000 annually by 2025, which would represent approximately 55% of all NCD-related deaths in the country. (Fong et al., 2023).

Despite ongoing efforts by the MOH to control CVD through various programmes and activities, the disease burden in Malaysia remains nearly twice that of advanced countries (Fong et al., 2023). The growing impact of CVD is further compounded by the increase in prevalence of risk factors such as hypercholesterolemia, hypertension, smoking, obesity, diabetes and alcohol use (IPH, 2020). These risk factors are becoming increasingly common, with the average age of onset for CVD in Malaysia being approximately 10 years younger than in advanced countries (Fong et al., 2023).

Moreover, Malaysia's aging population is expected to further amplify the CVD burden. By 2030, 14% of the population will be over 65 years old, an absolute increase of 7% from 2020 (Hamid, 2019). This demographic shift, combined with earlier onset of CVD, is likely to lead to a significant increase in mortality attributed to CVD in the coming years.

2.1.1 Ischemic Heart Disease (IHD) and Myocardial Infarction (MI)

IHD, also known as CAD, is primarily caused by the narrowing or blockage of coronary arteries due to the build-up of atherosclerotic plaques. These plaques, composed mainly of fats, cholesterol and other substances, lead to a reduction in coronary blood flow, thereby decreasing the supply of oxygen and nutrients to the heart muscle (Shao et al., 2020). One of the hallmark symptoms of IHD is angina pectoris, characterised by discomfort or pain in the chest, which may extend to the jaw, shoulder, back or arms (Hermiz & Sedhai, 2023). The severity of angina is often classified using the Canadian Cardiovascular Society classification (CCS-class), which correlates the angina threshold with the intensity of physical activity (Campeau, 1976). Additionally, patients with IHD may experience symptoms such as shortness of breath, fatigue, nausea and dizziness, especially during physical exertion or emotional stress.

IHD can present in various forms, including Chronic Coronary Syndrome (CCS) and ACS, depending on the stability of the atherosclerotic plaque and the extent of arterial blockage. In chronic cases, patients may experience stable angina, while acute cases can lead to more severe conditions such as MI or unstable angina (UA) (Knuuti et al., 2020). MI, occurs when there is a sudden and complete blockage of a coronary artery, leading to the necrosis of heart muscle tissue (Ojha & Dhamoon, 2023). This event is typically marked by a rise and/or fall in cardiac troponin levels, with at least one value exceeding

the 99th percentile of the upper reference limit (Knuuti et al., 2020; Thygesen et al., 2012).

MI is further classified into different subtypes based on its underlying cause and clinical presentation. For instance, a Type 1 MI is spontaneous and results from a primary coronary event such as plaque rupture, while a Type 2 MI is secondary to an imbalance between oxygen supply and demand in the myocardium (Thygesen et al., 2007). Patients diagnosed with MI are also categorised based on the presence or absence of ST-segment elevation on an electrocardiogram (ECG), leading to classifications of ST-segment elevation myocardial infarction (STEMI) and non-ST-segment elevation myocardial infarction (NSTEMI). STEMI requires urgent medical intervention to prevent severe complications such as arrhythmias and heart failure (HF), both of which can significantly impair the heart's ability to pump blood effectively (Avdikos et al., 2022).

Arrhythmias, which are irregularities in the heartbeat, can manifest as either tachyarrhythmias (rapid heart rate) or bradyarrhythmias (slow heart rate), both of which can be detected as irregular heartbeats (Desai & Hajouli, 2023). These complications, along with heart failure resulting from acute left ventricular (LV) dysfunction, highlight the severity of MI and the importance of early diagnosis and treatment using tools such as echocardiography to assess the extent of myocardial injury and LV function (Minicucci et al., 2011).

2.1.2 Stroke

Stroke is a severe neurological condition that occurs when blood supply to the brain is disrupted, leading to a lack of oxygen and essential nutrients that the brain needs to function (Tadi & Lui, 2023). This disruption can result from either a blockage in the blood vessels (ischemic stroke) or bleeding within or around the brain (haemorrhagic

stroke) (Kuriakose & Xiao, 2020; National Heart, Lung, and Blood Institute, 2023). Hence, strokes are broadly categorized into ischemic and haemorrhagic types. Ischemic stroke, the most common form, occurs when an artery supplying blood to the brain becomes blocked, typically due to a thrombus or embolus. This blockage leads to a reduction in blood flow and subsequent brain damage (Ibrahim & Murr, 2022). Ischemic strokes are further classified based on the underlying causes, including atherothrombosis of large vessels, small vessel disease, and embolism (Sacco et al., 2013; Mastrangelo et al., 2022). In Malaysia, ischemic stroke accounts for approximately 79.4% of all stroke cases, making it the predominant type, followed by haemorrhagic stroke at 18.2%, transient ischemic attacks (TIA) at 2%, and unclassified strokes at 0.4% (Aziz et al., 2015).

Diagnosing ischemic stroke involves a thorough evaluation of the patient's history and a physical examination, supported by diagnostic tests aimed at identifying the neurological deficit, the location, type, and cause of the lesion, as well as potential complications (Hui et al., 2024; NIH, 2023). Symptoms of ischemic stroke can vary widely depending on the affected area of the brain and the extent of the injury. However, common symptoms include sudden onset of weakness or numbness in the face, arm or leg, often on one side of the body, confusion, trouble speaking or understanding speech, visual disturbances, dizziness, loss of balance or coordination, and severe headache (Hui et al., 2024). Prompt diagnosis is critical, as early treatment can significantly improve outcomes. Nevertheless, diagnosing a stroke within the first few hours can be challenging due to factors such as atypical or evolving symptoms, patient distress, delayed imaging, or normal brain imaging findings (Hankey, 2017).

Haemorrhagic stroke, on the other hand, occurs when a blood vessel in the brain ruptures, leading to bleeding within or around the brain tissue. This type of stroke is further divided into intracerebral haemorrhage (ICH), where the bleeding occurs within the brain parenchyma, and subarachnoid haemorrhage (SAH), where the bleeding occurs in the space surrounding the brain (Unnithan et al., 2023). The causes of haemorrhagic stroke include hypertension, cerebral amyloid angiopathy, anticoagulant use, and vascular abnormalities. Symptoms of ICH may include sudden onset headache, vomiting, neck stiffness, elevated blood pressure, and abrupt neurological deficits. SAH, on the other hand, typically presents with a sudden and severe headache, along with vomiting, syncope, photophobia, neck stiffness, seizures, and altered consciousness (Unnithan et al., 2023; Rymer, 2011).

For diagnosing haemorrhagic stroke, a brain computed tomography (CT) scan is the most commonly used imaging tool due to its high sensitivity in detecting acute bleeding. Magnetic resonance imaging (MRI) may also be employed, particularly T2* susceptibility-weighted imaging and gradient echo sequences, which can detect previous haemorrhages and differentiate between primary haemorrhage and haemorrhage secondary to an infarct (Hermier & Nighoghossian, 2004). Blood tests, including assessments of bleeding time, clotting time, platelet count, prothrombin time (PT), and activated partial thromboplastin time (aPTT), are also critical in identifying potential bleeding disorders that could contribute to haemorrhagic stroke (Unnithan et al., 2023; Rountree et al., 2024).

2.2 CVD Risk Factors

Given the substantial impact of cardiovascular diseases, it is essential to identify and manage the contributing risk factors early, allowing for the timely implementation of interventional strategies. Risk factors for CVD can be broadly categorized into non-

modifiable factors such as age, gender and genetics, and modifiable factors, including lifestyle choices like diet, physical activity and smoking (Brown et al., 2023). While many CVD risk factors are universal, regional variations exist, influenced by genetic, cultural and environmental factors. This review will discuss common global CVD risk factors, with a focus on those specific to Asia and Malaysia, and highlight the most significant risk factors and their fluctuations over time.

Globally, CVD risk factors are well-documented, and they fall into several categories: demographic, physiological, behavioural and environmental. Demographic risk factors for CVD include age, gender, ethnicity, family history, education and income level. Age plays a critical role in the deterioration of cardiovascular functionality, significantly increasing the risk of CVD as individuals grow older. The prevalence of CVD, including conditions such as atherosclerosis, stroke, and myocardial infarction, rises markedly in both men and women with advancing age (Rodgers et al., 2019). According to the 2021 Heart Disease & Stroke Statistical Update Fact Sheet, based on data from 2015 to 2018, an estimated 126.9 million American adults (49.2%) had one or more types of CVD. In the 60–79-year-old age group, 77.5% of males and 75.4% of females had CVD, including hypertension (Virani et al., 2021). Among those aged 80 and older, the prevalence increases further, with 89.4% of males and 90.8% of females affected by CVD. This trend underscores the substantial burden that older adults present to the healthcare system, primarily due to the high prevalence of CVD in this population.

Research consistently shows that men generally have a higher absolute risk of developing CVD and experience higher rates of CVD-related deaths compared to women, particularly due to a higher propensity for coronary heart disease (CHD). Men tend to develop CVD at a younger age, with CHD mortality rates being significantly higher in middle-aged men (30–64 years) than in women of the same age group, where the

mortality rate ratio of men to women is approximately 4–5 (Bots et al., 2017; George et al., 2015). This difference persists, albeit to a lesser extent, into older age, where the ratio drops to around 2. Despite this, stroke mortality tends to be more similar between sexes, with men-to-women stroke mortality rate ratio of about 1.5–2 until old age (Bots et al., 2017). While CVD-related mortality rates have decreased in both men and women over the past few decades, the reduction has been more pronounced in men, particularly in age-standardized CHD mortality rates. However, after menopause, women's risk of CVD increases to a level comparable to that of men, likely due to significant changes in lipid profiles, which elevate the risk of conditions such as hypertension and diabetes (Benjamin et al., 2019). Ethnicity also plays a significant role, with studies indicating that Asian Indian men are more likely to have multiple CVD risk factors compared to other ethnic groups (Mohd Ghazali et al., 2015).

Key physiological risk factors include hypertension, hypercholesterolemia, diabetes mellitus, and obesity. Hypertension, defined as blood pressure levels of $\geq 130/80$ mmHg, is a major contributor to CVD, leading to complications such as heart failure, coronary artery disease, and stroke (Fuchs & Whelton, 2020). Hypercholesterolemia, or dyslipidemia, is characterized by elevated levels of total cholesterol and low-density lipoprotein cholesterol (LDL-C), contributing to atherosclerosis and increasing the risk of myocardial infarction and stroke (Alloubani et al., 2021). Diabetes mellitus, particularly type 2 diabetes, is another significant risk factor, leading to increased inflammation, oxidative stress and endothelial dysfunction, all of which accelerate atherosclerosis and increase the risk of coronary artery disease (Sapra & Bhandari, 2023; Martín-Timón et al., 2014). Obesity, particularly abdominal obesity characterized by a high waist-to-hip ratio, further exacerbates these risks by contributing to hypertension, dyslipidemia and insulin resistance (Bays et al., 2022).

Smoking and physical inactivity are critical behavioural risk factors for CVD. Smoking accelerates atherosclerosis by increasing inflammation, promoting thrombosis, and oxidizing LDL-C (Klein, 2022). Physical inactivity contributes to increased CVD risk by impairing metabolic parameters and weakening the cardiovascular system (Warren et al., 2010). Globally, physical inactivity is responsible for 7.2% of all-cause deaths and 7.6% of CVD deaths, according to a study by Katzmarzyk et al. (2022). The study also found that the contribution of physical inactivity to non-communicable diseases ranges from 1.6% for hypertension to 8.1% for dementia. Environmental risk factors include exposure to air pollution, such as particulate matter (PM_{2.5}), which has been linked to increased CVD mortality and hospital admissions (Feigin et al., 2021). Urbanization and economic development in many regions have led to higher exposure to such pollutants, further contributing to the global burden of CVD.

In Asia, there is a higher prevalence of stroke and CHD deaths compared to Western countries. CVD was the leading cause of death in Asia in 2019, accounting for approximately 10.8 million deaths, which represented about 35% of the total deaths in the region (Zhao, 2021; Roth et al., 2020). Notably, nearly 39% of these CVD deaths were premature, defined as deaths occurring before the age of 70. This proportion of premature deaths was significantly higher in Asia compared to the United States (23%) and Europe (22%). Most of these deaths were due to IHD (47%) and stroke (40%) (Zhao, 2021; Roth et al., 2020). From 1990 to 2019, the number of CVD deaths in Asia nearly doubled, from 5.6 million to 10.8 million, highlighting the increasing burden of CVD in the region. There are significant differences in the prevalence of stroke and CHD between South and East Asian populations as well. South Asians have higher rates of diabetes and insulin resistance, contributing to a greater risk of CHD, while East Asians, particularly those from countries other than Japan, have higher rates of hypertension and stroke (Zhang et al., 2022). Generally, CVD rates are higher in Central and Eastern Asia than in

Southern and Southeast Asia. Moreover, IHD and stroke are the most common causes of CVD in Asia, though their prevalence varies significantly across different regions (Zhao, 2021; Roth et al., 2020). IHD is the dominant cause of CVD deaths in Central, Western and Southern Asia, while stroke deaths are more common in Eastern and Southeastern Asia (Zhao, 2021; Roth et al., 2020). These differences highlight the importance of considering regional factors when assessing CVD risk.

Genetic factors play a crucial role in the prevalence of certain CVD risk factors in Asia. For instance, familial hypercholesterolemia (FH) is a genetic disorder more commonly observed in some Asian populations, leading to significantly elevated LDL-C levels and an increased risk of early-onset CVD (Jackson et al., 2021). Moreover, genetic predispositions to diabetes and metabolic syndrome are more prevalent in South Asian populations, further increasing their CVD risk (Amini et al., 2021; Wolf et al., 2022). Cultural practices and lifestyle choices also contribute to the unique CVD risk profile in Asia. Diets high in refined carbohydrates, salt and unhealthy fats, combined with low levels of physical activity, exacerbate the prevalence of risk factors such as obesity, hypertension and diabetes (Bhardwaj et al., 2016). Additionally, the high prevalence of smoking, particularly among men, further increases the risk of CVD in this region (Roth et al., 2020).

Malaysia's multi-ethnic population presents unique challenges in managing CVD risk. Ethnic Indians in Malaysia have the highest prevalence of abdominal obesity and metabolic syndrome, making them particularly vulnerable to CHD (Lim & Cheah, 2016). Bumiputera Sarawak, Malays and Indians have the highest rates of hypertension, hypercholesterolemia and diabetes, respectively (IPH, 2020). The National Cardiovascular Disease (NCVD) Database Registry, one of Malaysia's first prospective multi-center registries focused on the treatment and prevention of CAD, reported

significant ethnic disparities in the occurrence and management of ACS among high-risk groups. The registry, encompassing data from over 15 Ministry of Health hospitals, universities and the National Heart Institute, identified distinct risk factor profiles among the major ethnic groups in Malaysia (Lu & Nordin, 2013). For instance, Malays were reported to have a higher body mass index (BMI), Chinese individuals exhibited the highest rates of hypertension and hyperlipidaemia, and Indians showed a greater prevalence of diabetes mellitus (DM) and a family history of premature CAD (Lu & Nordin, 2013). Notable differences in clinical outcomes were also observed. While in-hospital mortality rates for STEMI were similar across ethnic groups, Malays had a higher in-hospital mortality rate for NSTEMI and UA compared to Chinese and Indians (Lu & Nordin, 2013). These variations highlight the need for targeted interventions that consider ethnic and cultural differences.

Hypertension, hypercholesterolemia and diabetes remain the three major CVD risk factors in Malaysia, affecting millions of people. The 2023 NHMS revealed that 33.3% of Malaysian adults, or nearly 7.5 million people, have high cholesterol. Additionally, 15.6% of adults, representing about 3.6 million people, have diabetes, while 29.2% (6.7 million) suffer from hypertension. The rising prevalence of these conditions highlights the increasing burden of CVD in the country. Moreover, more than half a million Malaysian adults, or 2.5% of the population, suffer from all four key NCDs — diabetes, hypertension, high cholesterol and obesity — placing them at significant risk of stroke or heart disease (IPH, 202).

Smoking and physical inactivity continue to be prevalent risk factors in Malaysia, particularly among men. According to the 2023 NHMS report, 19% of Malaysian adults are current smokers, with a significant gender disparity in smoking prevalence. Additionally, 5% of adults are current e-cigarette users, and 10.4% are current alcohol

drinkers, with 45.1% of drinkers engaging in binge drinking. Despite some improvements in physical activity levels, 29.9% of the population remains physically inactive, with sedentary behaviour reported in 49.9% of adults. These trends are particularly concerning among women, older adults and urban residents (IPH, 2024). Moreover, unhealthy dietary practices persist, contributing to the growing burden of CVD in Malaysia. The 2023 NHMS found that 95.1% of Malaysian adults do not consume the recommended five servings of fruits and vegetables per day, and 22.2% do not drink the recommended six glasses of water daily. The prevalence of overweight, obesity and abdominal obesity has increased by almost 10% since 2011, further exacerbating the nation's CVD risk profile (IPH, 2024). Recent studies explored that urbanization in Malaysia has led to increased exposure to environmental risk factors such as air pollution. Studies have shown a link between air pollution and increased CVD-related hospital admissions, further highlighting the impact of environmental factors on CVD risk in Malaysia (Song et al., 2022; Tajudin et al., 2019). The rapid urbanization and economic development in the country has also contributed to changes in lifestyle and dietary habits, increasing the prevalence of obesity, hypertension and diabetes.

2.3 CVD Risk Scores

Cardiovascular risk scores are essential tools in clinical practice, enabling healthcare providers to estimate an individual's likelihood of experiencing a cardiovascular event over a specified period. These tools play a critical role in primary prevention strategies, guiding decisions on whether to initiate interventions for risk factor management. Risk scores are typically presented as calculators, where clinicians input patient data to receive an estimated absolute risk, often expressed as a percentage. Developed using extensive datasets and statistical models, these scores are tailored to

specific populations and regions. This section will explore the most widely used CVD risk scores, with a particular focus on the FRS.

2.3.1 Framingham Risk Score (FRS)

The FRS is one of the most foundational and influential tools in CVD risk prediction. Originating from the Framingham Heart Study (FHS), the FRS has played a pivotal role in shaping modern preventive cardiology. The FHS began in 1948 as a long-term cohort study aimed at identifying common factors that contribute to cardiovascular disease, following the alarming increase in cardiovascular morbidity and mortality in the United States during the mid-20th century (Oppenheimer, 2011). The study initially recruited 5,209 adult residents of Framingham, Massachusetts, and extended over multiple decades, encompassing three generations of participants (Terry et al., 2007; Dawber et al., 1951).

The original purpose of the Framingham study was to investigate the epidemiology of cardiovascular disease and to understand the risk factors that contribute to its development. Over the years, the study led to the identification of several major cardiovascular risk factors, such as cigarette smoking, high blood cholesterol, high blood pressure, diabetes, atrial fibrillation, obesity and physical inactivity (Berger et al., 2010). These discoveries revolutionised the understanding of cardiovascular disease and provided a scientific basis for the development of risk prediction models like the FRS.

The first iteration of the Framingham Risk Score was introduced in 1976, designed to estimate the 8-year risk of developing cardiovascular disease (Kannel et al., 1976). This risk prediction model utilized a logistic regression approach to incorporate multiple variables: age, serum cholesterol levels, systolic blood pressure, smoking status, left ventricular hypertrophy (as determined by ECG) and glucose intolerance. The output

of the model was an estimated risk of developing any of several cardiovascular conditions, including coronary heart disease, atherothrombotic brain infarction (stroke), intermittent claudication and hypertensive heart disease (Kannel et al., 1976).

One of the key innovations of the Framingham Risk Score was its use of a single, unified risk function, allowing clinicians to estimate the overall risk of cardiovascular disease for an individual rather than focusing on specific types of cardiovascular events. This made the tool particularly useful in primary care settings, where it could guide decisions about preventive interventions, such as lifestyle modifications or the initiation of medications like statins or antihypertensives. However, the initial FRS was not designed for use in secondary or tertiary prevention, where the focus would be on preventing recurrent events in individuals with established cardiovascular disease.

As the Framingham study continued, researchers sought to refine the risk prediction model to improve its accuracy and applicability. In 1991, Anderson et al. expanded the Framingham Risk Profile by developing a set of equations to predict the risk of specific cardiovascular events, including myocardial infarction, CHD death, stroke and cardiovascular death. This approach recognized that different risk factors might have varying degrees of influence on different types of cardiovascular events. For example, hypertension was found to have a stronger association with stroke risk than with CHD, while cholesterol levels were more closely linked to CHD outcomes.

In 2008, D'Agostino et al. introduced a significant update to the Framingham Risk Score, creating a multivariable risk function that could estimate an individual's 10-year risk of developing general cardiovascular disease. This version of the FRS incorporated several key variables, including age, total cholesterol, high-density lipoprotein (HDL) cholesterol, systolic blood pressure, treatment for hypertension, smoking status and diabetes status. The updated model also introduced a non-laboratory version of the risk

function, which uses BMI instead of cholesterol levels, making it more accessible for use in settings without laboratory facilities. This non-laboratory model was particularly valuable for primary care settings, where it provided a practical tool for risk assessment without the need for blood tests.

The 2008 version of the FRS allowed for the calculation of both general cardiovascular disease risk and specific outcomes, such as coronary heart disease, cerebrovascular disease (including stroke), peripheral arterial disease and heart failure. The flexibility of this model made it a widely used tool not only in the United States but also in other countries that adopted or adapted it for their populations. The Framingham Risk Score has several strengths that have contributed to its widespread adoption. Its development was based on a well-characterized cohort with long-term follow-up, which provided robust data for model creation. The FRS also integrates multiple risk factors, offering a comprehensive assessment of cardiovascular risk that can be easily applied in clinical practice.

2.3.2 Other Risk Scores for CVD

The Revised Pooled Cohort Equations (RPCE) were developed by the American College of Cardiology (ACC) and the American Heart Association (AHA) as an enhancement to the FRS to address its limitations, particularly its outdated cohort data and lack of racial diversity. The RPCE were derived from a more contemporary and ethnically diverse population, utilizing data from studies such as the Atherosclerosis Risk in Communities (ARIC) and the Jackson Heart Study (JHS) (Goff et al., 2014). The RPCE estimates the 10-year risk of atherosclerotic cardiovascular disease (ASCVD) in individuals aged 40–79 who are free of ASCVD, with specific emphasis on African American and White populations. The equations incorporate variables such as age, total cholesterol, HDL cholesterol, systolic blood pressure, treatment for hypertension,

diabetes status and smoking status. A notable feature of the RPCE is the use of modern statistical techniques, including logistic regression with elastic net regularization, to improve accuracy and reduce overfitting, especially in racial subgroups that were underrepresented in earlier models. This resulted in more accurate risk estimations, particularly for African Americans, who had previously experienced significant underestimation of risk (Yadlowsky et al., 2018). The RPCE thus provides a more nuanced and reliable tool for cardiovascular risk assessment in the modern, diverse U.S. population.

The SCORE2 model is an evolution of the original SCORE project, designed to address the limitations of the FRS when applied to European populations. The original SCORE project was initiated in response to the overestimation of cardiovascular risk by the FRS in European countries with lower CHD prevalence. SCORE2 was developed using data from 45 modern cohort studies across 13 European countries, making it more representative of contemporary European populations (Crea, 2021). SCORE2 estimates the 10-year risk of both fatal and non-fatal CVD events in individuals aged 40–69 who have no history of CVD or diabetes. It includes variables such as age, gender, smoking status, systolic blood pressure, total cholesterol and HDL cholesterol. SCORE2 is calibrated to four different risk regions in Europe low, medium, high, and very high based on recent World Health Organization (WHO) data, allowing for tailored cardiovascular risk management according to regional risk levels. This model's ability to capture variations in population risk across Europe makes it a valuable tool for clinicians in tailoring CVD prevention strategies to specific regional contexts (SCORE2 working group and ESC Cardiovascular risk collaboration, 2021).

The WHO-CVD Risk Charts were developed to provide a global risk prediction tool that is applicable to low and middle-income countries, addressing the gap left by existing models that were primarily designed for high-income, predominantly Caucasian populations. The WHO charts are derived from data across 21 global regions, using information such as age, sex, smoking status, systolic blood pressure, diabetes history and total cholesterol levels. They aim to estimate the 10-year risk of fatal and non-fatal major cardiovascular events, including myocardial infarction and stroke (Kaptoge et al., 2019).

The WHO-CVD Risk Charts are available in both laboratory-based and non-laboratory-based formats, offering flexibility for use in resource-constrained settings. The laboratory-based charts utilize cholesterol and diabetes data, while the non-laboratory-based charts substitute these with BMI, making them accessible for communities with limited access to laboratory facilities. This dual approach ensures broader applicability across diverse healthcare environments, enhancing global efforts to manage and reduce the burden of cardiovascular disease, particularly in regions with limited healthcare infrastructure (WHO, 2020).

2.3.3 Limitations of Conventional CVD Risk Scores

One of the primary limitations of the FRS is its potential bias when used with diverse ethnic groups. The Framingham cohort, which primarily consisted of white middle-aged Americans, may not accurately reflect the risk profiles of individuals from different ethnic backgrounds (Zhang et al., 2022). Studies have shown that the FRS can overestimate or underestimate risk in non-Caucasian populations, leading to inaccurate risk stratification. For example, the FRS often overestimates CVD risk in Asian populations due to differences in baseline blood pressure, cholesterol levels, and the prevalence of cardiovascular events compared to the original Framingham cohort (Zhang et al., 2022; Liu et al., 2004; D'Agostino et al., 2001). This limitation underscores the

need for recalibration of the FRS or the development of new risk models tailored to specific populations (Selvarajah et al., 2014).

Age is another critical factor that affects the accuracy of the FRS. The model tends to underestimate CVD risk in younger individuals and overestimate it in older adults, particularly those over 75 years of age (Talha et al., 2024). This discrepancy arises because the original Framingham cohort included fewer very young or elderly participants, leading to a model that does not fully capture the risk dynamics in these age groups. Moreover, the FRS does not account for several emerging and lifestyle-related risk factors that have been increasingly recognized in recent years. Factors such as socioeconomic status, family history of premature coronary artery disease, and risk-enhancing are not included in the FRS (Gray, 2009). Gender differences also present a limitation for the FRS. The model has been shown to underestimate CHD risk in women, particularly younger women (Talha et al., 2024). This is partly due to the lower incidence of traditional risk factors such as hypertension and cholesterol in women compared to men.

While the RPCE and other risk scores like SCORE2 and the WHO-CVD Risk Charts have attempted to address some of the limitations of the FRS, they also have their own drawbacks. For instance, the RPCE, while more representative of diverse populations, still faces challenges such as underestimating risk in individuals with low socioeconomic status or those with a family history of premature cardiovascular events (Grundy et al., 2019). Additionally, the SCORE model, despite its recalibration for different European regions, can still misestimate risk in certain age groups, particularly older adults (van Trier et al., 2024). The WHO-CVD Risk Charts, while valuable for low- and middle-income countries, may lack precision when applied to populations with different baseline risk factors or access to healthcare and was also found to misestimate CVD risk in women (Selvarajah et al., 2014).

Another limitation of traditional CVD risk prediction tools like the Framingham FRS is their focus on a 10-year prediction horizon. While this long-term assessment provides valuable insights, it may not be adequate for identifying individuals who are at risk of developing CVD in the short to medium term. The need for short-term risk models is becoming increasingly recognized, particularly for those requiring immediate interventions. For instance, a study by Quispe et al. (2015) in Peruvian populations highlighted that approximately one-third of participants were classified as having a high short-term risk of cardiovascular events. Among those with low short-term risk, more than half were found to have a high lifetime risk, underscoring the necessity of short-term risk tools for ensuring timely and appropriate preventive measures.

Moreover, the dynamic nature of CVD risk factors further justifies the need for shorter prediction horizons. Risk factors such as hypertension, dyslipidemia and lifestyle-related conditions can fluctuate significantly over short periods due to interventions like dietary changes, physical activity, or medication adherence (Bays et al, 2021; NHAM, 2017). Shorter prediction terms align better with these dynamic changes, providing a more immediate and relevant assessment of an individual's current risk. This enables healthcare providers to tailor interventions more effectively, reducing the likelihood of missed opportunities for intervention and improving overall patient outcomes.

2.4 Artificial Intelligence (AI)

AI has evolved significantly since its inception in the 1950s by pioneers like John McCarthy, who aimed to simulate human intelligence through computational systems. AI, encompassing both machine learning (ML) and deep learning (DL), has grown into a field within computer science dedicated to developing systems capable of performing tasks traditionally requiring human intelligence, such as learning, reasoning, problem-solving, perception, language comprehension and decision-making. The application of AI

in healthcare, particularly in disease risk prediction, has rapidly advanced and is becoming increasingly crucial (Cai et al., 2024; Banerjee et al., 2021). AI reduces costs, increases labour efficiency, and supports clinical decision-making. These benefits are especially evident in precision medicine, where AI techniques like ML and DL are extensively utilized. Currently, these techniques are being explored to develop models that predict cardiovascular outcomes, such as hospital admissions and long-term mortality. Various studies have focused on secondary CVD prevention, highlighting AI's role in improving risk prediction and patient care (Aziz et al., 2021; Kasim et al., 2022). AI's robust data-processing capabilities and ability to model non-linear interactions within large, complex datasets have proven more effective than traditional methods, leading to more accurate predictions of cardiovascular outcomes. As healthcare data continues to grow in complexity, AI's advanced algorithms provide an essential advantage in extracting meaningful insights, enabling more personalized and precise risk assessments for CVD.

2.4.1 Machine Learning (ML)

Machine learning is a sub-discipline of AI that focuses on the development of algorithms capable of learning from data to make decisions or predictions. The fundamental principle of ML involves training a model on a dataset so that it can identify patterns and relationships within the data, enabling it to generalize and perform well on new, unseen data (Sarker, 2021). This iterative learning process allows ML models to continuously improve their performance, making them particularly valuable in complex problem-solving tasks across various domains, including healthcare.

ML can be broadly categorized into three types: supervised learning, unsupervised learning and reinforcement learning. Supervised learning is one of the most commonly used approaches, especially in healthcare, where it is applied to tasks such as

classification (predicting categorical outcomes), regression (predicting continuous values), and anomaly detection (identifying unusual patterns). In supervised learning, the model is trained on a labelled dataset, meaning that the input data is paired with the corresponding correct output. The model learns to map inputs to outputs by identifying patterns and correlations within the data, enabling it to predict outcomes for new data instances. Common supervised learning algorithms used in cardiovascular research include logistic regression, decision trees, random forests, support vector machines (SVMs), and ensemble methods (Cho et al., 2021; Jiang et al., 2020; Sajeev et al., 2021).

In cardiovascular risk prediction, machine learning has proven to be a powerful tool, providing more accurate and personalized risk assessments compared to traditional statistical methods (Liu et al., 2023). The following sections will explore some of the most commonly used machine learning algorithms in this field, which are also relevant to the present study.

2.4.1.1 Random Forest (RF)

Random Forest is a highly effective and flexible machine learning algorithm, widely recognized for its robustness in both classification and regression tasks. As an ensemble learning method, it leverages the technique known as bagging, or bootstrap aggregation, to enhance model accuracy and reduce overfitting. The process involves constructing multiple decision trees, each built from a subset of the training data generated through bootstrapping, where random sampling with replacement is used.

In the classification context, Random Forest builds several decision trees, as illustrated in Figure 2.1. Each tree is trained independently on a different subset of the data. Once all trees have been constructed, the model aggregates the predictions from each tree to determine the final output, typically through majority voting. This ensemble

approach allows Random Forest to create a model that is both powerful and resilient against overfitting.

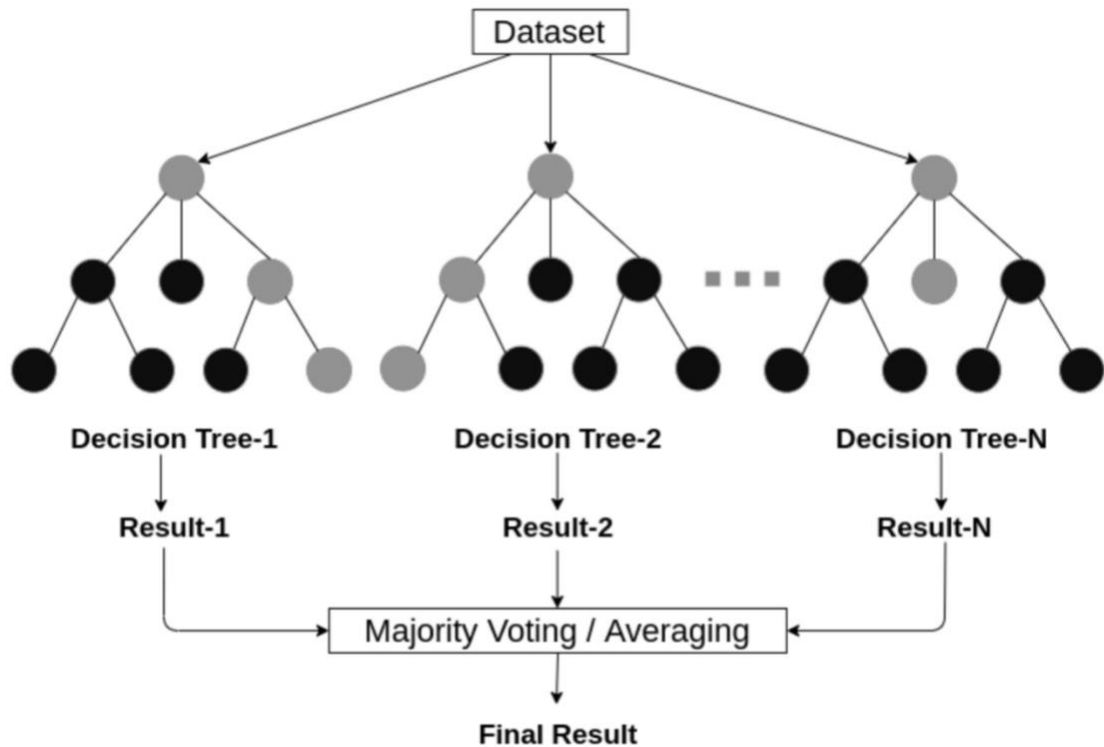


Figure 2. 1: Structure of a Random Forest model. Each decision tree is independently trained on a bootstrapped subset of the dataset. The final result is obtained through majority voting or averaging the outputs of all trees (Sarker, 2021).

One of the key strengths of Random Forest lies in its ability to introduce diversity during model development. Each decision tree in the forest is built using a random subset of features at each split, reducing both the feature space and the computational complexity. This randomization ensures that the individual trees are less correlated, leading to more accurate and stable aggregated predictions. Additionally, the parallelisable nature of Random Forest is a significant advantage. Because each decision tree operates independently, the algorithm can efficiently scale to large datasets, improving computational efficiency without compromising accuracy.

Another beneficial feature of Random Forest is the Out-Of-Bag (OOB) error estimation. When a bootstrap sample is created for training a tree, the data points not included in this sample form the OOB sample. These OOB samples are then used to validate the model, providing an internal method of cross-validation. The OOB error rate is calculated based on the proportion of incorrectly classified instances within the OOB samples, offering a reliable measure of model performance without the need for a separate validation dataset. This OOB estimate helps monitor and control overfitting, ensuring that the model generalises well to unseen data.

Two important hyperparameters in Random Forest are `mtry` and `ntree`. The `mtry` parameter, which determines the number of features considered for splitting at each node, is usually set to the square root of the total number of input features. However, adjusting this value based on the dataset's specific characteristics can optimize model performance. The `ntree` parameter dictates the number of decision trees in the forest. Although increasing the number of trees generally improves accuracy, the returns diminish after a certain point, and the computational cost can become excessive. Thus, tuning `ntree` is important to find a balance between accuracy and computational efficiency.

2.4.1.2 Support Vector Machine (SVM/SVC)

SVM is a robust and versatile algorithm primarily utilised for classification tasks, though it can also be adapted for regression and outlier detection. The core principle of SVM involves plotting data points in an n -dimensional space, where n represents the number of features. The algorithm then seeks to find the optimal hyperplane that separates the data into distinct classes (Suthaharan, 2016). This hyperplane is the decision boundary that maximizes the margin between the nearest data points from each class, known as support vectors. The wider this margin, the more robust the model is, as it reduces the likelihood of misclassification, particularly when dealing with unseen data. The diagram

in Figure 2.2 illustrates the concept of the optimal hyperplane, support vectors and the maximized margin.

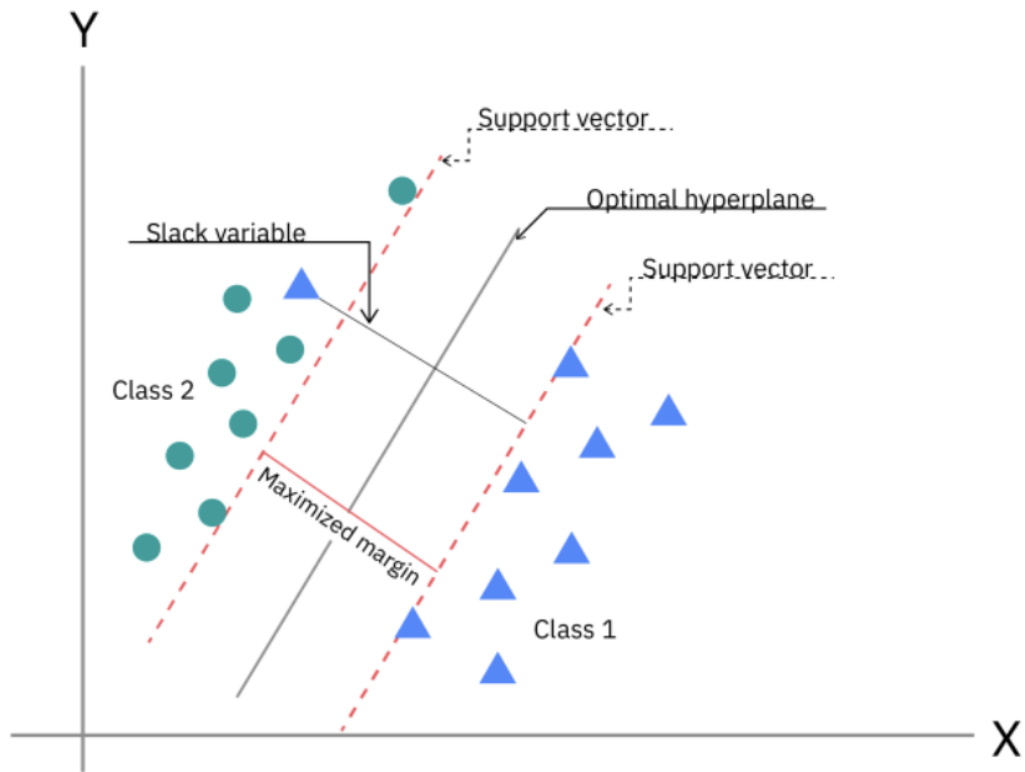


Figure 2. 2: Visualization of an SVM classification task showing the optimal hyperplane, support vectors, and maximized margin. (IBM, 2023)

One of the critical features of SVM is the kernel trick, which allows the algorithm to handle non-linear data effectively. The kernel trick transforms the original feature space into a higher-dimensional space, where a linear hyperplane can then separate the data points (Kecman, 2004). Common kernels include the linear kernel for linearly separable data, the polynomial kernel for capturing more complex relationships, and the Radial Basis Function (RBF) kernel, which is particularly effective in high-dimensional and non-linear contexts. By employing these kernels, SVM can classify data with intricate structures, making it highly applicable in various real-world scenarios.

Support Vector Classifier (SVC) is the specific implementation of SVM for classification tasks. While SVM refers to the broader algorithmic concept, SVC is the practical tool used for applying SVM to classification problems. The principles behind SVC align with those of SVM, including the use of support vectors and margin maximization. However, SVC also offers flexibility through hyperparameters that can be fine-tuned to enhance model performance. One of the crucial hyperparameters in SVC is the regularization parameter (C), which manages the trade-off between maximizing the margin and minimizing classification errors. A smaller C allows for a wider margin, which improves the model's ability to generalize, although it might result in some misclassifications on the training data. Conversely, a larger C value tightens the margin, potentially leading to overfitting as the model becomes more focused on minimizing errors in the training set. The gamma parameter, particularly relevant for non-linear kernels like RBF, influences the complexity of the decision boundary by controlling the impact of individual data points on the model's predictions. Adjusting these parameters appropriately can significantly affect the model's performance and its ability to generalize to new data.

2.4.1.3 K-Nearest Neighbours (KNN)

K-Nearest Neighbors is a widely used machine learning algorithm known for its simplicity and effectiveness in both classification and regression tasks. As a non-parametric, instance-based learning method, KNN operates by comparing a new data point to its nearest neighbors in the existing dataset, classifying it based on the majority vote among those neighbors. The parameter k represents the number of neighbors considered, and it is crucial to the algorithm's performance (Moldagulova & Sulaiman, 2017).

Figure 2.3 illustrates the impact of different values of k on the decision boundaries. A smaller k value, such as $k = 1$, results in a highly sensitive model that captures local patterns but is prone to overfitting. As k increases, the decision boundary becomes smoother, reducing sensitivity to noise but potentially causing underfitting if k is too large.

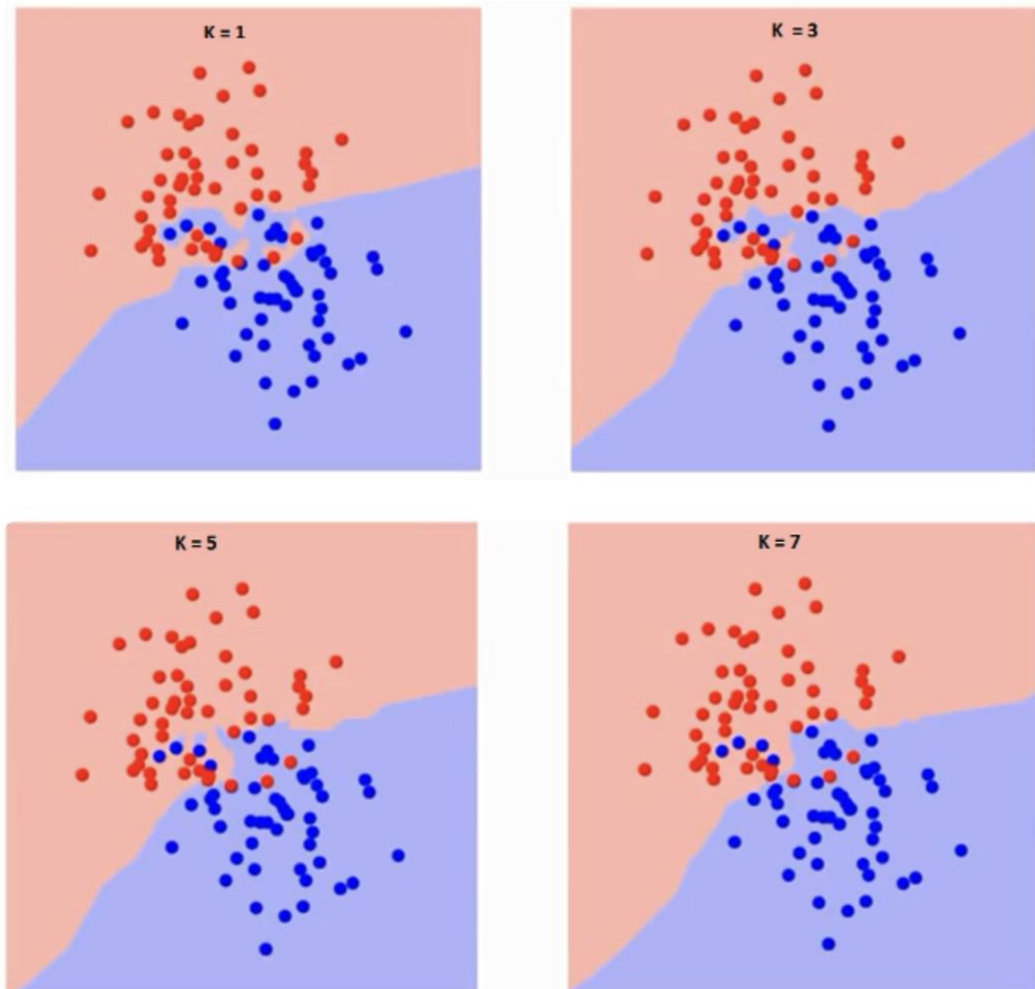


Figure 2. 3: Effect of varying k values on decision boundaries in KNN. As k increases, the boundary becomes smoother, which can prevent overfitting but may lead to underfitting if k is too high (Srivastava, 2024).

KNN's simplicity makes it easy to implement, and it requires no explicit training phase. Instead, the algorithm simply stores the training data and makes predictions in real-time by calculating the distances between the new data point and all training examples.

The distance metric typically used is Euclidean distance, though other metrics like Manhattan or Minkowski can be applied depending on the context. However, the choice of k and the distance metric is critical. A low k can make the model overly sensitive to noise in the data, while a high k can overly smooth the decision boundary, potentially leading to underfitting. As demonstrated in Figure 2.3, the decision boundary evolves with different values of k , showing the importance of selecting an optimal k .

KNN can be further enhanced by weighting the neighbors, giving more influence to closer points, which can improve accuracy in situations where proximity strongly indicates similarity. Despite its simplicity, KNN remains effective for many applications, especially when the proper k is chosen and the data is well-distributed. Its ease of use and implementation, especially in environments like Python and R, keeps it relevant for a wide range of applications, from text classification to economic forecasting (Moldagulova & Sulaiman, 2017).

2.4.1.4 Decision Tree (DT)

The Decision Tree algorithm is a prominent method in machine learning, particularly favoured for tasks involving classification and regression due to its simplicity and interpretability. The structure of a Decision Tree is hierarchical, starting with the root node, which represents the initial decision point that splits the dataset into two or more subsets based on specific features (Song & Lu, 2015). As the tree branches out, internal nodes, also known as decision or chance nodes, further divide the data, each representing a decision based on a particular feature. The ultimate goal of these splits is to create the most homogeneous subsets possible in relation to the target variable. This process of recursive partitioning continues until it reaches the leaf nodes, which represent the final outcomes or predictions. In classification tasks, these leaf nodes correspond to class labels, whereas in regression tasks, they represent continuous values (Song & Lu, 2015).

One of the defining strengths of Decision Trees is their interpretability. The tree's flowchart-like structure allows for easy visualization, making the decision-making process transparent and understandable, as illustrated in Figure 2.4. This characteristic is particularly valuable in domains where model interpretability is crucial, such as in healthcare or finance. Decision Trees are versatile and can handle both numerical and categorical data without the need for extensive preprocessing, such as feature scaling or normalization, which is often required by other algorithms like Support Vector Machines or K-Nearest Neighbors.

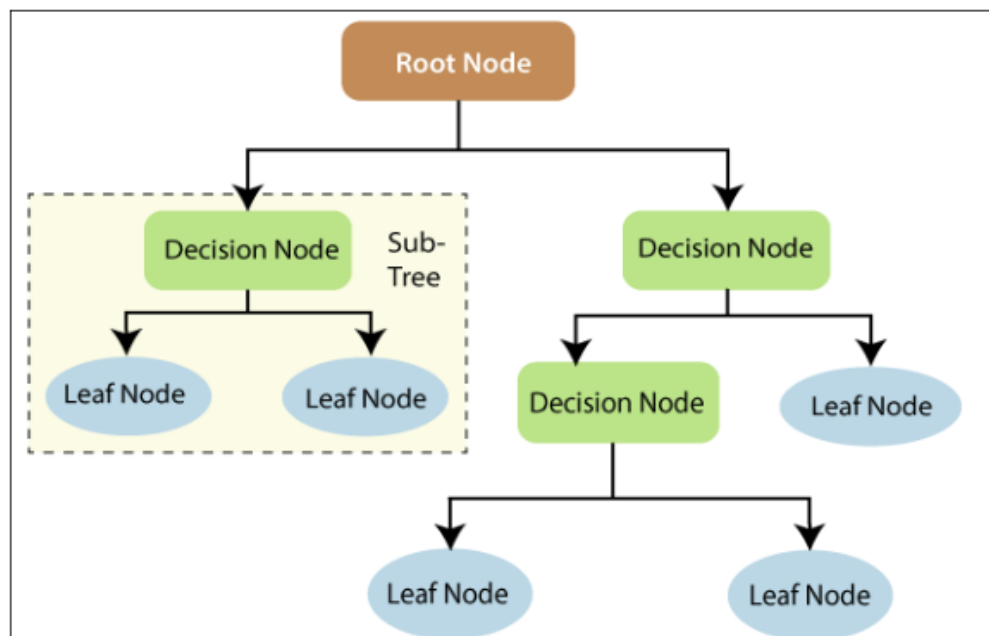


Figure 2. 4: Structure of a Decision Tree, illustrating the flow from the root node to leaf nodes through decision nodes (Jijo & Abdulazeez, 2021).

However, building an effective Decision Tree involves more than just growing the tree. The process of splitting nodes is guided by criteria like Gini impurity or entropy, aiming to maximize the homogeneity of the resulting child nodes. Yet, without proper constraints, a Decision Tree can grow excessively, capturing noise in the training data, which leads to overfitting. To counter this, stopping criteria are applied, such as limiting

the tree's depth, setting a minimum number of samples required to split a node, or defining the minimum number of samples in a leaf node. Despite these precautions, the resulting tree might still be too complex. In such cases, pruning techniques are employed. Pruning, which can be done either during the tree-building process (pre-pruning) or after the tree is fully grown (post-pruning), involves removing nodes that contribute little to the model's predictive power, thereby simplifying the model and improving its generalization to unseen data (Song & Lu, 2015).

2.4.1.5 Gradient Boosting Machine (GBM)

It operates by building an ensemble of decision trees in a sequential manner, where each subsequent tree aims to correct the errors made by the previous ones. This iterative process is guided by the principle of boosting, which combines weak learners into a robust model capable of making highly accurate predictions (Bentéjac et al., 2021). Unlike methods like Random Forest, where trees are built independently, GBM leverages the sequential correction of errors, with each tree focused on addressing the mistakes made by the ensemble up to that point. The core of GBM's approach is to fit a decision tree to the residual errors of the preceding model, minimizing the overall loss. This process allows GBM to capture complex patterns in the data by iteratively improving its predictions. For instance, if the first tree underestimates certain predictions, the next tree will be trained to adjust for those underestimations, progressively refining the model.

One of the distinguishing features of GBM is its ability to optimize a variety of loss functions, which makes it versatile for different types of predictive tasks, such as regression, classification and ranking (Bentéjac et al., 2021). Figure 2.5 illustrates how each decision tree in GBM is trained to minimize errors progressively, culminating in a model that is both powerful and highly adaptive to the data. Despite its powerful capabilities, GBM can be computationally demanding and requires careful tuning to

prevent overfitting, particularly because the sequential nature of the algorithm means that errors can propagate through the ensemble if not properly managed. However, when properly applied, GBM is a highly effective method for building predictive models, especially in scenarios involving complex and high-dimensional data.

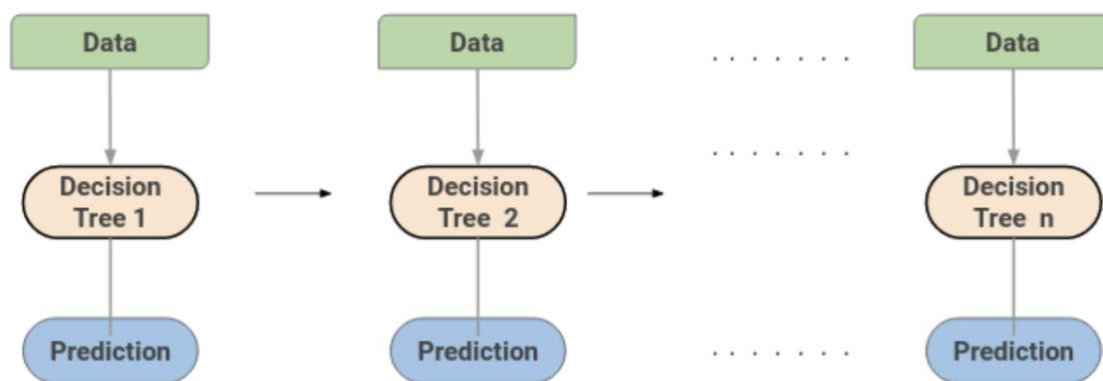


Figure 2. 5: Sequential construction of decision trees in GBM, where each tree is built to correct the errors of the previous ones. This process allows GBM to iteratively refine its predictions, leading to a robust and accurate model (Singh, 2024).

2.4.1.6 eXtreme Gradient Boosting (XGBoost)

XGBoost is an advanced implementation of the GBM algorithm, known for its efficiency and scalability. It builds an ensemble of decision trees sequentially, where each subsequent tree aims to correct the errors made by the previous ones, similar to the traditional GBM approach. However, XGBoost incorporates several enhancements that make it particularly powerful for complex machine learning tasks, especially when dealing with large-scale datasets.

One of the main advantages of XGBoost is its ability to handle overfitting effectively through the use of L1 and L2 regularization techniques (Chen & Guestrin, 2016). These techniques penalise the complexity of the model, encouraging the development of simpler, more generalizable models that perform well on new data. Additionally, XGBoost introduces a sparsity-aware algorithm, which is adept at managing datasets with many missing or zero values, making it suitable for a wide range of real-world applications.

XGBoost further distinguishes itself from standard gradient boosting by leveraging a multi-threaded approach, allowing it to utilise all available CPU cores (Ramraj et al., 2016). This parallelized tree construction significantly accelerates the training process, making XGBoost faster and more efficient than its predecessors. The algorithm also supports advanced techniques like weighted quantile sketch for approximate tree learning, enabling it to handle data with large cardinalities effectively. Furthermore, XGBoost automatically manages missing data during the training phase, enhancing its robustness and applicability.

2.4.1.7 Meta-Models Using Stacking

Stacking, also known as stacked generalization, is an advanced ensemble learning technique that combines the predictive capabilities of multiple machine learning models to create a more robust meta-model. Unlike traditional ensemble methods like bagging and boosting, which typically aggregate models of the same type, stacking allows for the integration of diverse classifiers. This diversity enables stacking to leverage the strengths of different models, thereby improving the overall predictive accuracy and generalization of the ensemble (Zian et al., 2021).

In stacking, a set of base learners (classifiers) is first trained on the dataset. The predictions generated by these base learners are then used as input features for a meta-classifier, which makes the final predictions (Patel, 2020). The meta-classifier, often referred to as the "meta-learner," is responsible for identifying the patterns and relationships among the predictions of the base learners. This hierarchical structure allows the meta-learner to correct for the biases and weaknesses of individual models, leading to improved performance.

The choice of base learners and the meta-classifier in stacking is crucial. Typically, base learners are selected based on their complementary strengths. For instance, decision trees, support vector machines, and gradient boosting models might be combined to capture different aspects of the data's complexity. The meta-classifier is often chosen from the same pool of models, although simpler models like linear regression or more complex models like gradient boosting machines can also be effective. Cross-validation is commonly employed to train the meta-classifier, ensuring that it is trained on predictions that are not overly optimistic due to overfitting.

Stacking has been shown to be particularly effective in cases where individual models have varying strengths and weaknesses. By combining these models, stacking can produce a meta-model that performs better than any single model on its own (Zian et al., 2021). This approach is especially useful in complex datasets with high dimensionality or non-linear relationships, where a single model might struggle to capture all the underlying patterns.

2.4.2 AI Model Development

Developing AI models for healthcare, particularly in predicting CVD risk, involves a complex, multidisciplinary approach that integrates domain knowledge from both medical and data science fields. The process of implementing machine learning methods in healthcare is often guided by a structured framework that ensures the robustness and accuracy of the predictions. Key stages in this process include data collection, preprocessing, model development, training, validation and evaluation (Goldstein et al., 2017).

A) Data Collection and Preprocessing

The initial phase of AI model development focuses on the acquisition and preparation of data. High-quality, relevant data is critical for the success of any ML model, and in healthcare, this data is typically sourced from electronic health records, clinical trials, and various medical databases. Raw healthcare data often contains inconsistencies, missing values, and noise, necessitating extensive preprocessing to ensure its suitability for analysis.

Preprocessing tasks include handling missing values through imputation techniques, normalizing and scaling features, and encoding categorical variables. Missing data can occur for various reasons, and the nature of the missing values—whether it is missing completely at random (MCAR), missing at random (MAR), or missing not at random (MNAR)—determines the appropriate handling strategy (Sterne et al., 2009). Imputation techniques are used to prepare the data for training, ensuring that the model is trained on a complete dataset (Rios et al., 2022).

B) Data Partitioning

Once the data is preprocessed, it is partitioned into training, validation, and testing sets. This division is essential to prevent overfitting and to ensure that the model generalizes well to new, unseen data. Studies have shown that splitting data into training and testing sets followed by cross-validation on the training set produces better results, especially with limited sample sizes (Singh et al., 2021). Commonly, 70-80% of the data is allocated for training, while the remaining 20-30% is used for testing and calibration.

C) Handling Class Imbalance

Real-world healthcare datasets often exhibit class imbalance, where one class (e.g., healthy patients) significantly outnumbers another (e.g., patients with CVD). This imbalance can lead to biased model predictions, favouring the majority class. To address this, data balancing techniques such as Random Under-Sampling (RUS), Random Over-Sampling (ROS), and Synthetic Minority Over-sampling Technique (SMOTE) are employed. These techniques adjust the class distribution to ensure that the model is exposed to a balanced dataset during training, thus improving its ability to correctly classify minority class instances (Werner de Vargas et al., 2023).

D) Feature Selection

After addressing class imbalance, the next crucial step is feature selection, which aims to reduce the dimensionality of the dataset by selecting only the most relevant features for the prediction task. This can be achieved using methods such as Recursive Feature Elimination (RFE) and SHAP-based backward elimination (BE) (Subramani et al., 2023; Liu et al., 2022). These methods help improve the model's performance by removing irrelevant or redundant features, thus enhancing both interpretability and computational efficiency.

E) Ensemble ML and Stacking

Following feature selection, various machine learning models are developed, including RF, XGBoost, DT, GBM, SVC and KNN. These models are often used as base learners in ensemble methods such as stacking, which combines their predictions to form a meta-model. Stacking is particularly effective in healthcare settings, as it leverages the strengths of different models, resulting in improved predictive performance and robustness (Zian et al., 2021).

Stacking involves training a set of base models on the original dataset and then using their predictions as inputs for a meta-classifier. This approach allows the meta-classifier to learn from the mistakes of the base models, thus improving the overall accuracy and generalization of the model. Cross-validation is often used in stacking to ensure that the meta-classifier is trained on unbiased predictions, further enhancing its robustness.

2.5 Model Evaluation Metrics

Model evaluation metrics are essential for assessing the performance of machine learning models, especially in classification tasks. These metrics help in understanding how well the model can predict outcomes and identify areas where improvements are needed. This section focuses on the various metrics used in binary classification, including the confusion matrix, accuracy, sensitivity, specificity, predictive values (PPV and NPV), Receiver Operating Characteristic (ROC) curve and AUC, and calibration metrics like the Brier score and Log Loss.

2.5.1 Accuracy

Accuracy is one of the most commonly used metrics for evaluating the performance of classification models. It is calculated as the ratio of correctly predicted instances, both true positives (TP) and true negatives (TN), to the total number of instances in the dataset (Vujovic, 2021). The formula for accuracy is expressed as:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

This metric provides a straightforward measure of how well a model performs in making predictions. However, it is important to recognize that accuracy can be misleading, particularly in cases where the dataset is imbalanced, meaning one class significantly outnumbers the other(s). For example, in a dataset with 95% negative cases and only 5% positive cases, a model that predicts all cases as negative would still achieve a high accuracy of 95%, despite failing to identify any of the positive cases. Therefore, while accuracy is a valuable metric, it should be interpreted with caution and supplemented with other metrics or considerations when working with imbalanced data or when the cost of false negatives and false positives is significant.

2.5.2 ROC - AUC

The ROC curve is a powerful tool widely used in machine learning for evaluating the performance of binary classification models. It graphically represents the trade-off between the true positive rate (sensitivity) and the false positive rate (1-specificity) across a range of threshold settings (Kumar & Indrayan, 2011). This characteristic makes the ROC curve particularly useful in scenarios where the balance between sensitivity and specificity needs to be understood and optimized.

The ROC curve is constructed by plotting sensitivity against 1-specificity for various threshold values. Unlike fixed-threshold metrics, the ROC curve provides a comprehensive view of the model's performance across all possible thresholds, thus avoiding the bias introduced by selecting a single arbitrary threshold. This approach is especially valuable in medical diagnostics and other fields where understanding the trade-off between true positives and false positives is crucial (Hajian-Tilaki, 2013).

One of the most critical aspects of the ROC curve is the AUC. The AUC is a scalar value that summarizes the overall performance of a model. It represents the probability that a randomly chosen positive instance is ranked higher than a randomly chosen negative instance (Vujovic, 2021). An AUC value ranges from 0 to 1, where a value of 0.5 indicates performance no better than random guessing, and a value of 1.0 represents perfect discrimination between the positive and negative classes. The ROC-AUC score is particularly advantageous because it is threshold-independent, making it a robust metric even when the class distribution is imbalanced (Richardson et al., 2024). This characteristic is especially important in fields like species distribution modeling, where the prevalence of certain species can vary widely.

2.5.3 Sensitivity (Recall/ True Positive Rate)

Sensitivity, also known as recall or the true positive rate, measures the proportion of actual positive cases that the model correctly identifies. It is particularly important in medical diagnostics, where failing to identify a positive case (false negative) can have serious consequences (Hicks et al., 2022). The formula for sensitivity is:

$$Sensitivity = \frac{TP}{TP + FN}$$

A high sensitivity indicates that the model is proficient at detecting positive instances, thereby minimizing the likelihood of false negatives. This is particularly vital in fields like healthcare, where failing to diagnose a condition, such as cancer, could lead to severe consequences (Marcus, 2019). The goal in such scenarios is to achieve a sensitivity as close to 1 as possible, which would mean that the model successfully identifies nearly all positive cases, ensuring that very few, if any, positive instances go undetected.

2.5.4 Specificity (True Negative Rate)

Specificity measures the proportion of actual negative cases that the model correctly identifies (Hicks et al., 2022). It is particularly relevant in situations where false positives can lead to unnecessary treatments or anxiety. Specificity is calculated as:

$$Specificity = \frac{TN}{TN + FP}$$

A model with high specificity is efficient in correctly classifying negative instances, thereby reducing the occurrence of false positives. This is especially important in situations where a false positive could result in unnecessary stress, treatment, or further testing, such as in the diagnosis of diseases like Alzheimer's (Diogo et al., 2022). Here, a high specificity ensures that individuals who do not have the disease are correctly identified, minimizing the risk of unnecessary interventions. In practice, achieving a balance between sensitivity and specificity is often necessary, as optimizing one can sometimes lead to a compromise in the other. For instance, increasing sensitivity might reduce specificity, and vice versa, depending on the threshold set for classification.

2.5.5 Prediction Values

Positive Predictive Value (PPV): PPV, also known as precision, measures the proportion of positive predictions that are true positives. It is given by:

$$PPV = \frac{TP}{TP + FP}$$

High PPV indicates that when the model predicts a positive result, it is likely correct. This is especially useful in clinical settings where the cost of false positives is high, as it ensures that positive predictions are reliable indicators of the actual condition (Parikh et al., 2008).

Negative Predictive Value (NPV): NPV measures the proportion of negative predictions that are true negatives, calculated as:

$$NPV = \frac{TN}{TN + FN}$$

High NPV indicates that a negative test result is likely correct, which is crucial for reassuring patients without the disease. PPV and NPV are influenced by the prevalence of the condition in the population, with PPV increasing and NPV decreasing as the prevalence of the disease increases (Hajian-Tilaki, 2013).

2.5.6 Confusion Matrix

The confusion matrix is an essential tool for evaluating the performance of classification models, particularly in binary classification tasks. It provides a detailed breakdown of how the model's predictions compare to the actual outcomes, allowing for a more nuanced understanding of the model's behaviour. The matrix is structured as a

table with four quadrants, each representing a different type of prediction outcome: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) as depicted in Figure 2.6 (Vujović, 2021).

TP: These are cases where the model correctly predicts the positive class. For example, if the model predicts that a patient has a disease and the patient indeed has the disease, this is counted as a TP.

TN: These are cases where the model correctly predicts the negative class. For example, if the model predicts that a patient does not have a disease and the patient indeed does not have the disease, this is counted as a TN.

FP: These occur when the model incorrectly predicts the positive class. This is also known as a Type I error. For example, if the model predicts that a patient has a disease when they do not, this is an FP.

FN: These occur when the model incorrectly predicts the negative class. This is known as a Type II error. For example, if the model predicts that a patient does not have a disease when they actually do, this is an FN (Vujovic, 2021).

Class designation		Actual class	
		True (1)	False (0)
Predicted class	Positive (1)	TP	FP
	Negative (0)	FN	TN

Figure 2. 6: Confusion matrix for the binary classification problem (Vujović, 2021).

The confusion matrix is particularly valuable because it allows for the calculation of several important performance metrics, including accuracy, sensitivity, specificity, and predictive values (PPV and NPV). In binary classification, the matrix is typically 2x2, but

it can be expanded to accommodate multi-class problems, forming a $k \times k$ matrix for k classes (Caelen, 2017).

In diagnostic modelling, the confusion matrix serves as a foundation for understanding the effectiveness of a classifier. For example, sensitivity (or recall) can be derived from the ratio of TP to the sum of TP and FN, giving an indication of how well the model identifies positive cases (Pal et al., 2022). Similarly, specificity can be calculated from the ratio of TN to the sum of TN and FP, reflecting the model's ability to correctly identify negative cases.

The confusion matrix's structure makes it an indispensable tool in evaluating classification models, especially in medical diagnostics, where the costs of false positives and false negatives can be substantial. For instance, in the prediction of myocardial infarction outcomes, a false negative (failing to predict an event that actually occurs) may have far more serious consequences than a false positive, making the careful analysis of the confusion matrix critical in such contexts (Wallert et al., 2017).

2.5.7 Calibration

Calibration metrics assess the agreement between predicted probabilities and observed outcomes. Two key calibration metrics are the Brier Score and Log Loss.

Brier Score: The Brier Score measures the mean squared difference between predicted probabilities and actual outcomes. A lower Brier Score indicates better calibration, meaning the predicted probabilities more closely match the observed outcomes. The Brier Score is particularly useful in probabilistic predictions where the accuracy of predicted probabilities is important (Yang et al., 2022).

Log Loss: Log Loss, also known as logistic loss or cross-entropy loss, penalizes confident but incorrect predictions more heavily. It is calculated as:

$$\text{Log Loss} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

Where y_i is the actual label (0 or 1), and p_i is the predicted probability for class 1. Lower log loss values indicate better model performance, particularly in scenarios where accurate probability estimation is crucial (Srivastava, 2024).

In addition to these metrics, Isotonic Regression is often employed to improve the calibration of predicted probabilities. Isotonic Regression is a non-parametric technique that ensures that predicted probabilities are monotonic, meaning they increase or stay the same as the model's confidence increases (Sarkar, 2019). This method is particularly useful when a model's probabilities are not well-calibrated, either being too extreme or not extreme enough. By applying Isotonic Regression, the calibration of a model can be significantly improved, making the predicted probabilities more reliable, which is especially important in high-stakes decision-making scenarios. Incorporating calibration techniques like Isotonic Regression alongside metrics such as the Brier Score and Log Loss ensures that a model not only performs well in terms of accuracy but also provides reliable probability estimates.

2.5.8 Net Reclassification Improvement (NRI)

The NRI is a statistical tool used to evaluate the added predictive ability of a new marker or model compared to an existing one. It quantifies how well the new model reclassifies subjects into correct risk categories, thereby assessing the improvement in prediction offered by the new model (Pencina et al., 2008, 2011). The NRI is calculated as:

$$NRI = [P(up|event) - P(down|event)] + [P(down|nonevent) - P(up|nonevent)]$$

Where, $P(up|event)$ is the proportion of events reclassified into higher risk categories.

$P(down|event)$ is the proportion of events reclassified into lower risk categories.

$P(down|nonevent)$ is the proportion of non-events reclassified into lower risk categories.

$P(up|nonevent)$ is the proportion of non-events reclassified into higher risk categories.

The NRI is particularly useful in clinical studies where new biomarkers or variables are introduced, allowing researchers to quantify the improvement in risk prediction beyond traditional risk prediction models (Fan et al., 2023).

2.6 Machine Learning Applications in CVD Risk Prediction

The integration of machine learning into healthcare has catalysed a significant shift in how cardiovascular diseases are predicted and managed. Cardiovascular diseases remain the leading cause of mortality worldwide, making early detection and accurate risk assessment crucial. Machine learning offers powerful tools capable of analysing vast and diverse datasets to uncover patterns that may be indicative of future cardiovascular events. This section reviews the current body of literature on the use of various machine learning algorithms in predicting cardiovascular disease risk, highlighting their performance across different populations.

Table 2. 1: Summary of previous studies that incorporate ML in the CVD risk

Title	Data	No. of Cases	Outcome variable	No. of selected features	ML model Performances
Cardiovascular Disease (CVD) Prediction using Machine Learning Algorithms (Marbaniang et al., 2020)	Kaggle Dataset	70,000	Presence or absence of cardiovascular disease	13	Accuracy KNN: 72.91% RF: 72.12% NB: 60.1% DT: 63.59% SVM: 65.23% LDA: 70.54%
Machine Learning Models for Cardiovascular Disease Events Prediction (Tsarapatsani et al., 2022)	Ludwigshafen Risk and Cardiovascular Health (LURIC) dataset	2,943 patients (484 annotated as dead due to CVD)	Death caused by CVD within ten years	20	AUC LR: 72.97% SVM: 70.36% XGB: 72.02% NB: 66.22% RF: 66.69% AdaBoost: 68.44%
Risk prediction of cardiovascular disease using machine learning classifiers (Pal et al., 2022)	University of California Irvine repository data	303	CVD event 1 = patient having CVD, 0 = patient not having CVD	13	AUC K-NN: 86.21% MLP: 86.41%
Heart Disease Risk Prediction Using Machine Learning Classifiers with Attribute Evaluators (Reddy et al., 2021)	Cleveland heart dataset from the UCI machine learning repository	303	Risk of developing heart disease	14	Bagging with LR: ROC Area 0.91
Cardiovascular disease risk prediction using	UK Biobank	423604	First fatal or non-	7	AUC SVM: 0.709 RF: 0.730

automated machine learning: A prospective study of 423,604 UK Biobank participants (Alaa et al., 2019)			fatal CVD event		Neural Networks: 0.755 AdaBoost: 0.759 Gradient Boosting: 0.769 AutoPrognosis (7 features): 0.744 AutoPrognosis (369 non-lab features): 0.761 AutoPrognosis (104 lab features): 0.735 AutoPrognosis (all features): 0.7774
Prediction of Cardiovascular Disease Using Machine Learning Algorithms (Dinesh et al., 2018)	Cleveland, Hungarian, Switzerland, Long Beach VA heart disease database	920	Cardiovascular Disease	14	Accuracy: LR: 0.916. RF: 0.895. Naïve Bayes: 0.910 Gradient boosting: 0.907 SVM: 0.883
Machine learning and atherosclerotic cardiovascular disease risk prediction in a multi-ethnic population (Ward et al., 2020)	Community-based outpatient healthcare system in Northern California.	262,923	5-year ASCVD event	20	AUC (on PCE-eligible cohort): LRL2: 0.749 GBM: 0.779 RF: 0.773 LRLasso: 784 XGBoost: 0.784 PCE: 0.775 AUC (for full patient cohort): LRL2: 0.808 GBM: 0.835 RF: 0.831 LRLasso: 0.825 XGBoost: 0.830
Cardiovascular Risk Prediction Using	Employee health data collected by	155108	6-year CAD or ASCVD event	6	AUC (CAD): RSF: >0.84

Machine Learning in A Large Japanese Cohort (Matheson et al., 2022)	Toshiba Corporation (2011 to 2017)				AUC(ASCVD): RSF: >0.82
Predicting Australian Adults at High Risk of Cardiovascular Disease Mortality Using Standard Risk Factors and Machine Learning (Sajeev et al., 2021)	The Northwest Adelaide Health Study (NWAHS), the Australian Diabetes, Obesity, and Lifestyle study, and the Melbourne Collaborative Cohort Study (MCCS)	NWAHS: 3654 AusDiab: 10150 MCCS: 32611	15-year CVD mortality	8	NWAHS AUC: FRS: 0.837 LR: 0.874 LDA: 0.874 SVM: 0.873 RF: 0.854 AusDiab AUC: FRS: 0.850 LR: 0.900 LDA: 0.901 SVM: 0.902 RF: 0.891 MCCS: AUC: FRS: 0.754 LR: 0.753 LDA: 0.758 SVM: 0.758 RF: 0.781 Combined: AUC: FRS: 0.802 LR: 0.852 LDA: 0.852 SVM: 0.851 RF: 0.832
Development of machine learning-based models to predict 10-year risk of cardiovascular disease: a prospective cohort study. (You et al., 2023)	UK Biobank	473611	10-year CVD event	10	AUC: LCGB: 0.762 XGBoost: 0.761 ANN: 757 SVM: 0.756 RF: 0.755 LR: 0.752 KNN: 0.750

Study of Cardiovascular Disease Prediction Model Based on Random Forest in Eastern China (Yang et al., 2020)	Cohort data from Zhejiang province, China	25,231	3-year risk of CVD	30	AUC: RF: 0.7872; AdaBoost: 0.7862; Bagged Trees: 0.7448; Naïve Bayes: 0.7074; CART: 0.7025; Multivariate Regression: 0.7143 (Benchmark)
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Chapter 3: RESEARCH METHODOLOGY

3.1 Study Framework

The study framework involves developing two predictive models using patient data from the REDISCOVER database, focusing on a three-year prediction window. Model 1 is designed to predict CVD and includes calibration and NRI analysis to enhance prediction reliability and assess reclassification improvements over the FRS. Model 2 focuses on distinguishing between MI and stroke. The process includes feature selection using techniques like RFE and SHAP-based backward elimination, data balancing through RUS, ROS, or SMOTE, and model development with various machine learning algorithms, including ensemble methods. The models undergo training using k-fold cross-validation, and the best model is selected based on performance metrics such as AUC, accuracy, and confusion matrix results (Figure 3.1)

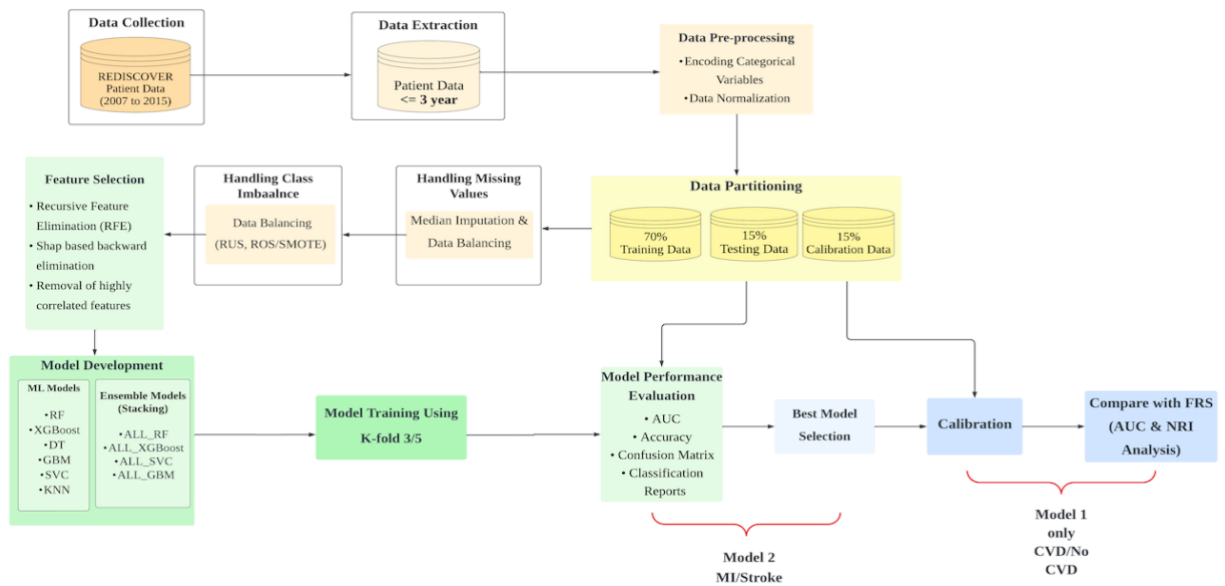


Figure 3. 1: Study Framework

3.2 Dataset

The dataset used in this study is sourced from the Responding to Increasing Cardiovascular Disease Prevalence (REDISCOVER) Study, a long-term observational research project initiated by Universiti Teknologi MARA (UiTM) in 2007. This study, conducted in various regions across Malaysia, aims to investigate the links between cardiovascular diseases and various social, behavioural, genetic, and environmental factors within diverse Malaysian communities.

Malaysia's population is known for its rich ethnic diversity, with significant groups originating from Indo-China, South Asia, and East Asia. To capture this diversity, the REDISCOVER Study recruited participants from eight states across Malaysia, including Selangor, Kuala Lumpur, Kelantan, Pahang, Terengganu, Negeri Sembilan, Sabah, and Sarawak. These regions were chosen to ensure that the sample represented the major ethnic groups in the country, including Malay, Chinese, Indian, and indigenous populations from Sabah and Sarawak. Both urban and rural areas were included to provide a comprehensive understanding of cardiovascular health across different environments.

Recruitment efforts were made through local community leaders and direct invitations to households, targeting individuals aged 30 years and above. Participants were invited to screening sessions, with an emphasis on fasting for 8 hours before attending. The response rates varied between 60% and 70% across different recruitment areas, leading to a total enrolment of 12,617 participants, all of whom provided written informed consent. The initial recruitment phase took place between 2007 and 2015, with follow-up assessments scheduled every three years. On average, participants were monitored over a period of 12.05 years.

The dataset collected from these participants includes an extensive range of 202 features divided into 11 categories: demographics, medical history, family medical history, current medications, anthropometry, clinical measurements, ECG measurements, spirometry measurements, echocardiogram measurements, ankle-brachial index (ABI) measurements, neighborhood walkability assessments, and physical activity questionnaires. The primary outcomes of interest in this study are fatal and non-fatal cardiovascular events occurring within ten years of the baseline assessment. These events are tracked through hospital records and national death certificates, with follow-up assessments conducted every three years.

Cardiovascular events recorded in the REDISCOVER Study include both fatal and non-fatal outcomes. Fatal events encompass a range of conditions, including MI, stroke, congestive heart failure, arterial aneurysm rupture, pulmonary embolism, arrhythmia-related deaths, deaths following cardiovascular procedures, and other fatal cardiovascular incidents. Non-fatal events cover MI, stroke, congestive heart failure, unstable angina, and atrial fibrillation (AF), both during and outside of medical procedures.

The following table outlines the 48 features specifically analysed in this study:

Table 3. 1: Feature involved in this study

Variables	Names	Data Type
Sociodemographic		
area	Survey Area	categorical
gender	Gender	categorical
age	Age	continuous
Current Medication		
asthma_med	Asthma Medication	binary_categorical

Variables	Names	Data Type
chol_med	Cholesterol Medication	binary_categorical
stroke_med	Stroke Medication	binary_categorical
diabetes_med	Diabetes Medication	binary_categorical
hpt_med	Hypertension Medication	binary_categorical
Family Medical History		
hpt_f	Hypertension Family	categorical
diabetes_f	Diabetes Family	categorical
stroke_f	Stroke Family	categorical
cancer_f	Cancer Family	categorical
heartdisease_f	Heart Disease Family	categorical
Clinical Measurements		
menopause	Menopause	binary_categorical
smoking	Smoking	categorical
glucose_baseline	Glucose Level	continuous
tc_baseline	Total Cholesterol	continuous
tg_baseline	Triglyceride	continuous
hdl_baseline	HDL	continuous
ldl_baseline	LDL	continuous
bps	Systolic BP	continuous
bpd	Diastolic BP	continuous
pulse	Pulse	continuous
Anthropometric measurements		
handgrip_dominant	Handgrip Strength (Dominant)	continuous
handgrip_nondominant	Handgrip Strength (Non-dominant)	continuous
rightarm	Right Arm Circumference	continuous
rightcalf	Right Calf Circumference	continuous
head	Head Circumference	continuous

Variables	Names	Data Type
weight	Weight	continuous
height	Height	continuous
bmi	BMI	continuous
hip	Hip Circumference	continuous
waist	Waist Circumference	continuous
whr	Waist Hip Ratio	continuous
Spirometry Measurements		
FEV1	Forced Expiratory Volume in 1 Second	continuous
FVC	Forced Vital Capacity	continuous
PERF	Peak Expiratory Flow Rate	continuous
Medical History		
diabetes_b	Diabetes	binary_categorical
highchol_b	High Cholesterol	binary_categorical
hpt_b	Hypertension	binary_categorical
heartdisease_b	Heart Disease	binary_categorical
hepatitis_b	Hepatitis	binary_categorical
cancer_b	Cancer	binary_categorical
copd_b	Chronic obstructive pulmonary disease	binary_categorical
asthma_b	Asthma	binary_categorical
tuberculosis_b	Tuberculosis	binary_categorical
malaria_b	Malaria	binary_categorical
stroke_b	Stroke	binary_categorical

3.3 Python Packages

In this study, Python serves as the primary programming language employed for both data preprocessing and the development of machine learning models. The Python libraries and packages utilized within this research are comprehensively detailed in the appendix, offering insights into their respective functions and roles. (See Appendix for full list of Python packages & Modules used).

3.4 Data Pre-processing

In the REDISCOVER dataset, a total of 12,617 cases were initially available for analysis. For the preprocessing stage, a refined subset of 11,807 cases was selected based on specific inclusion criteria to ensure data quality and relevance for the machine learning models being developed. These criteria included:

- A) Patients aged 18 and above: This criterion ensured that the analysis focused on adults, who are the primary demographic of interest in cardiovascular disease studies.
- B) Follow-up period of up to 3 years: The study included patients with a follow-up period of 3 years to maintain a consistent timeframe for tracking outcomes.
- C) Features with less than 30% missing values: To maintain the reliability of the dataset, only features with a missing value rate below 30% were included.

3.4.1 Data Cleaning Process for Both Models

The initial data preparation involved several key steps, applied across both models:

- A) Recode the recruitment year: The recruitment years were converted from integer format to a standard year format for clarity and consistency.
- B) Include patients aged 18 and above: Participants younger than 18 years were excluded, ensuring that the analysis focused solely on adult patients.
- C) Rearrange outcome and follow-up dates: The outcomes were consolidated into a single 'outcome' column, and follow-up dates were standardized into a 'follow_up_date' column.
- D) Remove unused outcome columns: The various cardiovascular outcomes were recoded into distinct classes:
 - 1. Class 1: Included Fatal and Non-fatal MI, Fatal and Non-fatal ACS, Fatal and Non-fatal Angina, and Fatal Sakit Jantung.
 - 2. Class 2: Included Fatal and Non-fatal Stroke.
 - 3. Class 0: Represented no cardiovascular events.
- E) Exclude features with more than 30% missing values: Any feature exceeding the 30% missing data threshold was removed to maintain data integrity.
- F) Remove specific patient records: The record for patient "tr31_AE224" was excluded due to missing follow-up date information.
- G) Add "year_diff" column: This column was introduced to indicate the follow-up years for each patient, which was critical for time-related analysis.

Model 1: CVD vs. No CVD Preprocessing

For the first model, which focused on predicting the occurrence of any cardiovascular disease (CVD) versus no CVD, the preprocessing involved several additional steps:

- A) Binary outcome recoding: The dataset was filtered to include only those patients who had been followed up for a maximum of three years. The outcomes were recoded into a binary format where any cardiovascular event was marked as '1' (CVD) and no events were marked as '0' (No CVD).
- B) Duplicate removal: The dataset was cleaned to remove any duplicate records to avoid bias in the model.
- C) Binary and categorical encoding: Binary features were mapped to 0 and 1 values, while categorical variables were label-encoded to be in a numerical format.
- D) Normalization: Continuous features were normalized using MinMaxScaler to ensure uniformity across the dataset.

Model 2: MI vs. Stroke Preprocessing

The preprocessing for the second model, which aimed to differentiate between myocardial infarction (MI) and stroke patients, required a different approach:

- A) Outcome differentiation: The dataset was again filtered for a three-year follow-up period. This time, however, outcomes were divided into three distinct classes: '1' for MI, '2' for stroke, and '0' for no cardiovascular events.
- B) Outcome recoding: The outcome classes were further refined, with MI being recoded as '0', stroke as '1', and no cardiovascular events as '2'.

C) Data cleaning and encoding: Similar to Model 1, duplicate records were removed, binary features were mapped to 0 and 1 values, and categorical variables were label-encoded. Continuous variables were normalized to ensure consistency across the dataset.

Table 3. 2: Missing Value Percentage

Feature	Missing Value Percentage (%)
Area	0.00
gender	0.00
age	0.00
asthma_med	0.36
chol_med	0.82
stroke_med	1.81
diabetes_med	0.32
hpt_med	0.54
hpt_f	0.75
diabetes_f	0.74
stroke_f	0.84
cancer_f	0.93
heartdisease_f	0.88
menopause	0.00
smoking	1.30
glucose_baseline	6.17
tc_baseline	4.45
tg_baseline	4.49
hdl_baseline	4.45
ldl_baseline	4.68
bps	0.86
bpd	0.86
pulse	1.51
handgrip_dominant	9.93
handgrip_nondominant	9.93
rightarm	1.57
rightcalf	1.53
head	1.99
weight	0.93
height	1.00
bmi	1.02
hip	2.58
waist	2.55
whr	1.21
FEV1	12.85

Feature	Missing Value Percentage (%)
FVC	12.90
PERF	12.87
diabetes_b	0.26
highchol_b	14.90
hpt_b	0.15
heartdisease_b	0.36
hepatitis_b	21.66
cancer_b	0.40
copd_b	22.16
asthma_b	0.36
tuberculosis_b	21.83
malaria_b	22.00
stroke_b	0.29

3.4.2 Data Splitting

In preparation for the development of the machine learning models, the dataset underwent a careful data splitting process to ensure the models were trained, tested, and, in some cases, calibrated effectively. For Model 1, which focuses on predicting the presence or absence of CVD, the dataset was divided into three distinct parts. The majority, 70% of the data, was allocated to the training set, which was used to train the machine learning algorithm and allow it to learn patterns from the data. A further 15% of the dataset was set aside as the testing set, which served as unseen data to evaluate the model's performance and ensure it could generalize well to new data. Additionally, 15% of the dataset was designated as the calibration set (Chi et al., 2022; Jalili et al., 2021). This calibration set was crucial for fine-tuning the model's probability predictions, ensuring they were well-calibrated and reflective of real-world probabilities.

For Model 2, which is designed to differentiate between MI and stroke patients, the data was split into two parts. A significant portion, 70% of the data, was used for training, providing the model with ample information to learn from the data. The

remaining 30% was reserved as a testing set, used to assess the model's ability to correctly distinguish between MI and stroke. This approach ensured that the model could be reliably tested on unseen data, which is essential for evaluating its predictive accuracy (Gholamy, 2018).

Throughout the data splitting process for both models, stratified sampling was employed to maintain the distribution of the target variable across the different subsets. This stratification ensured that the proportions of the different classes (e.g., CVD vs. no CVD, MI vs. stroke) remained consistent in the training, testing, and calibration datasets. By doing so, the representativeness of the data was preserved, preventing any potential bias that could arise from imbalanced classes. This methodical approach to data splitting was fundamental in ensuring that the resulting models were robust, accurate, and generalizable to new data.

3.4.3 Data Imputation

In the preprocessing stage of this study, addressing missing values in the dataset was crucial to ensure the integrity and reliability of the subsequent machine learning models. To handle missing data, a simple yet effective imputation technique was employed—median imputation.

Median imputation involves replacing missing values with the median value of the corresponding feature in the dataset. This method was chosen for its robustness, particularly in dealing with outliers, as the median is less affected by extreme values compared to the mean. Previously studies have confirmed the performance of median imputation in CVD prediction tasks. Berkelmans et al. (2022), demonstrated that median imputation's validity and robustness were comparable to more complex methods when applied in large-scale cardiovascular disease risk prediction models. The study

highlighted that, provided key predictor variables like age were not missing, median imputation performed similarly to more sophisticated techniques, making it a reliable choice for this study.

The imputation process was carried out separately for the training, testing, and calibration datasets. By fitting the imputer on each dataset individually, the integrity of the data in terms of distribution and statistical properties was preserved within each split. This method ensures that no data leakage occurs between the training and testing datasets, which is essential for the validity of the model evaluation process.

3.4.4 Data Balancing

Imbalanced datasets can lead to models that are skewed in favoUr of the more prevalent class, potentially underperforming when predicting minority class instances.

To address the issue of class imbalance, a combination of RUS and a balancing technique, either ROS or SMOTE, was employed. The process began with RUS, where the majority class was reduced to a more manageable size. This initial step helped to mitigate the impact of the imbalance by reducing the dominance of the majority class in the training set (Budi Alamsyah et al., 2021).

Following RUS, the dataset was further balanced using either ROS or SMOTE, depending on the selected method.

A) ROS: This method was used to oversample the minority class until the class distribution was more balanced. ROS ensures that each class, except the majority, is oversampled, thereby reducing the imbalance without introducing synthetic data.

B) SMOTE: Generates synthetic samples for the minority class by interpolating between existing minority class instances. This technique not only balances the dataset but also enhances the diversity of the minority class by creating new, plausible instances.

3.5 Feature Selection

Feature selection was a crucial step to enhance the performance and efficiency of the machine learning models. Given the high dimensionality of the dataset, selecting the most relevant features was essential to improve model accuracy and reduce computational complexity. Four different methods were employed for feature selection: RFE, Backward Elimination based on SHAP values and removal of highly correlated features.

RFE is a widely used technique that recursively removes the least important features to select the most significant ones for model training. In this study, RFE was performed using a Random Forest classifier as the estimator. The process involved fitting the model to the training data and iteratively eliminating the least important features based on the model's feature importance rankings. RFE was configured to select the top 10 features. After running the RFE process, the selected features were saved, and the datasets (training, testing, and calibration) were transformed to include only these selected features. This method ensured that the most predictive features were retained for model training, enhancing the model's focus on the most relevant data (Theerthagiri & Vidya, 2022).

In this study, a novel approach to feature selection was employed by combining SHAP (SHapley Additive exPlanations) values with Backward Elimination. SHAP values offer a detailed insight into the impact of each feature on the model's predictions, particularly in complex, non-linear models such as Random Forests. By leveraging these

values, we aimed to systematically eliminate less impactful features while retaining those most crucial for accurate predictions. The process began with an initial model run, where SHAP values were calculated for all features in the training dataset. These SHAP values were then used to rank the features according to their importance. Instead of relying on traditional statistical methods for Backward Elimination, this approach iteratively removed the features with the lowest SHAP values, i.e., those contributing the least to the model's predictive power. This method allowed for a more nuanced and data-driven understanding of feature importance, ensuring that the final feature set comprised only the most significant features as determined by the model's internal logic. The selected features were subsequently used to create reduced datasets, streamlining the data and focusing the model training on the most impactful variables. This approach not only reduced model complexity but also enhanced the model's interpretability and predictive accuracy by retaining the features most relevant to the outcome of interest (Subramani et al., 2023; Liu et al., 2022).

Highly correlated features can introduce redundancy and multi-collinearity into the model, which can negatively impact model performance. To address this, a correlation threshold of 0.9 was set, and any features with a correlation coefficient exceeding this threshold were identified. The correlation matrix was generated to visualize the relationships between features, and a heatmap was created to highlight highly correlated pairs. Features that were highly correlated were removed from the dataset to reduce redundancy and ensure that the model was not biased by multiple features representing the same information (Mesquita & Marques, 2024). This step was particularly important for maintaining the model's interpretability and ensuring that each feature contributed unique information to the predictions.

3.6 Model Development and Evaluation

In the model development phase of this study, a range of machine learning algorithms were applied to predict cardiovascular outcomes using various feature-selected datasets. The process began with training individual models using six different algorithms: RF, XGBoost, DT, GBM, SVC, and KNN. These algorithms were chosen due to previous literature on CVD risk prediction, which has demonstrated their effectiveness in similar predictive modelling tasks (Tsarapatsani et al., 2022; Dinesh et al., 2018; You et al., 2023; Yang et al., 2020). Each model was trained on datasets derived from different feature selection techniques, including the original dataset containing all 48 features. To ensure robustness and avoid overfitting, cross-validation was employed, with both 3-fold and 5-fold splits being used depending on the specific dataset and model configuration.

In addition to the individual models, ensemble models were developed to leverage the strengths of multiple algorithms. These ensemble models involved stacking several base classifiers, with a meta-classifier making the final predictions. For instance, the ensemble models included configurations such as ALL_RF, where Random Forest served as the meta-classifier, and ALL_XGBoost, where XGBoost was the meta-classifier. Other ensemble configurations involved using SVC and GBM as meta-classifiers. The selection of these ensemble strategies was informed by their proven effectiveness in previous studies focused on ensemble learning in predictive modelling (Nath & Sahu, 2019; Rustamov et al., 2023; Ghasemieh et al., 2024). These ensemble models were also subjected to the same rigorous cross-validation process as the individual models to ensure consistency in evaluation.

During cross-validation, each dataset was split into multiple folds, and the models were trained on a subset of the data and validated on the remaining portion. This process was repeated several times to average the results, which helped to reduce variance and provide a more accurate assessment of model performance. The models were evaluated based on several key metrics, including accuracy, precision, negative predictive value, sensitivity, specificity, and the area under the ROC curve (AUC-ROC) (Peng et al., 2023). These metrics were calculated for both the training and testing sets to ensure a comprehensive evaluation of the models' effectiveness.

For Model 2, the evaluation focused specifically on metrics such as AUC, accuracy, sensitivity, specificity, and the confusion matrix. These metrics were crucial in assessing how well Model 2 distinguished between the classes, particularly for predicting myocardial infarction versus stroke. The use of these evaluation metrics ensured that the model's ability to correctly identify true positives and true negatives was thoroughly assessed, providing a robust comparison of its performance against other models.

3.7 Calibration

For Model 1, calibration was performed to ensure that the predicted probabilities of the models accurately reflected the true likelihood of outcomes. The calibration process involved the use of isotonic regression for both the RF model and the FRS model (You et al., 2023). The calibration process aimed to adjust the probability outputs of the models so that they matched the observed outcomes more closely. The calibration process began with the non-calibrated models. Metrics such as Brier score, log loss, and ROC AUC were calculated for both the non-calibrated RF model and the non-calibrated FRS model (Srivastava, 2024; Yang et al., 2022). Additionally, the NRI was calculated to assess how well the models reclassified patients into correct risk categories compared to the non-

calibrated FRS model. Calibration curves were also plotted to visualize the calibration performance of the models.

Following the calibration, the same metrics were recalculated for the calibrated models. The results indicated whether the calibration improved the accuracy and reliability of the probability predictions. The calibrated models were compared to determine which model provided the most reliable predictions. Calibration curves were plotted again to assess the improvement in model calibration. By integrating calibration into the evaluation process, the study ensured that the model's probability outputs were more accurate and reliable, which is crucial for clinical decision-making and risk prediction in cardiovascular disease.

Chapter 4: RESULTS

4.1 Feature Selection

In this study, feature selection techniques were employed to enhance the performance of two distinct predictive models: one for CVD versus no CVD (Model 1), and another for MI versus stroke (Model 2). These techniques aimed to identify the most significant predictors among the available variables, ensuring that the models focused on the most relevant features for accurate and reliable predictions.

4.1.1 Model 1: CVD vs No CVD

For Model 1, which aimed to predict the likelihood of CVD versus no CVD, three feature selection methods were applied: the FRS, SHAP-based BE, and RFE. Each method resulted in a different set of features. The FRS method identified 7 features, the SHAP-based BE technique selected 26 features, and the RFE method selected 10 features. These selected features formed the foundation for building the predictive models. Table 4.1 below presents a comparison of the features selected by each method. The selected features for Model 1 demonstrated the relevance of variables like systolic BP, age, and HDL, which are consistent with known risk factors for CVD. Notably, systolic BP, age, and HDL were common predictors identified across all three feature selection methods, highlighting their critical role in predicting cardiovascular disease. The SHAP-based BE method selected a broader range of features, providing a more comprehensive perspective on potential risk factors.

Table 4. 1: Model 1 Selected Features and Methods

FRS Selected Features (7 features)	Shap based BE (26 features)		RFE selected features (10 features)
1. Sex	1. Area	14. Diabetes	1. Area
2. Age	2. Systolic BP	15. Heartdisease	2. Age
3. LDL	3. Age	16. Height	3. Systolic BP
4. HDL	4. Waist Hip Ratio	17. Hpt_f	4. Diastolic BP
5. Systolic bp	5. Hpt_med	18. Glucose_baseline	5. Waist
6. Diabetes	6. Gender	19. Bmi	6. Waist Hip Ratio
7. Smoking	7. FVC	20. Handgrip_nondominant	7. Glucose baseline
	8. HDL	21. Smoking	8. TG baseline
	9. Hpt_b	22. Rightcalf	9. HDL
	10. Waist	23. Head	10. Forced Expiratory Volume in 1 Second
	11. TG_baseline	24. Rightarm	
	12. Diastolic BP	25. Diabetes_f	
	13. FEV1	26. Ldl_baseline	

4.1.2 Model 2: MI vs Stroke

For Model 2, which focused on distinguishing between MI and stroke, a different set of features was selected using the same SHAP-based BE and RFE methods. This model did not include the FRS as a comparison point. The SHAP-based BE method selected 14 features, and the RFE method selected 10 features. Table 4.2 below presents the features selected by each method for Model 2. Several predictors were consistently identified as significant across both the SHAP-based BE and RFE methods. These common predictors included diastolic BP, systolic BP, HDL, glucose baseline, height, rightcalf, TG baseline, waist, and TC baseline.

Table 4. 2: Model 2 Selected Features and Methods

Shap Based BE (14 features)	RFE selected features (10 features)
1. Diastolic BP	1. Height
2. Systolic BP	2. Systolic BP
3. HDL	3. Diastolic BP
4. Glucose_baseline	4. Pulse
5. Height	5. Waist
6. Rightcalf	6. Glucose baseline
7. FVC	7. TC baseline
8. Waist Hip Ratio	8. TG baseline
9. TG_baseline	9. HDL
10. Age	10. Rightcalf
11. Waist	
12. Diabetes_med	
13. TC_baseline	
14. PERF	

4.2 Model Performance

The performance of two predictive models, Model 1 (CVD vs. No CVD) and Model 2 (MI vs. Stroke), was evaluated using a variety of machine learning algorithms. For Model 1, the RF model utilizing 26 features selected by the SHAP-based BE method emerged as the top performer. This model demonstrated an AUC of 0.859 and an accuracy of 0.811 on the test set, alongside strong metrics in PPV (0.047), NPV (0.996), sensitivity

(0.762), and specificity (0.800). The minimal AUC difference of 0.024 between the training and test sets indicated that the model did not suffer from significant overfitting, which underscores its robustness and generalizability in predicting cardiovascular disease.

In addition to RF, other algorithms such as GBM and SVC also performed well with SHAP-based BE-selected features. Ensemble models like ALL_RF and ALL_XGBoost, which integrated multiple base classifiers, showed promising results, with AUCs close to or slightly above 0.780. However, these models had higher AUC differences, indicating that while they excelled in specific metrics, their overall stability might not be as strong as the RF model.

Table 4. 3: Model 1 Performance

Feature selection method	Model	Feature number	Test						Train		AUC Difference
			Accuracy	AUC	PPV	NPV	Sensitivity	Specificity	Accuracy	AUC	
No feature selection	RF	48	0.815	0.823	0.037	0.994	0.571	0.818	0.798	0.888	0.066
	XGBoost	48	0.815	0.812	0.040	0.994	0.619	0.818	0.813	0.886	0.074
	DT	48	0.722	0.624	0.023	0.992	0.524	0.725	0.743	0.743	0.118
	GBM	48	0.817	0.827	0.043	0.995	0.667	0.819	0.758	0.836	0.009
	SVC	48	0.734	0.787	0.030	0.994	0.667	0.734	0.810	0.871	0.085
	KNN	48	0.630	0.751	0.023	0.994	0.714	0.629	0.738	0.800	0.049
	ALL_RF	48	0.797	0.781	0.039	0.995	0.667	0.799	1.000	1.000	0.219
	ALL_XGBoost	48	0.814	0.826	0.048	0.996	0.762	0.815	1.000	1.000	0.174
	ALL_SVC	48	0.830	0.821	0.047	0.995	0.667	0.832	1.000	1.000	0.179
	ALL_GBM	48	0.815	0.791	0.034	0.993	0.524	0.819	1.000	1.000	0.209
Recursive Feature Elimination	RF	10	0.786	0.802	0.037	0.995	0.667	0.787	0.790	0.865	0.062
	XGBoost	10	0.774	0.793	0.035	0.995	0.667	0.775	0.783	0.846	0.053
	DT	10	0.734	0.654	0.026	0.993	0.571	0.736	0.723	0.723	0.069

	GBM	10	0.762	0.790	0.031	0.994	0.619	0.764	0.763	0.835	0.044
	SVC	10	0.726	0.807	0.029	0.994	0.667	0.726	0.770	0.831	0.024
	KNN	10	0.694	0.719	0.024	0.993	0.619	0.695	0.740	0.814	0.095
	ALL_RF	10	0.737	0.745	0.034	0.996	0.762	0.736	0.898	0.910	0.165
	ALL_XGBoost	10	0.783	0.781	0.034	0.994	0.619	0.785	0.850	0.965	0.184
	ALL_SVC	10	0.790	0.768	0.035	0.994	0.619	0.792	0.883	0.986	0.218
	ALL_GBM	10	0.779	0.769	0.036	0.995	0.667	0.780	0.925	1.000	0.231
Shap based Backward Elimination	RF	26	0.811	0.859	0.047	0.996	0.762	0.811	0.800	0.883	0.024
	XGBoost	26	0.796	0.844	0.041	0.996	0.714	0.797	0.775	0.864	0.020
	DT	26	0.694	0.657	0.024	0.993	0.619	0.695	0.710	0.710	0.053
	GBM	26	0.771	0.821	0.035	0.995	0.667	0.772	0.790	0.859	0.039
	SVC	26	0.767	0.823	0.034	0.995	0.667	0.768	0.785	0.849	0.027
	KNN	26	0.713	0.761	0.030	0.995	0.714	0.713	0.745	0.812	0.051
	ALL_RF	26	0.785	0.780	0.042	0.996	0.762	0.785	1.000	1.000	0.220
	ALL_XGBoost	26	0.814	0.715	0.040	0.994	0.619	0.816	0.983	0.998	0.283
	ALL_SVC	26	0.772	0.820	0.044	0.998	0.857	0.771	0.978	1.000	0.180
	ALL_GBM	26	0.768	0.818	0.039	0.996	0.762	0.768	1.000	1.000	0.182
Removed highly correlated features	RF	45	0.826	0.851	0.049	0.996	0.714	0.828	0.805	0.887	0.036
	XGBoost	45	0.826	0.815	0.043	0.994	0.619	0.829	0.795	0.878	0.063
	DT	45	0.719	0.623	0.023	0.992	0.524	0.722	0.728	0.728	0.105
	GBM	45	0.818	0.826	0.041	0.994	0.619	0.821	0.780	0.853	0.027
	SVC	45	0.731	0.786	0.030	0.994	0.667	0.732	0.808	0.872	0.086
	KNN	45	0.642	0.719	0.024	0.995	0.714	0.641	0.738	0.795	0.076
	ALL_RF	45	0.831	0.787	0.041	0.994	0.571	0.835	1.000	1.000	0.213
	ALL_XGBoost	45	0.784	0.768	0.039	0.996	0.714	0.785	0.948	1.000	0.232
	ALL_SVC	45	0.762	0.782	0.036	0.995	0.714	0.763	0.948	1.000	0.218
	ALL_GBM	45	0.816	0.849	0.046	0.996	0.714	0.817	1.000	1.000	0.151

For Model 2, the ALL_RF ensemble model, utilizing 14 features selected through SHAP-based BE, proved to be the best at distinguishing between myocardial infarction (MI) and stroke. This model achieved an impressive AUC of 0.944, with an accuracy of 0.833, and maintained high sensitivity (0.875) and specificity (0.800). The AUC difference of just 0.030 between the training and test sets further highlighted the model's reliability and resistance to overfitting, making it an ideal candidate for practical application in clinical settings. Other models in Model 2, such as ALL_XGBoost and ALL_GBM, also performed well but fell short of the ALL_RF model's high standards, both in terms of AUC and stability, as indicated by their relatively higher AUC differences.

Table 4. 4: Model 2 Performance

Feature selection method	Model	Feature number	Test						Train		AUC Difference
			Accuracy	AUC	PPV	NPV	Sensitivity	Specificity	Accuracy	AUC	
No feature selection	RF	48	0.667	0.802	0.667	0.667	0.667	0.667	0.602	0.645	-0.158
	XGBoost	48	0.667	0.691	0.636	0.714	0.778	0.556	0.545	0.561	-0.130
	DT	48	0.556	0.556	0.571	0.545	0.444	0.667	0.648	0.648	0.092
	GBM	48	0.667	0.741	0.636	0.714	0.778	0.556	0.591	0.619	-0.122
	SVC	48	0.667	0.704	0.714	0.636	0.556	0.778	0.602	0.593	-0.111
	KNN	48	0.667	0.667	0.667	0.667	0.667	0.667	0.625	0.629	-0.038
	ALL_RF	48	0.556	0.463	0.571	0.545	0.444	0.667	0.875	0.875	0.412
	ALL_XGBoost	48	0.722	0.722	0.833	0.667	0.556	0.889	0.864	0.792	0.070
	ALL_SVC	48	0.611	0.623	0.750	0.571	0.333	0.889	0.875	0.985	0.361
	ALL_GBM	48	0.722	0.685	0.750	0.700	0.667	0.778	0.750	0.740	0.054
Recursive Feature Elimination	RF	10	0.667	0.821	0.636	0.714	0.778	0.556	0.659	0.661	-0.160
	XGBoost	10	0.667	0.716	0.667	0.667	0.667	0.667	0.591	0.599	-0.117
	DT	10	0.500	0.500	0.500	0.500	0.444	0.556	0.693	0.693	0.193
	GBM	10	0.667	0.741	0.667	0.667	0.667	0.667	0.659	0.648	-0.092

	SVC	10	0.667	0.852	0.667	0.667	0.667	0.667	0.591	0.561	-0.291
	KNN	10	0.667	0.710	0.636	0.714	0.778	0.556	0.545	0.574	-0.136
	ALL_RF	10	0.778	0.765	0.778	0.778	0.778	0.778	1.000	1.000	0.235
	ALL_XGBoost	10	0.556	0.630	0.556	0.556	0.556	0.556	0.977	1.000	0.370
	ALL_SVC	10	0.611	0.753	0.625	0.600	0.556	0.667	0.943	1.000	0.247
	ALL_GBM	10	0.722	0.753	0.700	0.750	0.778	0.667	0.943	0.899	0.146
Shap based Backward Elimination	RF	14	0.778	0.784	0.778	0.778	0.778	0.778	0.659	0.696	-0.088
	XGBoost	14	0.722	0.753	0.700	0.750	0.778	0.667	0.568	0.570	-0.183
	DT	14	0.500	0.500	0.500	0.500	0.444	0.556	0.580	0.580	0.080
	GBM	14	0.722	0.753	0.700	0.750	0.778	0.667	0.557	0.620	-0.133
	SVC	14	0.722	0.815	0.750	0.700	0.667	0.778	0.648	0.608	-0.207
	KNN	14	0.611	0.722	0.750	0.571	0.333	0.889	0.602	0.606	-0.116
	ALL_RF	14	0.833	0.944	0.875	0.800	0.778	0.889	0.943	0.974	0.030
	ALL_XGBoost	14	0.778	0.747	0.727	0.857	0.889	0.667	0.920	0.868	0.121
	ALL_SVC	14	0.611	0.772	0.667	0.583	0.444	0.778	0.943	0.994	0.222
	ALL_GBM	14	0.333	0.216	0.200	0.385	0.111	0.556	0.807	0.772	0.556
Removed highly correlated features	RF	46	0.667	0.821	0.667	0.667	0.667	0.667	0.625	0.643	-0.178
	XGBoost	46	0.667	0.593	0.636	0.714	0.778	0.556	0.545	0.564	-0.029
	DT	46	0.611	0.611	0.667	0.583	0.444	0.778	0.693	0.693	0.082
	GBM	46	0.667	0.790	0.636	0.714	0.778	0.556	0.557	0.597	-0.194
	SVC	46	0.722	0.704	0.750	0.700	0.667	0.778	0.625	0.582	-0.122
	KNN	46	0.778	0.772	0.778	0.778	0.778	0.778	0.580	0.575	-0.197
	ALL_RF	46	0.889	0.889	1.000	0.818	0.778	1.000	0.966	0.932	0.043
	ALL_XGBoost	46	0.611	0.691	0.600	0.625	0.667	0.556	0.852	0.868	0.176
	ALL_SVC	46	0.667	0.772	0.636	0.714	0.778	0.556	0.920	1.000	0.228
	ALL_GBM	46	0.500	0.407	0.500	0.500	0.444	0.556	0.852	0.898	0.490

4.3 Model Interpretability

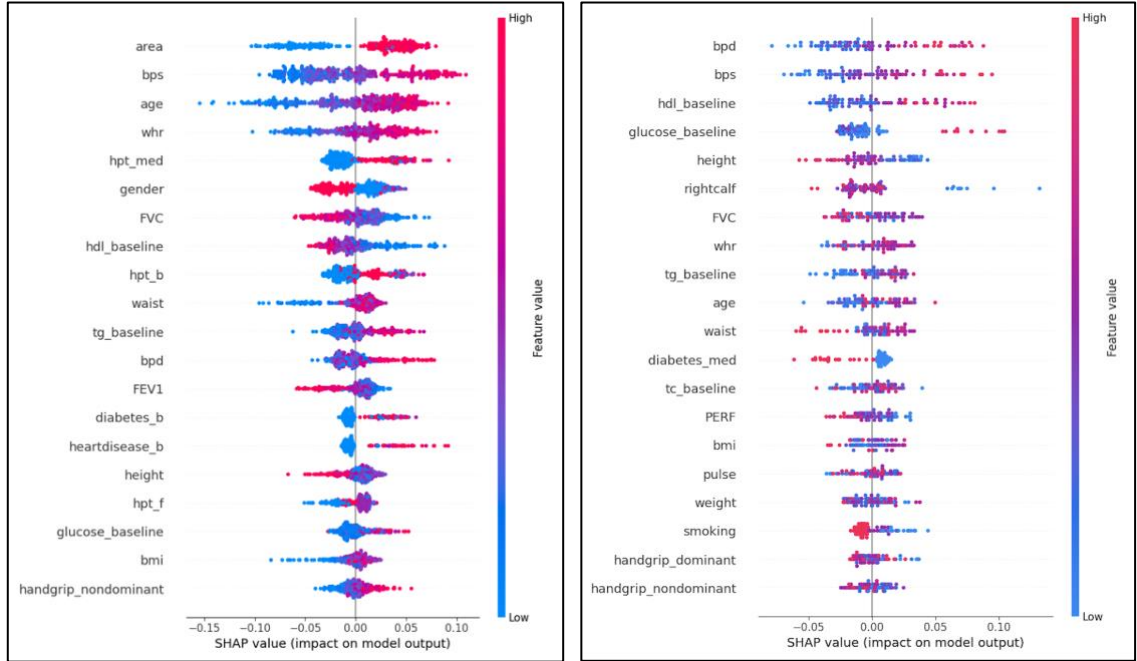


Figure 4. 1: SHAP Summary Plots for Model 1 (CVD vs No CVD) and Model 2 (Myocardial Infarction vs Stroke)

The SHAP analysis provided insights into the key features driving the predictions for both models. For Model 1, which predicted CVD versus no CVD, the SHAP summary plot (Figure 4.1, left panel) identified significant predictors such as systolic blood pressure (bps), age, and HDL levels. These factors were critical in determining the likelihood of a cardiovascular event, with higher systolic blood pressure and age being particularly influential. Other important variables included area, hypertension medication (hpt_med), and Forced Vital Capacity (FVC), which further highlight the multidimensional nature of CVD risk involving demographic, clinical, and lifestyle components.

In Model 2, which focused on predicting myocardial infarction versus stroke, the SHAP summary plot (Figure 4.1, right panel) highlighted diastolic blood pressure (bpd), systolic blood pressure (bps), HDL baseline, glucose baseline, and height as key predictors. The analysis revealed that higher values of systolic and diastolic blood

pressure significantly increased the risk of myocardial infarction over stroke, while HDL and glucose levels played crucial roles in the model's decision-making process. Features like right calf circumference (rightcalf) and waist-hip ratio (whr) also contributed to the model's predictions, indicating that both clinical measurements and anthropometric factors are essential in differentiating between these two cardiovascular outcomes.

4.4 Model 1 Comparisons

For the comparison between the RF model and the FRS model for CVD Risk prediction using Model 1, the analysis focused on several key metrics, including AUC, accuracy, confusion matrix results, and the NRI. These comparisons were performed to evaluate the discrimination ability and the potential improvements in classification that the RF model could offer over the traditional FRS model.

4.4.1 Discrimination Metrics

The RF model achieved a higher ROC AUC of 0.858 compared to 0.834 for the FRS model, indicating slightly better discriminatory power. However, the FRS model exhibited a higher accuracy of 0.987 compared to 0.810 for the RF model. The confusion matrix for the RF model showed 322 false positives and 16 false negatives, while the FRS model had 23 false positives and no false negatives, reflecting its higher accuracy but also highlighting the potential trade-offs between sensitivity and specificity in these models.

Table 4. 5: Comparison of Discrimination Metrics Between the Random Forest (RF) Model and the Framingham Risk Score (FRS) Model for CVD Prediction

Metric	RF Model	FRS Model																		
ROC AUC	0.858	0.834																		
Accuracy	0.810	0.987																		
Confusion Matrix	<table> <tr> <td></td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1384</td><td>322</td></tr> <tr> <td>1</td><td>5</td><td>16</td></tr> </table>		0	1	0	1384	322	1	5	16	<table> <tr> <td></td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1683</td><td>23</td></tr> <tr> <td>1</td><td>0</td><td>21</td></tr> </table>		0	1	0	1683	23	1	0	21
	0	1																		
0	1384	322																		
1	5	16																		
	0	1																		
0	1683	23																		
1	0	21																		

4.4.2 Calibration

Calibration of the RF model was performed to enhance the reliability of its probability predictions. Post-calibration results showed significant improvements in both the Brier Score and Log Loss (Table 3). Specifically, the Brier Score decreased from 0.1422 in the non-calibrated model to 0.0118 in the calibrated model, and the Log Loss reduced from 0.4402 to 0.0572. These improvements, along with the slightly decreased but still strong AUC of 0.8367, reflect the effectiveness of the calibration process.

Table 4. 6: Comparison of Non-Calibrated and Calibrated RF Model Performance Metrics

Metric	Non-calibrated RF Model	Calibrated RF Model	p-value	Significance
Brier Score	0.142	0.011	0.0	Significant
Log Loss	0.440	0.057	0.0	Significant
ROC AUC	0.858	0.836	4.23×10^{-57}	Significant

4.4.3 Net Reclassification Index

The NRI analysis at the 0.2 threshold further supports the RF model's superiority. The RF model reclassified a significant portion of patients into more appropriate risk categories, particularly evident at this threshold where the RF model showed an NRI value of 0.259, indicating a substantial 25% improvement in patient stratification (Table 4.7). This finding underscores the RF model's capability to provide better targeted risk predictions compared to the FRS model.

The calibration process further refined the predictive accuracy of the RF model. Post-calibration, the NRI values were analysed at various thresholds, including 0.05, 0.1, and 0.2, to compare the performance of the calibrated RF model against the calibrated FRS model (Table 4.8). At the 0.05 threshold, the calibrated RF model showed a modest improvement (19%) with an NRI of 0.194. However, at the 0.2 threshold, the NRI value dropped to 0.0, indicating that, at this level, the calibrated RF model and the calibrated FRS model reclassified patients similarly.

Table 4. 7: Net Reclassification Index (NRI) Comparison between RF Model and FRS Model across Different Risk Thresholds

Threshold	Events upward	Events downward	Non events upward	Non events downward	Total events	Total Non events	NRI
0.1	13	0	1308	0	21	1706	-0.147
0.2	19	0	1102	0	21	1706	0.258
0.3	19	0	855	0	21	1706	0.403
0.5	16	0	321	0	21	1706	0.573

**Table 4. 8: Net Reclassification Index (NRI) Comparison between Calibrated RF
Model and Calibrated FRS Model across Different Risk Thresholds**

Threshold	Events upward	Events downward	Non events upward	Non events downward	Total events	Total Non events	NRI
0.05	5	0	81	7	21	1706	0.194
0.1	0	1	0	5	21	1706	-0.044
0.2	0	0	0	0	21	1706	0.0

Chapter 5: DISCUSSION

This study provides significant insights into the potential of ML models to predict CVD risks within a three-year horizon, focusing on the Malaysian population. The models developed, Model 1 for predicting CVD versus no CVD and Model 2 for differentiating between MI and stroke, demonstrated strong predictive performance, emphasizing the importance of short-term CVD risk assessment.

Model 1: CVD vs. No CVD

Model 1, designed to predict the likelihood of CVD versus no CVD, employed various feature selection techniques, with the RF model utilizing 26 features selected through SHAP-based BE emerging as the top performer. The RF model achieved an AUC of 0.859 and an accuracy of 0.811 on the test set, indicating strong predictive power. The minimal AUC difference between the training and test sets further highlights the model's robustness and generalizability.

The comparison between the RF model and the FRS model revealed that while the RF model achieved a higher AUC (0.858 vs. 0.834), the FRS model demonstrated higher accuracy (0.987 vs. 0.810). This suggests that the FRS, despite its simplicity, remains a highly accurate tool for predicting CVD in this population, though it may be outperformed by ML models in terms of overall discriminatory power.

The selected features for Model 1, including systolic blood pressure, HDL cholesterol, glucose levels, and age, are consistent with well-established CVD risk factors. Literature has consistently highlighted these variables as significant contributors to cardiovascular health, directly influencing outcomes such as blood pressure and lipid metabolism (Bays et al., 2021; Dolezel et al., 2021). A study by Kırboğa and Küçüksille

(2023) further supports these findings, identifying systolic blood pressure, cholesterol, and age as critical predictors of cardiovascular diseases through machine learning models.

In the Malaysian context, Moh Azahar et al. (2022) identified key predictors for high cardiovascular risk using the FRS in rural communities, including waist-hip ratio (WHR), hypertension, smoking, diabetes, and high LDL cholesterol. These predictors closely align with the features highlighted in our SHAP summary plot for Model 1, such as systolic blood pressure (bps), WHR, age, and diabetes, all of which show strong influence on the model's predictions for CVD risk. These findings reinforce the relevance of the selected features in Model 1, demonstrating that ML models can capture the complex interplay of risk factors specific to the Malaysian population.

Model 2: MI vs. Stroke

Model 2, which focused on predicting MI versus stroke, showed promising results with the ALL_RF ensemble model, utilizing 14 features selected through SHAP-based BE, achieving an AUC of 0.944 and an accuracy of 0.833. This model's high sensitivity (0.875) and specificity (0.800) underscore its effectiveness in differentiating between these two critical cardiovascular outcomes.

The SHAP summary plot highlighted diastolic blood pressure (bpd), systolic blood pressure (bps), HDL baseline, glucose baseline, and height as key predictors. The analysis revealed that higher values of systolic and diastolic blood pressure significantly increased the risk of myocardial infarction over stroke, aligning with well-established risk factors for these conditions. Hypertension is a well-known contributor to both MI and stroke, with it being one of the most significant modifiable risk factors for stroke (Boehme et al., 2018). Similarly, dyslipidemia and diabetes mellitus have been

consistently associated with an increased risk of myocardial infarction (Mechanic et al., 2023).

Additionally, features such as right calf circumference (rightcalf) and waist-hip ratio were found to contribute to the model's predictions, indicating that both clinical measurements and anthropometric factors are essential in differentiating between these two cardiovascular outcomes. The waist-hip ratio, in particular, has been shown to significantly increase the risk of MI. A meta-analysis of 12 case-control studies demonstrated that a high WHR is associated with a more than twofold increased risk of MI (Cao et al., 2018).

The evaluation of Model 1 (CVD vs No CVD) through the NRI at various thresholds highlights the importance of selecting appropriate cut-offs for risk stratification, both before and after model calibration.

The NRI is a valuable metric used to assess the improvement in risk prediction models by examining how well new models reclassify individuals into appropriate risk categories compared to existing models like the FRS. This analysis was performed at various thresholds (e.g., 0.05, 0.10, 0.20), each representing different risk cut-offs that are clinically relevant in CVD risk stratification.

In Table 4.7, which compares the RF model and the FRS across different thresholds before calibration, the use of a 20% threshold is particularly relevant. The 20% threshold, showing a positive NRI of 0.258, is commonly used in clinical practice to identify high-risk individuals who might benefit from more aggressive preventive measures. This positive NRI indicates that the RF model is more effective in reclassifying patients into appropriate risk categories at this level, supporting its use in clinical decision-making for high-risk patients.

In contrast, Table 4.8 shows the NRI comparison post-calibration between the calibrated RF model and the calibrated FRS model. Here, the 0.05 threshold is noteworthy due to its relatively positive NRI (0.194). This lower threshold is justified post-calibration because calibration tends to improve the model's sensitivity, making it more accurate in identifying individuals who might benefit from early interventions, even if they are at lower risk. This approach is particularly important in clinical settings where the goal is to maximize sensitivity, ensuring that fewer high-risk individuals are missed, thus preventing adverse outcomes.

Calibrating the model enhances its ability to align predicted probabilities with actual outcomes, making the lower 0.05 threshold more effective in identifying patients who might otherwise be overlooked at higher thresholds. This improvement in sensitivity, although sometimes at the cost of specificity, is crucial in managing CVD risks where early detection and intervention are essential.

In the context of CVD management, short-term prediction models, such as the ones developed in this study for predicting CVD versus no CVD and differentiating between MI and stroke, are crucial for immediate clinical decision-making. These models allow healthcare providers to identify high-risk patients within a shorter timeframe (e.g., three years), enabling timely interventions that can prevent adverse outcomes.

Short-term predictions are particularly relevant in rapidly changing clinical scenarios, where a patient's risk factors might fluctuate over a short period due to lifestyle changes, treatment compliance, or emerging comorbidities. Furthermore, focusing on short-term predictions ensures that the models remain sensitive to recent changes in patient data, which can lead to more accurate and responsive risk stratification. This is particularly critical in managing chronic conditions like CVD, where ongoing monitoring

and timely adjustments in treatment plans are essential to prevent severe cardiovascular events such as MI and stroke.

Chapter 6: CONCLUSION

This study demonstrates the effectiveness of ML models in predicting CVD within a 3-year period for the Malaysian population. The models developed address the limitations of the FRS, particularly in short to medium-term risk prediction. The RF model, utilizing SHAP-based backward elimination for feature selection, showed strong performance with a high AUC and accuracy. The minimal difference between the training and test sets indicates the model's reliability and low risk of overfitting. The comparison between the RF model and the FRS highlighted that while the FRS maintained higher accuracy, the ML model provided better overall discrimination in predicting CVD risk. These results suggest that ML models can improve risk stratification in populations with specific genetic and environmental factors like those in Malaysia. The focus on short-term predictions is important for clinical practice, where early detection and timely intervention are essential in managing CVD. The ML models were also effective in distinguishing between myocardial infarction and stroke, demonstrating their potential utility in clinical decision-making.

REFERENCES

- Alaa, A. M., Bolton, T., Di Angelantonio, E., Rudd, J. H. F., & van der Schaar, M. (2019). Cardiovascular disease risk prediction using automated machine learning: A prospective study of 423,604 UK Biobank participants. *PloS One*, 14(5). <https://doi.org/10.1371/journal.pone.0213653>
- Alloubani, A., Nimer, R., & Samara, R. (2021). Relationship between hyperlipidemia, cardiovascular disease, and stroke: A systematic review. *Current Cardiology Reviews*, 17(6). <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8950504/>
- Amini, M., Zayeri, F., & Salehi, M. (2021). Trend analysis of cardiovascular disease mortality, incidence, and mortality-to-incidence ratio: Results from global burden of disease study 2017. *BMC Public Health*, 21(1), 401. <https://doi.org/10.1186/s12889-021-10429-0>
- Anderson, K. M., Odell, P. M., Wilson, P. W., & Kannel, W. B. (1991). Cardiovascular disease risk profiles. *American Heart Journal*, 121, 293-298. <https://pubmed.ncbi.nlm.nih.gov/1985385/>
- Avdikos, G., Michas, G., & Smith, S. W. (2022). From Q/Non-Q myocardial infarction to STEMI/NSTEMI: Why it's time to consider another simplified dichotomy; a narrative literature review. *Archives of Academic Emergency Medicine*, 10(1). <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9676707/>
- Aziz, F., Malek, S., Ibrahim, K. S., Shariff, R. E. R., Wan Ahmad, W. A., Ali, R. M., Liu, K. T., Selvaraj, G., & Kasim, S. (2021). Short- And long-term mortality prediction after an acute ST-elevation myocardial infarction (STEMI) in Asians: A machine learning approach. *PLoS ONE*, 16(8). <https://doi.org/10.1371/journal.pone.0254894>
- Aziz, Z. A., Lee, Y. Y. L., Ngah, B. A., Sidek, N. N., Looi, I., Hanip, M. R., & Basri, H. B. (2015). Acute Stroke Registry Malaysia, 2010-2014: Results from the National Neurology Registry. *Journal of Stroke and Cerebrovascular Diseases*, 24(12), 2701–2709. <https://doi.org/10.1016/j.jstrokecerebrovasdis.2015.07.025>
- Banerjee, A., Chen, S., Fatemifar, G., Zeina, M., Lumbers, R. T., Mielke, J., Gill, S., Kotecha, D., Freitag, D. F., Denaxas, S., & Hemingway, H. (2021). Machine learning for subtype definition and risk prediction in heart failure, acute coronary syndromes, and atrial fibrillation: Systematic review of validity and clinical utility. *BMC Medicine*, 19(1), 85. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8022365/>
- Bays, H. E., Agarwala, A., German, C., Satish, P., Iluyomade, A., Dudum, R., Thakkar, A., Rifai, M. A., Mehta, A., Thobani, A., Al-Saiegh, Y., Nelson, A. J., Sheth, S., & Toth, P. P.

- (2022). Ten things to know about ten cardiovascular disease risk factors – 2022. *American Journal of Preventive Cardiology*, 10. <https://doi.org/10.1016/j.ajpc.2022.100342>
- Bays, H. E., Taub, P. R., Epstein, E., Michos, E. D., Ferraro, R. A., Bailey, A. L., Kelli, H. M., Ferdinand, K. C., Echols, M. R., Weintraub, H., Bostrom, J., Johnson, H. M., Hoppe, K. K., Shapiro, M. D., German, C. A., Virani, S. S., Hussain, A., Ballantyne, C. M., Agha, A. M., & Toth, P. P. (2021). Ten things to know about ten cardiovascular disease risk factors. *American Journal of Preventive Cardiology*, 5, 100149. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8315386/>
- Benjamin, E. J., Muntner, P., Alonso, A., Bittencourt, M. S., Callaway, C. W., Carson, A. P., Chamberlain, A. M., Chang, A. R., Cheng, S., Das, S. R., Delling, F. N., Djousse, L., Elkind, M. S. V., Ferguson, J. F., Fornage, M., Jordan, L. C., Khan, S. S., Kissela, B. M., Knutson, K. L., ... Virani, S. S. (2019). Heart disease and stroke statistics—2019 update: A report from the American Heart Association. *Circulation*, 139(10). <https://doi.org/10.1161/CIR.0000000000000659>
- Bentéjac, C., Csörgő, A., & Martínez-Muñoz, G. (2021). A comparative analysis of gradient boosting algorithms. *Artificial Intelligence Review*, 54(3), 1937–1967. <https://doi.org/10.1007/s10462-020-09896-5>
- Berger, J. S., Jordan, C. O., Lloyd-Jones, D., & Blumenthal, R. S. (2010). Screening for cardiovascular risk in asymptomatic patients. *Journal of the American College of Cardiology*, 55(12), 1169–1177. <https://www.sciencedirect.com/science/article/pii/S0735109710001762>
- Berkelmans, G. F. N., Read, S. H., Gudbjörnsdottir, S., Wild, S. H., Franzen, S., van der Graaf, Y., Eliasson, B., Visseren, F. L. J., Paynter, N. P., & Dorresteyn, J. A. N. (2022). Population median imputation was noninferior to complex approaches for imputing missing values in cardiovascular prediction models in clinical practice. *Journal of Clinical Epidemiology*, 145, 70–80. <https://pubmed.ncbi.nlm.nih.gov/35066115/>
- Bhardwaj, B., O'Keefe, E. L., & O'Keefe, J. H. (2016). Death by carbs: Added sugars and refined carbohydrates cause diabetes and cardiovascular disease in Asian Indians. *Missouri Medicine*, 113(5), 395–400. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6139832/>
- Bloom, D. E., Chisholm, D., Jané-Llopis, E., Prettnner, K., Stein, A., & Feigl, A. (2011, October). *From burden to “best buys”: Reducing the economic impact of non-communicable diseases in low- and middle-income countries* (PGDA Working Paper No. 75). World Health Organization & World Economic Forum. https://www.hsph.harvard.edu/pgda/wp-content/uploads/sites/1288/2013/10/PGDA_WP_75.pdf

- Boehme, A. K., Esenwa, C., & Elkind, M. S. V. (2017). Stroke risk factors, genetics, and prevention. *Circulation Research*, 120(3), 472-495. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5321635/>
- Bots, S. H., Peters, S. A. E., & Woodward, M. (2017). Sex differences in coronary heart disease and stroke mortality: A global assessment of the effect of ageing between 1980 and 2010. *BMJ Global Health*, 2(2). <https://gh.bmj.com/content/2/2/e000298>
- Brown, J. C., Gerhardt, T. E., & Kwon, E. (2023). Risk Factors for Coronary Artery Disease. In *StatPearls*. StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK554410/>
- Budi Alamsyah, A. R., Rahma, S., Sri Belinda, N., & Setiawan, A. (2021). SMOTE and Nearmiss methods for disease classification with unbalanced data case study: IFLS 5. In *Proceedings of the International Conference on Data Science and Official Statistics (ICDSOS) 2021* (Vol. 2021, No. 1). https://www.researchgate.net/publication/357576714_SMOTE_and_Nearmiss_Methods_for_Disease_Classification_with_Unbalanced_Data_Case_Study_IFLS_5
- Caelen, O. (2017). A Bayesian interpretation of the confusion matrix. *Annals of Mathematics and Artificial Intelligence*, 81(3), 429–450. <https://doi.org/10.1007/s10472-017-9564-8>
- Cai, Y., Cai, Y. Q., Tang, L. Y., Wang, Y. H., Gong, M., Jing, T. C., Li, H. J., Li-Ling, J., Hu, W., Yin, Z., Gong, D. X., & Zhang, G. W. (2024). Artificial intelligence in the risk prediction models of cardiovascular disease and development of an independent validation screening tool: A systematic review. *BMC Medicine*, 22(1), 56. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10845808/>
- Campeau, L. (1976). Letter: Grading of angina pectoris. *Circulation*, 54(3), 522-523. <https://www.ahajournals.org/doi/10.1161/circ.54.3.947585>
- Cao, Q., Yu, S., Xiong, W., Li, Y., Li, H., Li, J., & Li, F. (2018, July). Waist-hip ratio as a predictor of myocardial infarction risk: A systematic review and meta-analysis. *Medicine (Baltimore)*, 97(30), e11639. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6078643/>
- Chen, T., & Guestrin, C. (2016). *XGBoost: A scalable tree boosting system*. <https://arxiv.org/abs/1603.02754>
- Chia, Y. C., Gray, S. Y. W., Ching, S. M., Lim, H. M., & Chinna, K. (2015). Validation of the Framingham general cardiovascular risk score in a multiethnic Asian population: A retrospective cohort study. *BMJ Open*, 5(5). <https://doi.org/10.1136/bmjopen-2014-007324>

- Chi, J., Liu, Y., Wang, V., & Yan, J. (2022). Performance analysis of three kinds of neural networks in the classification of mask images. *Journal of Physics: Conference Series*, 2181, 012032. <https://iopscience.iop.org/article/10.1088/1742-6596/2181/1/012032/pdf>
- Cho, S. Y., Kim, S. H., Kang, S. H., Lee, K. J., Choi, D., Kang, S., Park, S. J., Kim, T., Yoon, C. H., Youn, T. J., & Chae, I. H. (2021). Pre-existing and machine learning-based models for cardiovascular risk prediction. *Scientific Reports*, 11(1). <https://doi.org/10.1038/s41598-021-88257-w>
- Crea, F. (2021). The new SCORE2 risk prediction algorithms and the growing challenge of risk factors not captured by traditional risk scores. *European Heart Journal*, 42(25), 2403–2407. <https://academic.oup.com/eurheartj/article/42/25/2403/6312863>
- D'Agostino, R. B., Sr., Vasan, R. S., Pencina, M. J., Wolf, P. A., Cobain, M., Massaro, J. M., & Kannel, W. B. (2008). General cardiovascular risk profile for use in primary care: The Framingham Heart Study. *Circulation*, 117(6), 743-753. <https://pubmed.ncbi.nlm.nih.gov/18212285/>
- D'Agostino, R. B., Grundy, S., Sullivan, L. M., & Wilson, P., for the CHD Risk Prediction Group. (2001). Validation of the Framingham coronary heart disease prediction scores: Results of a multiple ethnic groups investigation. *JAMA*, 286(2), 180–187. <https://jamanetwork.com/journals/jama/fullarticle/193997>
- Dawber, T. R., Meadors, G. F., & Moore, F. E., Jr. (1951). Epidemiological approaches to heart disease: The Framingham Study. *American Journal of Public Health and the Nation's Health*, 41(3), 279-281. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1525365/>
- de Almeida, A. J. P. O., de Almeida Rezende, M. S., Dantas, S. H., de Lima Silva, S., de Oliveira, J. C. P. L., de Lourdes Assunção Araújo de Azevedo, F., Alves, R. M. F. R., de Menezes, G. M. S., Dos Santos, P. F., Gonçalves, T. A. F., Schini-Kerth, V. B., & de Medeiros, I. A. (2020). Unveiling the role of inflammation and oxidative stress on age-related cardiovascular diseases. *Oxidative Medicine and Cellular Longevity*. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7232723/>
- Department of Statistics Malaysia. (2023). *Statistics on Causes of Death, Malaysia, 2023*. Department of Statistics Malaysia. <https://www.dosm.gov.my/portal-main/release-content/statistics-on-causes-of-death-malaysia-2023>
- Desai, D. S., & Hajouli, S. (2023). Arrhythmias. In *StatPearls*. StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK558923/>
- Di Cesare, M., Perel, P., Taylor, S., Kabudula, C., Bixby, H., Gaziano, T. A., McGhie, D. V., Mwangi, J., Pervan, B., Narula, J., Pineiro, D., & Pinto, F. J. (2024). The heart of the

- Dinesh, K. G., Arumugaraj, K., Santhosh, K. D., & Mareeswari, V. (2018). Prediction of Cardiovascular Disease Using Machine Learning Algorithms. *2018 International Conference on Current Trends towards Converging Technologies (ICCTCT)*, 1–7. <https://doi.org/10.1109/ICCTCT.2018.8550857>
- Diogo, V. S., Ferreira, H. A., & Prata, D. (2022). Early diagnosis of Alzheimer's disease using machine learning: A multi-diagnostic, generalizable approach. *Alzheimer's Research & Therapy*, 14(1), 107. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9347083/>
- Dolezel, D., McLeod, A., & Fulton, L. (2021). Examining predictors of myocardial infarction. *International Journal of Environmental Research and Public Health*, 18(21), 11284. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8583114/>
- Fan, Z., Du, Z., Fu, J., Zhou, Y., Zhang, P., Shi, C., & Sun, Y. (2023). Comparing the performance of machine learning and conventional models for predicting atherosclerotic cardiovascular disease in a general Chinese population. *BMC Medical Informatics and Decision Making*, 23(1), 134. <https://bmcmedinformdecismak.biomedcentral.com/articles/10.1186/s12911-023-02242-z>
- Feigin, V. L., Stark, B. A., Johnson, C. O., Roth, G. A., Bisignano, C., Abady, G. G., Abbasifard, M., Abbasi-Kangevari, M., Abd-Allah, F., Abedi, V., Abualhasan, A., Abu-Rmeileh, N. M. E., Abushouk, A. I., Adebayo, O. M., Agarwal, G., Agasthi, P., Ahinkorah, B. O., Ahmad, S., Ahmadi, S., ... Murray, C. J.L. (2021). Global, regional, and national burden of stroke and its risk factors, 1990-2019: A systematic analysis for the Global Burden of Disease Study 2019. *The Lancet Neurology*, 20(10), 1–26. [https://doi.org/10.1016/S1474-4422\(21\)00252-0](https://doi.org/10.1016/S1474-4422(21)00252-0)
- Fong, A., Wan Azman Wan Ahmad, W. A., Azhari Rosman, A., Feisul Mustapha, F., Zanariah Hussein, Z., Sri Wahyu Taher, S., Asrul Akmal Shafie, A., Khicha, N., Kim, P., & Tan, H. H. (2023). *Heart Matters: The rising burden of cardiovascular disease in Malaysia and potential touchpoints for interventions*. IQVIA. <https://www.malaysianheart.org/management.file/doc/202312231133720341.pdf>
- Fuchs, F. D., & Whelton, P. K. (2020). High blood pressure and cardiovascular disease. *Hypertension*, 75(2), 285-292. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10243231/>
- Gaziano, T. A., Bitton, A., Anand, S., Abrahams-Gessel, S., & Murphy, A. (2010). Growing epidemic of coronary heart disease in low- and middle-income countries. *Current*

- GBD 2019 Diseases and Injuries Collaborators. (2020). Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019: A systematic analysis for the Global Burden of Disease Study 2019. *The Lancet*, 396(10258), 1204–1222. [https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(20\)30925-9/fulltext](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(20)30925-9/fulltext)
- George, J., Rapsomaniki, E., Pujades-Rodriguez, M., Shah, A. D., Denaxas, S., Herrett, E., Smeeth, L., Timmis, A., & Hemingway, H. (2015). How does cardiovascular disease first present in women and men?: Incidence of 12 cardiovascular diseases in a contemporary cohort of 1,937,360 people. *Circulation*, 132(14). <https://www.ahajournals.org/doi/full/10.1161/CIRCULATIONAHA.114.013797>
- Ghasemieh, A., Lloyed, A., Bahrami, P., Vajar, P., & Kashef, R. (2023). A novel machine learning model with stacking ensemble learner for predicting emergency readmission of heart-disease patients. *Decision Analytics Journal*, 7, 100242. <https://doi.org/10.1016/j.dajour.2023.100242>
- Gheorghe, A., Griffiths, U., Murphy, A., Legido-Quigley, H., Lamptey, P., & Perel, P. (2018). The economic burden of cardiovascular disease and hypertension in low- and middle-income countries: A systematic review. *BMC Public Health*, 18(1), 975. <https://doi.org/10.1186/s12889-018-5806-x>
- Gholamy, A., Kreinovich, V., & Kosheleva, O. (2018). Why 70/30 or 80/20 Relation Between Training and Testing Sets: A Pedagogical Explanation. *Departmental Technical Reports (CS)*. https://scholarworks.utep.edu/cs_techrep/1209
- Goff, D. C., Lloyd-Jones, D. M., Bennett, G., Coady, S., D’Agostino, R. B., Gibbons, R., Greenland, P., Lackland, D. T., Levy, D., O’Donnell, C. J., Robinson, J. G., Schwartz, J. S., Shero, S. T., Smith, S. C., Sorlie, P., Stone, N. J., & Wilson, P. W. F. (2014). 2013 ACC/AHA guideline on the assessment of cardiovascular risk: A report of the American college of cardiology/American heart association task force on practice guidelines. *Journal of the American College of Cardiology*, 63(25 PART B), 2935–2959. <https://doi.org/10.1016/j.jacc.2013.11.005>
- Goldstein, B. A., Navar, A. M., & Carter, R. E. (2017). Moving beyond regression techniques in cardiovascular risk prediction: applying machine learning to address analytic challenges. *European Heart Journal*, 38(23), 1805–1814. <https://pubmed.ncbi.nlm.nih.gov/27436868/>
- Gray, D. (2009). Risk assessment gone mad: The rise of risk evaluation and mass public deception. *British Journal of Cardiology*, 16, 117–118.

<https://bjcardio.co.uk/2009/05/risk-assessment-gone-mad-the-rise-of-risk-evaluation-and-mass-public-deception/>

Grundy, S. M., Stone, N. J., Bailey, A. L., Beam, C., Birtcher, K. K., Blumenthal, R. S., Braun, L. T., de Ferranti, S., Faiella-Tommasino, J., Forman, D. E., Goldberg, R., Heidenreich, P. A., Hlatky, M. A., Jones, D. W., Lloyd-Jones, D., Lopez-Pajares, N., Ndumele, C. E., Orringer, C. E., Peralta, C. A., ... Yeboah, J. (2019). 2018 AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA Guideline on the Management of Blood Cholesterol: Executive Summary: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice¹¹⁵ Guidelines. *Journal of the American College of Cardiology*, 73(24), 3168–3209. <https://doi.org/10.1016/j.jacc.2018.11.002>

Hajian-Tilaki, K. (2013). Receiver operating characteristic (ROC) curve analysis for medical diagnostic test evaluation. *Caspian Journal of Internal Medicine*, 4(2), 627-635. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3755824/>

Hamid, T. A. (2019). *Ageing in Malaysia* [PowerPoint slides]. Malaysian Research Institute on Ageing. https://www.un.org/development/desa/pd/sites/www.un.org.development.desa.pd/files/unpd_egm_201902_s3_tengkuaizanhamid.pdf

Hankey, G. J. (2017). Stroke. *The Lancet*, 389(10069), 641–654. [https://doi.org/10.1016/S0140-6736\(16\)30962-X](https://doi.org/10.1016/S0140-6736(16)30962-X)

Health Facts 2021. Health Informatics Centre Planning Division, Ministry of Health Malaysia; 2021 [cited 2024Aug16]. Available from: https://www.moh.gov.my/moh/resources/Penerbitan/Penerbitan%20Utama/HEALTH%20FACTS/Health_Facts_2021.pdf

Hermier, M., & Nighoghossian, N. (2004). Contribution of susceptibility-weighted imaging to acute stroke assessment. *Stroke*, 35(8). <https://doi.org/10.1161/01.STR.0000133341.74387.9>

Hermiz, C., & Sedhai, Y. R. (2023). Angina. In *StatPearls*. StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK557672/>

Hicks, S. A., Strümke, I., Thambawita, V., Hammou, M., Riegler, M. A., Halvorsen, P., & Parasa, S. (2022). On evaluation metrics for medical applications of artificial intelligence. *Scientific Reports*, 12, 5979. <https://www.nature.com/articles/s41598-022-09954-8>

Hui, C., Tadi, P., Khan Suheb, M. Z., & Patti, L. (2024). Ischemic Stroke. In *StatPearls*. StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK499997/>

IBM, (2023). *What are SVMs?* <https://www.ibm.com/topics/support-vector-machine>

Ibrahim, F., & Murr, N. I. (2022). Embolic Stroke. In *StatPearls*. StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK564351/>

Institute for Public Health 2020. National Health and Morbidity Survey (NHMS) 2019: Non-Communicable Diseases, Healthcare Demand, and Health Literacy: Key Findings. National Institutes of Health (NIH) Ministry of Health Malaysia; 2020 [cited 2024Aug15]. Available from: https://iptk.moh.gov.my/images/technical_report/2020/4_Infographic_Booklet_NHMS_2019_-_English.pdf

Institute for Public Health 2024. National Health and Morbidity Survey (NHMS) 2023: Non-Communicable Diseases and Healthcare Demand: Key Findings. National Institutes of Health (NIH). Ministry of Health Malaysia. <https://iku.nih.gov.my/images/nhms2023/key-findings-nhms-2023.pdf>

Institute for Public Health 2024. National Health and Morbidity Survey (NHMS) 2013: Non-Communicable Diseases and Healthcare Demand. National Institutes of Health (NIH). Ministry of Health Malaysia. <https://iku.nih.gov.my/images/nhms2023/fact-sheet-nhms-2023.pdf>

Jackson, C. L., Zordok, M., & Kullo, I. J. (2021). Familial hypercholesterolemia in Southeast and East Asia. *American Journal of Preventive Cardiology*, 6. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8315601/>

Jalili, M., Ehrampoush, M. H., Mokhtari, M., Ebrahimi, A. A., Mazidi, F., Abbasi, F., & Karimi, H. (2021). Ambient air pollution and cardiovascular disease rate: An ANN modeling study in Yazd, Central Iran. *Scientific Reports*, 11, 16937. <https://www.nature.com/articles/s41598-021-94925-8>

Jiang, T., Gradus, J. L., & Rosellini, A. J. (2020). Supervised machine learning: A brief primer. *Behavior Therapy*, 51(5), 675-687. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7431677/>

Jijo, B. T., & Abdulazeez, A. M. (2021). Classification based on decision tree algorithm for machine learning. *Journal of Applied Science and Technology Trends*, 2(1), 20–28. <https://www.jastt.org/index.php/jasttpath/article/view/65/24>

Kannel, W. B., McGee, D., & Gordon, T. (1976). A general cardiovascular risk profile: The Framingham study. *The American Journal of Cardiology*, 38(1), 46-51. <https://www.sciencedirect.com/science/article/abs/pii/0002914976900618>

- Kaptoge, S., Pennells, L., De Bacquer, D., Cooney, M. T., Kavousi, M., Stevens, G., Riley, L. M., Savin, S., Khan, T., Altay, S., Amouyel, P., Assmann, G., Bell, S., Ben-Shlomo, Y., Berkman, L., Beulens, J. W., Björkelund, C., Blaha, M., Blazer, D. G., ... Di Angelantonio, E. (2019). World Health Organization cardiovascular disease risk charts: revised models to estimate risk in 21 global regions. *The Lancet Global Health*, 7(10). [https://doi.org/10.1016/S2214-109X\(19\)30318-3](https://doi.org/10.1016/S2214-109X(19)30318-3)
- Kasim, S., Malek, S., Song, C., Ahmad, W. A. W., Fong, A., Ibrahim, K. S., Safiruz, M. S., Aziz, F., Hiew, J. H., & Ibrahim, N. (2022). In-hospital mortality risk stratification of Asian ACS patients with artificial intelligence algorithm. *PLoS ONE*, 17(12). <https://doi.org/10.1371/journal.pone.0278944>
- Kasim, S. S., Ibrahim, N., Malek, S., Ibrahim, K. S., Aziz, M. F., Song, C., Chia, Y. C., Ramli, A. S., Negishi, K., & Nasir, N. M. (2023). Validation of the general Framingham Risk Score (FRS), SCORE2, revised PCE and WHO CVD risk scores in an Asian population. *The Lancet Regional Health – Western Pacific*, 35. <https://doi.org/10.1016/j.lanwpc.2023.100742>
- Katzmarzyk, P. T., Friedenreich, C., Shiroma, E. J., & Lee, I. M. (2022). Physical inactivity and non-communicable disease burden in low-income, middle-income, and high-income countries. *British Journal of Sports Medicine*, 56(2), 101-106. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8478970/>
- Kecman, V. (2004). Support vector machines – An introduction. In *Studies in Fuzziness and Soft Computing* (Vol. 177, pp. 1–47). Springer-Verlag Berlin Heidelberg. <https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=09a8e0be3aff0c8ed0e91167609814a0e9215481>
- Khaw, W.-F., & Yong, H. Y. (2024). Top ten causes of death in Malaysia for the years 2013 and 2019. *The Lancet Regional Health – Western Pacific*, 45. [https://www.thelancet.com/journals/lanwpc/article/PIIS2666-6065\(24\)00060-9/fulltext](https://www.thelancet.com/journals/lanwpc/article/PIIS2666-6065(24)00060-9/fulltext)
- Kırboğa, K. K., & Küçükşille, E. U. (2023). Identifying cardiovascular disease risk factors in adults with explainable artificial intelligence. *Anatolian Journal of Cardiology*, 27(11), 657-663. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10621606/>
- Klein, L. W. (2022). Pathophysiologic mechanisms of tobacco smoke producing atherosclerosis. *Current Cardiology Reviews*, 18(6). <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9893148/>
- Knuuti, J., Wijns, W., Saraste, A., Capodanno, D., Barbato, E., Funck-Brentano, C., Prescott, E., Storey, R. F., Deaton, C., Cuisset, T., Agewall, S., Dickstein, K., Edvardsen, T., Escaned, J., Gersh, B. J., Svitil, P., Gilard, M., Hasdai, D., Hatala, R., Mahfoud, F., Masip, J., Muneretto, C., Valgimigli, M., Achenbach, S., Bax, J. J., & ESC Scientific Document

- Group. (2020). 2019 ESC guidelines for the diagnosis and management of chronic coronary syndromes: The Task Force for the diagnosis and management of chronic coronary syndromes of the European Society of Cardiology (ESC). *European Heart Journal*, 41(3), 407–477. <https://academic.oup.com/eurheartj/article/41/3/407/5556137>
- Kuhn, M., & Johnson, K. (2013). Over-Fitting and Model Tuning. In *Applied predictive modeling*. Springer. <https://link.springer.com/book/10.1007/978-1-4614-6849-3>
- Kumar, R., & Indrayan, A. (2011). Receiver operating characteristic (ROC) curve for medical researchers. *Indian Pediatrics*, 48(4), 277-287. <https://www.indianpediatrics.net/apr2011/277.pdf>
- Kuriakose, D., & Xiao, Z. (2020). Pathophysiology and Treatment of Stroke: Present Status and Future Perspectives. *International Journal of Molecular Sciences*, 21(20), 7609. <https://doi.org/10.3390/ijms21207609>
- Lim, K. G., & Cheah, W. K. (2016). A Review of Metabolic Syndrome Research in Malaysia. 71, 20-28. <https://pubmed.ncbi.nlm.nih.gov/27801385/>
- Liu, J., Dong, X., Zhao, H., & Tian, Y. (2022). Predictive classifier for cardiovascular disease based on stacking model fusion. *Processes*, 10(4), 749. <https://www.mdpi.com/2227-9717/10/4/749>
- Liu, J., Hong, Y., D'Agostino, R. B., Sr., et al. (2004). Predictive value for the Chinese population of the Framingham CHD risk assessment tool compared with the Chinese multi-provincial cohort study. *JAMA*, 291(21), 2591–2599. <https://jamanetwork.com/journals/jama/article-abstract/198843>
- Liu, W., Laranjo, L., Klimis, H., Chiang, J., Yue, J., Marschner, S., Quiroz, J. C., Jorm, L., & Chow, C. K. (2023). Machine-learning versus traditional approaches for atherosclerotic cardiovascular risk prognostication in primary prevention cohorts: A systematic review and meta-analysis. *European Heart Journal - Quality of Care and Clinical Outcomes*, 9(4), 310-322. <https://pubmed.ncbi.nlm.nih.gov/36869800/>
- Lu, H. T., & Nordin, R. B. (2013). Ethnic differences in the occurrence of acute coronary syndrome: Results of the Malaysian National Cardiovascular Disease (NCVD) Database Registry (March 2006 - February 2010). *BMC Cardiovascular Disorders*, 13, 97. <https://doi.org/10.1186/1471-2261-13-97>
- Machta, S., Gauthier, V., Ferrières, J., Montaye, M., Huo Yung Kai, S., Gbokou, S., Biasch, K., Moitry, M., Amouyel, P., Dallongeville, J., & Meirhaeghe, A. (2022). Comparison of clinical profiles and care for patients with incident versus recurrent acute coronary

syndromes in France: Data from the MONICA registries. *PLoS ONE*, 17(2). <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0263589>

Marbaniang, I. A., Choudhury, N. A., & Moulik, S. (2020). Cardiovascular disease (CVD) prediction using machine learning algorithms. In *Proceedings of the 2020 IEEE 17th India Council International Conference (INDICON)* (pp. 1-6). IEEE. <https://ieeexplore.ieee.org/document/9342297>

Marcus, P. M. (2019). Assessment of cancer screening: A primer. In *Performance measures* (Chap. 3). Bethesda, MD: National Cancer Institute (US). <https://www.ncbi.nlm.nih.gov/books/NBK550207/>

Martín-Timón, I., Sevillano-Collantes, C., Segura-Galindo, A., & Del Cañizo-Gómez, F. J. (2014). Type 2 diabetes and cardiovascular disease: Have all risk factors the same strength? *World Journal of Diabetes*, 5(4), 444-470. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4127581/>

Mastrangelo, M., Giordo, L., Ricciardi, G., De Michele, M., Toni, D., & Leuzzi, V. (2022). Acute ischemic stroke in childhood: A comprehensive review. *European Journal of Pediatrics*, 181(1), 45–58. <https://doi.org/10.1007/s00431-021-04212-X>

Matheson, M. B., Kato, Y., Baba, S., Cox, C., Lima, J. A. C., & Ambale-Venkatesh, B. (2022). Cardiovascular Risk Prediction Using Machine Learning in a Large Japanese Cohort. *Circulation Reports*, 4(12), 595–603. <https://doi.org/10.1253/circrep.CR-22-0101>

Mechanic, O. J., Gavin, M., & Grossman, S. A. (2023). Acute myocardial infarction. In *StatPearls* (2024 ed.). StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK459269/>

Mesquita, F., & Marques, G. (2024). An explainable machine learning approach for automated medical decision support of heart disease. *Data & Knowledge Engineering*, 153, 102339. <https://doi.org/10.1016/j.datak.2024.102339>

Minicucci, M. F., Azevedo, P. S., Polegato, B. F., Paiva, S. A., & Zornoff, L. A. (2011). Heart failure after myocardial infarction: Clinical implications and treatment. *Clinical Cardiology*, 34(7), 410–414. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6652587/>

Mocumbi, A. O. (2024). Cardiovascular Health Care in Low- and Middle-Income Countries. *Circulation*, 149(8), 557-559. <https://www.ahajournals.org/doi/10.1161/CIRCULATIONAHA.123.065717>

Mohd Azahar, N. M. Z., Krishnapillai, A., Che Hassan, M. R., Dasiman, R., & Yusoff, K. (2022). Predictors of cardiovascular disease using Framingham risk score in a rural population.

Malaysian Journal of Medicine and Health Sciences, 18(SUPP8), 340-347.
https://medic.upm.edu.my/upload/dokumen/2022070112475643_1430.pdf

Mohd Ghazali, S., Seman, Z., Cheong, K. C., Hock, L. K., Manickam, M., Kuay, L. K., Yusoff, A. F., Mustafa, F. I., & Mustafa, A. N. (2015). Sociodemographic factors associated with multiple cardiovascular risk factors among Malaysian adults. *BMC Public Health*, 15(1).
<https://doi.org/10.1186/s12889-015-1432-z>

Moldagulova, A., & Sulaiman, R. B. (2017). Using KNN algorithm for classification of textual documents. In *Proceedings of the 2017 8th International Conference on Information Technology (ICIT)* (pp. 665–671). IEEE.
<https://ieeexplore.ieee.org/abstract/document/8079924>

Nath, A., & Sahu, G. K. (2019). Exploiting ensemble learning to improve prediction of phospholipidosis inducing potential. *Journal of Theoretical Biology*, 479, 37-47.
<https://doi.org/10.1016/j.jtbi.2019.07.009>

National Institute of Health. (2023). *Stroke*. <https://www.nhlbi.nih.gov/health/stroke/diagnosis>

National Institute of Health. (2022). *What Is Atherosclerosis?*
<https://www.nhlbi.nih.gov/health/atherosclerosis>

National Health Association of Malaysia. (2017). *Clinical Practice Guidelines on Primary & Secondary Prevention of Cardiovascular Disease 2017*.
<https://www.malaysianheart.org/publication/clinical-practice-guidelines/p/primary-secondary-prevention-of-cardiovascular-disease-2017>

National Heart, Lung, and Blood Institute. (2023, May 26). *What Is a Stroke?*
<https://www.nhlbi.nih.gov/health/stroke>

Ojha, N., & Dhamoon, A. S. (2023). Myocardial Infarction. In *StatPearls*. StatPearls Publishing.
<https://www.ncbi.nlm.nih.gov/books/NBK537076/>

Olvera Lopez, E., Ballard, B. D., & Jan, A. (2023). *Cardiovascular disease*. In *StatPearls*. StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK535419/>

Oppenheimer, G. M. (2005). Becoming the Framingham Study 1947–1950. *American Journal of Public Health*, 95(4), 602-610. <https://pubmed.ncbi.nlm.nih.gov/15798116/>

- Pal, M., Parija, S., Panda, G., Dhama, K., & Mohapatra, R. K. (2022). Risk prediction of cardiovascular disease using machine learning classifiers. *Open Medicine (Warsaw)*, 17(1), 1100-1113. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9206502/>
- Parikh, R., Mathai, A., Parikh, S., Chandra Sekhar, G., & Thomas, R. (2008). Understanding and using sensitivity, specificity and predictive values. *Indian Journal of Ophthalmology*, 56(1), 45–50. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2636062/>
- Patel, A. (2020). *Stacking-ensemble meta algorithms for improving predictions*. ML Research Lab. Medium. <https://medium.com/ml-research-lab/stacking-ensemble-meta-algorithms-for-improve-predictions-f4b4cf3b9237>
- Pencina, M. J., D'Agostino Sr, R. B., & Steyerberg, E. W. (2011). Extensions of net reclassification improvement calculations to measure usefulness of new biomarkers. *Statistics in Medicine*, 30(1), 11–21. <https://pubmed.ncbi.nlm.nih.gov/21204120/>
- Pencina, M. J., D'Agostino Sr, R. B., D'Agostino Jr, R. B., & Vasan, R. S. (2008). Evaluating the added predictive ability of a new marker: from area under the ROC curve to reclassification and beyond. *Statistics in Medicine*, 27(2), 157–172. <https://pubmed.ncbi.nlm.nih.gov/17569110/>
- Peng, M., Hou, F., Cheng, Z., Shen, T., Liu, K., Zhao, C., & Zheng, W. (2023). Prediction of cardiovascular disease risk based on major contributing features. *Scientific Reports*, 13, 4778. <https://www.nature.com/articles/s41598-023-31870->
- Putri, N. F. A., Sazzli, S. K., Sorayya, M., Nurulain, I., Nafiza, M. N., & Najmin, A. B. (2023). Comparing the performance of the FRS, machine learning, and stacked ensemble learning in estimating the 10-year CVD risk in the Asian population. *International Journal of Cardiology*, 393(Supplement), 131483. <https://www.sciencedirect.com/science/article/abs/pii/S0167527323015450>
- Quispe, R., Bazo-Alvarez, J. C., Burroughs Peña, M. S., Poterico, J. A., Gilman, R. H., Checkley, W., Bernabé-Ortiz, A., Huffman, M. D., & Miranda, J. J. (2015). Distribution of Short-Term and Lifetime Predicted Risks of Cardiovascular Diseases in Peruvian Adults. *Journal of the American Heart Association*, 4(8). <https://www.ahajournals.org/doi/full/10.1161/JAHA.115.002112>
- Ramraj, S., Uzir, N., Sunil, R., & Banerjee, S. (2016). Experimenting XGBoost algorithm for prediction and classification of different datasets. *International Journal of Control Theory and Applications*, 9(40), 1-10. https://www.researchgate.net/profile/Shatadeep-Banerjee/publication/318132203_Experimenting_XGBoost_Algorithm_for_Prediction_and_Classification_of_Different_Datasets/links/595b89b0458515117741a571/Experimenting-XGBoost-Algorithm-for-Prediction-and-Classification-of-Different-Datasets.pdf

- Reddy, K. V. V., Elamvazuthi, I., Abd Aziz, A., Paramasivam, S., Chua, H. N., & Pranavanand, S. (2021). Heart disease risk prediction using machine learning classifiers with attribute evaluators. *Applied Sciences*, 11(18), 8352. <https://www.mdpi.com/2076-3417/11/18/8352>
- Richardson, E., Trevizani, R., Greenbaum, J. A., Carter, H., Nielsen, M., & Peters, B. (2024). The receiver operating characteristic curve accurately assesses imbalanced datasets. *Patterns*, 5(6), 100994. https://www.sciencedirect.com/science/article/pii/S2666389924001090?ssrnid=4655233&dgcid=SSRN_redirect_SD
- Rios, R., Miller, R. J. H., Manral, N., Sharir, T., Einstein, A. J., Fish, M. B., Ruddy, T. D., Kaufmann, P. A., Sinusas, A. J., Miller, E. J., Bateman, T. M., Dorbala, S., Di Carli, M., Van Kriekinge, S. D., Kavanagh, P. B., Parekh, T., Liang, J. X., Dey, D., Berman, D. S., & Slomka, P. J. (2022). Handling missing values in machine learning to predict patient-specific risk of adverse cardiac events: Insights from REFINE SPECT registry. *Computers in Biology and Medicine*, 145, 105449. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9117456/>
- Rodgers, J. L., Jones, J., Bolleddu, S. I., Vanthenapalli, S., Rodgers, L. E., Shah, K., Karia, K., & Panguluri, S. K. (2019). Cardiovascular risks associated with gender and aging. *Journal of Cardiovascular Development and Disease*, 6(2). <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6616540/>
- Roth, G. A., Mensah, G. A., Johnson, C. O., Addolorato, G., Ammirati, E., Baddour, L. M., Barengo, N. C., Beaton, A. Z., Benjamin, E. J., Benziger, C. P., Bonny, A., Brauer, M., Brodmann, M., Cahill, T. J., Carapetis, J., Catapano, A. L., Chugh, S. S., Cooper, L. T., Coresh, J., & Criqui, M., ... Fuster, V. (2020). Global burden of cardiovascular diseases and risk factors, 1990-2019. *Journal of the American College of Cardiology*, 76(25), 2982-3021. <https://doi.org/10.1016/j.jacc.2020.11.010>
- Rountree, K. M., Yaker, Z., & Lopez, P. P. (2024). Partial Thromboplastin Time. In *StatPearls*. StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK507772/>
- Rustamov, Z., Rustamov, J., Sultana, M. S., Ywei, J., Balakrishnan, V., & Zaki, N. (2023). Cardiovascular disease prediction using ensemble learning techniques: A stacking approach. In *Proceedings of the 2023 19th IEEE International Colloquium on Signal Processing & Its Applications (CSPA)* (pp. 93-98). IEEE. <https://ieeexplore.ieee.org/document/10087730>
- Rymer, M. M. (2011). Hemorrhagic stroke: Intracerebral hemorrhage. *Missouri Medicine*, 108(1), 50–54. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6188453/>

- Sacco, R. L., Kasner, S. E., Broderick, J. P., Caplan, L. R., Connors, J. J., Culebras, A., Elkind, M. S. V., George, M. G., Hamdan, A. D., Higashida, R. T., Hoh, B. L., Janis, L. S., Kase, C. S., Kleindorfer, D. O., Lee, J. M., Moseley, M. E., Peterson, E. D., Turan, T. N., Valderrama, A. L., & Vinters, H. V. (2013). An updated definition of stroke for the 21st century: A statement for healthcare professionals from the American heart association/American stroke association. *Stroke*, 44(7), 2064–2089. <https://doi.org/10.1161/STR.0b013e318296aeca>
- Sajeev, S., Champion, S., Beleigoli, A., Chew, D., Reed, R. L., Magliano, D. J., Shaw, J. E., Milne, R. L., Appleton, S., Gill, T. K., & Maeder, A. (2021). Predicting australian adults at high risk of cardiovascular disease mortality using standard risk factors and machine learning. *International Journal of Environmental Research and Public Health*, 18(6), 1–14. <https://doi.org/10.3390/ijerph18063187>
- Sapra, A., & Bhandari, P. (2023). Diabetes. In *StatPearls*. StatPearls Publishing. <http://www.ncbi.nlm.nih.gov/books/NBK551501/>
- Sarkar, S. (2019). Calibration techniques and its importance in machine learning. *Medium*. <https://kingsubham27.medium.com/calibration-techniques-and-its-importance-in-machine-learning-71bec997b661>
- Sarker, I. H. (2021). Machine learning: Algorithms, real-world applications and research directions. *SN Computer Science*, 2, 160. <https://link.springer.com/article/10.1007/s42979-021-00592-x#citeas>
- SCORE2 Working Group & ESC Cardiovascular Risk Collaboration. (2021). SCORE2 risk prediction algorithms: New models to estimate 10-year risk of cardiovascular disease in Europe. *European Heart Journal*, 42(25), 2439–2454. <https://doi.org/10.1093/eurheartj/ehab309>
- Selvarajah, S., Kaur, G., Haniff, J., Cheong, K. C., Hiong, T. G., Van Der Graaf, Y., & Bots, M. L. (2014). Comparison of the Framingham Risk Score, SCORE and WHO/ISH cardiovascular risk prediction models in an Asian population. *International Journal of Cardiology*, 176(1), 211–218. <https://doi.org/10.1016/j.ijcard.2014.07.066>
- Shao, C., Wang, J., Tian, J., & Tang, Y. D. (2020). Coronary artery disease: From mechanism to clinical practice. In M. Wang (Ed.), *Coronary artery disease: Therapeutics and drug discovery* (Vol. 1177, pp. 1-36). Springer.
- Simon, S., Mandair, D., Albakri, A., Fohner, A., Simon, N., Lange, L., Biggs, M., Mukamal, K., Psaty, B., & Rosenberg, M. (2022). The impact of time horizon on classification accuracy: Application of machine learning to prediction of incident coronary heart disease. *JMIR Cardio*, 6(2), e38040. <https://doi.org/10.2196/38040>

- Singh, A. (2024). *4 Boosting Algorithms You Should Know: GBM, XGBoost, LightGBM & CatBoost*. AnalyticsVidhya. <https://www.analyticsvidhya.com/blog/2020/02/4-boosting-algorithms-machine-learning/#h-gradient-boosting-machine-gbm>
- Singh, V., Pencina, M., Einstein, A. J., et al. (2021). Impact of train/test sample regimen on performance estimate stability of machine learning in cardiovascular imaging. *Scientific Reports*, 11, 14490. <https://www.nature.com/articles/s41598-021-93651-5>
- Song, C., Kasim, S. S., Malek, S., Ibrahim, K. S., Sulaiman, N. M. N., Negishi, K., Hamidi, M. H. M., Aziz, M. F., & Ibrahim, N. (2022). Effects of air pollution towards hospital admission prediction of Asian patients with acute coronary syndrome (ACS) using LSTM method. *International Journal of Cardiology*, 369, 16. <https://doi.org/10.1016/j.ijcard.2022.10.059>
- Song, Y. Y., & Lu, Y. (2015). Decision tree methods: Applications for classification and prediction. *Shanghai Archives of Psychiatry*, 27(2), 130–135. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4466856/>
- Srivastava, T. (2024). *Guide to K-Nearest Neighbors Algorithm in Machine Learning*. AnalyticsVidhya. <https://www.analyticsvidhya.com/blog/2018/03/introduction-k-neighbours-algorithm-clustering/>
- Sterne, J. A., White, I. R., Carlin, J. B., Spratt, M., Royston, P., Kenward, M. G., Wood, A. M., & Carpenter, J. R. (2009). Multiple imputation for missing data in epidemiological and clinical research: Potential and pitfalls. *BMJ*, 338. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2714692/>
- Subramani, S., Varshney, N., Anand, M. V., Soudagar, M. E. M., Al-keridis, L. A., Upadhyay, T. K., Alshammari, N., Saeed, M., Subramanian, K., Anbarasu, K., & Rohini, K. (2023). Cardiovascular diseases prediction by machine learning incorporation with deep learning. *Frontiers in Medicine*, 10. <https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2023.1150933/full>
- Suthaharan, S. (2016). Support vector machine. In *Machine learning models and algorithms for big data classification*, 36. Springer. https://link.springer.com/chapter/10.1007/978-1-4899-7641-3_9
- Tadi, P., & Lui, F. (2023). Acute Stroke. In *StatPearls*. StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK535369/>
- Tajudin, M. A. B. A., Khan, M. F., Mahiyuddin, W. R. W., Hod, R., Latif, M. T., Hamid, A. H., Rahman, S. A., & Sahani, M. (2019). Risk of concentrations of major air pollutants on the prevalence of cardiovascular and respiratory diseases in the urbanized area of Kuala

Lumpur, Malaysia. *Ecotoxicology and Environmental Safety*, 171, 290-300.
<https://pubmed.ncbi.nlm.nih.gov/30612017/>

Talha, I., Elkhoudri, N., & Hilali, A. (2024). Major limitations of cardiovascular risk scores. *Cardiovascular Therapeutics*, 2024.
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10917477/>

Terry, D. F., Evans, J. C., Pencina, M. J., Murabito, J. M., Vasan, R. S., Wolf, P. A., Kelly-Hayes, M., Levy, D., D'Agostino, R. B., Sr., & Benjamin, E. J. (2007). Characteristics of Framingham Offspring participants with long-lived parents. *Archives of Internal Medicine*, 167(5), 438–444.
<https://jamanetwork.com/journals/jamainternalmedicine/fullarticle/769638>

Theerthagiri, P., & Vidya, J. (2022). Cardiovascular disease prediction using recursive feature elimination and gradient boosting classification techniques. *Expert Systems*, 39(9).
<https://onlinelibrary.wiley.com/doi/abs/10.1111/exsy.13064>

The Direct Healthcare Cost of Non-Communicable Diseases in Malaysia (2022). Ministry of Health Malaysia; [cited 2024]4Aug15] institute of health. Available from:
https://www.moh.gov.my/moh/resources/Penerbitan/Rujukan/NCD/NCD_Laporan/HEALTH-COST_of_NCDs-7a-WEB.pdf

The Impact of Non-Communicable Diseases and their Risk Factors on Malaysia's Gross Domestic Product. Non-Communicable Disease Section, Ministry of Health Malaysia; 2020 [cited 2024Aug15]. Available from: https://leadncdmalaysia.com/wp-content/uploads/2020/10/NCD-Economic-Webinar_Moodie-M.pdf

The Impact of Non-Communicable Diseases and their Risk Factors on Malaysia's Gross Domestic Product. Non-Communicable Disease Section, Ministry of Health Malaysia; 2020 [cited 2024Aug15]. Available from: https://leadncdmalaysia.com/wp-content/uploads/2020/10/NCD-Economic-Webinar_Moodie-M.pdf

Tsarapatsani, K., Sakellarios, A. I., Pezoulas, V. C., Tsakanikas, V. D., Kleber, M. E., Marz, W., Michalis, L. K., & Fotiadis, D. I. (2022). Machine learning models for cardiovascular disease events prediction. In *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)* (pp. 1066-1069). IEEE.
<https://pubmed.ncbi.nlm.nih.gov/36085658/>

Thygesen, K., Alpert, J. S., & White, H. D. (2007). Universal definition of myocardial infarction. *Circulation*, 116(22), 2634–2653.
<https://doi.org/10.1161/CIRCULATIONAHA.107.187397>

- Thygesen, K., Mair, J., Giannitsis, E., Mueller, C., Lindahl, B., Blankenberg, S., Huber, K., Plebani, M., Biasucci, L. M., Tubaro, M., Collinson, P., Venge, P., Hasin, Y., Galvani, M., Koenig, W., Hamm, C., Alpert, J. S., Katus, H., & Jaffe, A. S. (2012). How to use high-sensitivity cardiac troponins in acute cardiac care. *European Heart Journal*, 33(18), 2252–2257. <https://doi.org/10.1093/eurheartj/ehs154>
- Unnithan, A. K. A., Das, J. M., & Mehta, P. (2023). Hemorrhagic Stroke. In *StatPearls*. StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK559173/>
- van Trier, T. J., Snaterse, M., Boekholdt, S. M., Scholte op Reimer, W. J. M., Hageman, S. H. J., Visseren, F. L. J., Dorresteijn, J. A. N., Peters, R. J. G., & Jørstad, H. T. (2024). Validation of Systematic Coronary Risk Evaluation 2 (SCORE2) and SCORE2-Older Persons in the EPIC-Norfolk prospective population cohort. *European Journal of Preventive Cardiology*, 31(2), 182–189. <https://doi.org/10.1093/eurjpc/zwad318>
- Virani, S. S., Alonso, A., Aparicio, H. J., Benjamin, E. J., Bittencourt, M. S., Callaway, C. W., Carson, A. P., Chamberlain, A. M., Cheng, S., Delling, F. N., Elkind, M. S. V., Evenson, K. R., Ferguson, J. F., Gupta, D. K., Khan, S. S., Kissela, B. M., Knutson, K. L., Lee, C. D., Lewis, T. T., ... Tsao, C. W. (2021). Heart disease and stroke statistics—2021 update: A report from the American Heart Association. *Circulation*. https://professional.heart.org/-/media/PHD-Files-2/Science-News/2/2021-Heart-and-Stroke-Stat-Update/2021_Stat_Update_factsheet_Older_and_CVD.pdf
- Vujovic, Ž. Đ. (2021). Classification model evaluation metrics. *International Journal of Advanced Computer Science and Applications*, 12(6). <http://dx.doi.org/10.14569/IJACSA.2021.0120670>
- Wallert, J., Tomasoni, M., Madison, G., & Held, C. (2017). Predicting two-year survival versus non-survival after first myocardial infarction using machine learning and Swedish national register data. *BMC Medical Informatics and Decision Making*, 17(1), 99. <https://doi.org/10.1186/s12911-017-0500-y>
- Ward, A., Sarraju, A., Chung, S., Li, J., Harrington, R., Heidenreich, P., Palaniappan, L., Scheinker, D., & Rodriguez, F. (2020). Machine learning and atherosclerotic cardiovascular disease risk prediction in a multi-ethnic population. *NPJ Digital Medicine*, 3, 125. <https://doi.org/10.1038/s41746-020-00331-1>
- Warren, T. Y., Barry, V., Hooker, S. P., Sui, X., Church, T. S., & Blair, S. N. (2010). Sedentary behaviors increase risk of cardiovascular disease mortality in men. *Medicine and Science in Sports and Exercise*, 42(5), 879-885. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2857522/>

- Weng, S. F., Reps, J., Kai, J., Garibaldi, J. M., & Qureshi, N. (2017). Can machine- learning improve cardiovascular risk prediction using routine clinical data? *PLOS ONE*, 12(4), e0174944. <https://doi.org/10.1371/journal.pone.0174944>
- Werner de Vargas, V., Schneider Aranda, J. A., Dos Santos Costa, R., da Silva Pereira, P. R., & Victória Barbosa, J. L. (2023). Imbalanced data preprocessing techniques for machine learning: A systematic mapping study. *Knowledge and Information Systems*, 65(1), 31-57. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9645765/>
- Wolf, R. M., Nagpal, M., & Magge, S. N. (2021). Diabetes and cardiometabolic risk in South Asian youth: A review. *Pediatric Diabetes*, 22(1), 52-66. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8191592/>
- World Health Organization. (2021). *Cardiovascular diseases (CVDs)*. World Health Organization. [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds))
- World Health Organization. (2020). *HEARTS technical package for cardiovascular disease management in primary health care: Risk based CVD management*. <https://www.who.int/publications/i/item/9789240001367>
- Yadlowsky, S., Hayward, R. A., Sussman, J. B., McClelland, R. L., Min, Y. I., & Basu, S. (2018). Clinical implications of revised pooled Cohort equations for estimating atherosclerotic cardiovascular disease risk. *Annals of Internal Medicine*, 169(1), 20–29. <https://pubmed.ncbi.nlm.nih.gov/29868850/>
- Yang, L., Wu, H., Jin, X., Zheng, P., Hu, S., Xu, X., Yu, W., & Yan, J. (2020). Study of cardiovascular disease prediction model based on random forest in eastern China. *Scientific Reports*, 10, 5245. <https://www.nature.com/articles/s41598-020-62133-5>
- Yang, W., Jiang, J., Schnellinger, E. M., Kimmel, S. E., & Guo, W. (2022). Modified Brier score for evaluating prediction accuracy for binary outcomes. *Statistical Methods in Medical Research*, 31(12), 2287-2296. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9691523/>
- Yew, S. Q., Chia, Y. C., & Theodorakis, M. (2023). Assessing 10-year cardiovascular disease risk in Malaysians with type 2 diabetes mellitus: Framingham cardiovascular versus United Kingdom prospective diabetes study equations. *Journal of Clinical Medicine*, 31(7). <https://doi.org/10.1177/10105395198734>
- You, J., Guo, Y., Kang, J. J., Wang, H. F., Yang, M., Feng, J. F., Yu, T., & Cheng, W. (2023). Development of machine learning-based models to predict 10-year risk of cardiovascular disease: A prospective cohort study. *Stroke & Vascular Neurology*, 8(6), 475-485. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10800279/>

- Zhang, Y., Miao, H., Chia, Y. C., Buranakitjaroen, P., Siddique, S., Shin, J., Turana, Y., Park, S., Tsoi, K., Chen, C. H., Cheng, H. M., Li, Y., Minh, H. V., Nagai, M., Nailles, J., Sison, J., Soenarta, A. A., Sogunuru, G. P., Sukonthasarn, A., ... Wang, J. (2022). Cardiovascular risk assessment tools in Asia. *Journal of Clinical Hypertension (Greenwich)*, 24(4), 369-377. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8989745/>
- Zhao, D. (2021). Epidemiological features of cardiovascular disease in Asia. *JACC Asia*, 1(1), 1-13. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9627928/>
- Zian, S., Kareem, S. A., & Varathan, K. D. (2021). An empirical evaluation of stacked ensembles with different meta-learners in imbalanced classification. *IEEE Access*, 9, 87434–87452. <https://ieeexplore.ieee.org/abstract/document/9452051>

APPENDIX A

Python Packages & Modules Used

Package/ Module	Functions
pandas	Data manipulation, reading CSV files, grouping data, handling missing values, and exporting results.
numpy	Numerical operations, including random seed setting for reproducibility
os	File and directory operations, joining file paths
missingno	Visualizing missing data to identify patterns of missingness.
matplotlib.pyplot	Creating and saving visualizations such as heatmaps and bar plots.
seaborn	Advanced data visualization, especially for statistical graphics.
joblib	Efficient serialization and de-serialization of Python objects, used for saving and loading models and datasets
sklearn.preprocessing	Data preprocessing utilities, including LabelEncoder for encoding categorical labels and MinMaxScaler for normalizing data.
sklearn.model_selection	Utilities for model selection, including train_test_split for splitting datasets and cross_val_score for evaluating models using cross-validation.
sklearn.impute	Imputation of missing data, specifically SimpleImputer and IterativeImputer.
sklearn.feature_selection	Feature selection techniques, such as RFE
sklearn.ensemble	Implementation of ensemble algorithms, specifically RandomForestClassifier
statsmodels.regression	Provides tools for estimating and evaluating statistical models, particularly OLS (Ordinary Least Squares) for linear regression

Package/ Module	Functions
statsmodels.tools	Tools for statistical models, including add_constant for adding a constant to a regression model.
imblearn.under_sampling	Undersampling techniques to balance class distribution, specifically RandomUnderSampler
imblearn.over_sampling	Oversampling techniques to balance class distribution, including SMOTE and RandomOverSampler
shap	Explainability of machine learning models through SHAP (SHapley Additive exPlanations) values, useful for understanding the impact of each feature
pickle	Serializing and de-serializing Python object structures, such as saving and loading custom objects like lists.
imblearn.pipeline.Pipeline	Creating machine learning pipelines that can handle imbalanced datasets
xgboost	Implementation of the XGBoost algorithm for classification tasks.
sklearn.neighbors.KNeighborsClassifier	K-Nearest Neighbors algorithm for classification.
sklearn.svm.SVC	Support Vector Classifier (SVC) for classification with probability estimates.
sklearn.tree.DecisionTreeClassifier	Decision Tree algorithm for classification
sklearn.ensemble.RandomForestClassifier	Random Forest algorithm for classification.
sklearn.ensemble.GradientBoostingClassifier	Gradient Boosting Machine (GBM) for classification.
mlxtend.classifier.StackingCVClassifier	Stacking classifier with cross-validation for ensemble learning

<code>sklearn.model_selection.cross_val_predict</code>	Generating cross-validated estimates for classification predictions.
<code>sklearn.metrics</code>	Evaluation metrics such as accuracy, confusion matrix, and ROC AUC score.
<code>math</code>	Mathematical functions, specifically used to calculate hazard rates and convert 10-year risk to 3-year risk.
<code>scipy.stats.ttest_ind</code>	Statistical hypothesis testing, specifically for performing t-tests to compare the means of two independent samples.
<code>sklearn.calibration.calibration_curve</code>	Generating calibration curves to assess the accuracy of probabilistic predictions.
<code>sklearn.calibration.CalibratedClassifierCV</code>	Calibration of probabilistic classifiers using methods like isotonic regression

APPENDIX B

Codes Used in This Project

1. Data Descriptive Analysis

```
DATA_DIR = '/Users/malithidesilva/fyp'
RESULT_DIR = f'/Users/malithidesilva/fyp/analysis'

import pandas as pd
import os

features_list = pd.read_csv(os.path.join(DATA_DIR, 'features_REDISCOVER.csv'))
raw_data = pd.read_csv(os.path.join(DATA_DIR,
'REDISCOVER_Rearranged_Cleaned_withNA_withFRS.csv'))

# Separate feature, metadata, outcome from df
features = features_list[features_list['Types'] == 'features']
metadata = features_list[features_list['Types'] == 'metadata']
outcomes = features_list[features_list['Types'] == 'outcome']

# Separate data according to stats type: categorical, binary, continuous and ordinal
categorical_data = features[features['Stats Type'] == 'categorical']['Variables'].to_list()
binary_data = features[features['Stats Type'] == 'binary_categorical']['Variables'].to_list()
continuous_data = features[features['Stats Type'] == 'continuous']['Variables'].to_list()
ordinal_data = features[features['Stats Type'] == 'ordinal']['Variables'].to_list()

# Fill Null value with 999
# raw_data.fillna(999, inplace=True)

raw_data_lessThan3Years = raw_data[raw_data['year_diff'] <= 3]

raw_data_lessThan3Years = raw_data_lessThan3Years.drop_duplicates()

# %% [markdown]
# ### Missing values rate analysis
#
# No Significant Correlation: If no significant correlation is found (and this is consistent across several
variables), it may suggest that the data is MCAR.
#
```

```
# Significant Correlation: If there is a significant correlation, especially with a small p-value, the missing data may be MAR, depending on the observed data.
```

```
# %%
```

```
!pip install missingno
```

```
# %%
```

```
missing_data_rates = raw_data.isnull().mean() * 100
```

```
missing_data_rates.to_csv(os.path.join(RESULT_DIR, 'missing_data_rates.csv'))
```

```
# print('Missing Data Rate:', missing_data_rates)
```

```
import missingno as msno
```

```
import matplotlib.pyplot as plt
```

```
# msno.bar(raw_data)
```

```
# plt.show()
```

```
msno.heatmap(raw_data)
```

```
plt.savefig(os.path.join(RESULT_DIR, 'missing_rate_heatmap.png'))
```

```
# %%
```

```
import pandas as pd
```

```
# Assuming 'df' is your DataFrame and 'class_label' is the class variable
```

```
grouped = raw_data.groupby('outcome') # Group data by class
```

```
def missing_rate_by_group(group):
```

```
    return group.isnull().mean() * 100 # Calculate missing rate as a percentage
```

```
# Apply the function to each group
```

```
missing_rates_by_class = grouped.apply(missing_rate_by_group)
```

```
missing_rates_by_class.to_csv(os.path.join(RESULT_DIR, 'missing_rate_by_class.csv'))
```

```
# %%
```

```
missing_rates_by_class
```

```
# %%
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```

# Plotting
fig, ax = plt.subplots(figsize=(12, 8))
missing_rates_by_class.iloc[:, 25:].T.plot(kind='bar', ax=ax) # Transpose to switch rows and columns
for better visualization

plt.title('Missing Data Percentage by Class for Each Feature')
plt.ylabel('Percentage Missing')
plt.xlabel('Features')
plt.legend(title='Class')
plt.savefig(os.path.join(RESULT_DIR, 'missing_rate_by_class_bar_remaining.png'))

```

2. Model Preparation

```

import os
import pandas as pd
import joblib

random_seed = 1234
DATA_DIR = '/Users/malithidesilva/fyp'
RESULT_DIR = '/Users/malithidesilva/fyp/model_1'

features_list = pd.read_csv(os.path.join(DATA_DIR, 'features_REDISCOVER.csv'))
raw_data = pd.read_csv(os.path.join(DATA_DIR,
'REDISCOVER_Rearranged_Cleaned_withNA_withFRS.csv'))

# Separate feature, metadata, outcome from df
features = features_list[features_list['Types'] == 'features']
metadata = features_list[features_list['Types'] == 'metadata']
outcomes = features_list[features_list['Types'] == 'outcome']

os.makedirs(RESULT_DIR, exist_ok=True)

# %% [markdown]
# Data Preparation

# %%
features_excluded = []

# Separate data according to stats type: categorical, binary, continuous and ordinal
categorical_data = features[features['Stats Type'] == 'categorical']['Variables'].to_list()

```

```

binary_data = features[features['Stats Type'] == 'binary_categorical']['Variables'].to_list()
continuous_data = features[features['Stats Type'] == 'continuous']['Variables'].to_list()
ordinal_data = features[features['Stats Type'] == 'ordinal']['Variables'].to_list()

# Fill Null value with 999
# raw_data.fillna(999, inplace=True)

raw_data_lessThan3Years = raw_data[raw_data['year_diff'] <= 3]
# raw_data_lessThan3Years = raw_data[raw_data['outcome'] < 3]
raw_data_lessThan3Years.loc[raw_data_lessThan3Years['outcome'] > 1, 'outcome'] = 1
raw_data_lessThan3Years = raw_data_lessThan3Years.drop_duplicates()
# raw_data_lessThan3Years = raw_data_lessThan3Years.drop(columns=features_excluded)
raw_data_lessThan3Years = raw_data_lessThan3Years.iloc[:, 2:]

# %%
class_0_data = raw_data_lessThan3Years[raw_data_lessThan3Years['outcome']==0].dropna()
# class_0_data = class_0_data.sample(n=500)
class_1_data = raw_data_lessThan3Years[raw_data_lessThan3Years['outcome']==1]

print('No CVD:', class_0_data.shape, 'CVD:', class_1_data.shape)

encoded_raw_data = pd.concat([class_0_data, class_1_data], axis=0)

encoded_raw_data['outcome'].astype('category')

# Map binary data into 0 and 1
for variable in binary_data:
    if variable == 'menopause':
        encoded_raw_data.loc[:, variable] = encoded_raw_data[variable].map({1: 1, 2: 0, 3:0})
    else:
        encoded_raw_data.loc[:, variable] = encoded_raw_data[variable].map({1: 1, 2: 0})

# Label encoding for categorical columns
from sklearn.preprocessing import LabelEncoder
label_encoder = LabelEncoder()

for variable in categorical_data:
    encoded_raw_data.loc[:, variable] = label_encoder.fit_transform(encoded_raw_data[variable])
    encoded_raw_data.loc[:, variable] = encoded_raw_data[variable].astype('category')

# Normalize continuous datam

```

```

from sklearn.preprocessing import MinMaxScaler, StandardScaler
scaler = MinMaxScaler()

def get_normalized_data(data):
    scaler = MinMaxScaler()
    scaled_data = data.copy() # Create a copy of the data to avoid modifying the original DataFrame

    for variable in continuous_data:
        # Reshape data to 2D array for the scaler
        data_array = data[[variable]].values.reshape(-1, 1)
        # Apply the scaler
        scaled_data.loc[:, variable] = scaler.fit_transform(data_array)

    return scaled_data

# %% [markdown]
# ## Stratified Splitting
#
# 70% training, 15% testing, 15% calibration

# %%
split_raw_data = encoded_raw_data

from sklearn.model_selection import train_test_split

excluded_col_for_training = outcomes["Variables"].to_list()
excluded_col_for_training.extend(metadata["Variables"].to_list())

# Split the data into features and target
X = split_raw_data.drop(excluded_col_for_training, axis=1)
y = split_raw_data["outcome"]

X_train, X_temp, y_train, y_temp = train_test_split(X, y, test_size=0.3, random_state=random_seed,
stratify=y)
X_test, X_calib, y_test, y_calib = train_test_split(X_temp, y_temp, test_size=0.5,
random_state=random_seed, stratify=y_temp)

print(f'----- Splitting Data: -----')
print(f'Original: {X.shape} splitted into -> test: {X_test.shape}, train: {X_train.shape}, calib:
{X_calib.shape}')
print(f'Testing set (%): {X_test.shape[0] / X.shape[0]}')

```

```

print(f'Training set (%): {X_train.shape[0] / X.shape[0]}')
print(f'Calibration set (%): {X_calib.shape[0] / X.shape[0]}')

print(pd.DataFrame([y_train.value_counts(), y_test.value_counts(), y_calib.value_counts()],
index=['train', 'test', 'calib']))

joblib.dump(
    {
        'X_train': X_train,
        'y_train': y_train,
        'X_test': X_test,
        'y_test': y_test,
        'X_calib': X_calib,
        'y_calib': y_calib,
    }, os.path.join(RESULT_DIR, 'train_test_data.pkl'))

frs_df = split_raw_data[['FRS_score_coeff3', 'FRS_outcome', 'FRS_score']]
X_train_frs = frs_df.loc[X_train.index]
X_test_frs = frs_df.loc[X_test.index]
X_calib_frs = frs_df.loc[X_calib.index]
y_train_frs = frs_df.loc[y_train.index]
y_test_frs = frs_df.loc[y_test.index]
y_calib_frs = frs_df.loc[y_calib.index]

joblib.dump(
    {
        'X_train_frs': X_train_frs,
        'X_test_frs': X_test_frs,
        'X_calib_frs': X_calib_frs,
        'y_train_frs': y_train_frs,
        'y_test_frs': y_test_frs,
        'y_calib_frs': y_calib_frs,
    }, os.path.join(RESULT_DIR, 'frs_data.pkl'))

# %% [markdown]
# ## Imputation

# %%
from sklearn.experimental import enable_iterative_imputer
from sklearn.impute import SimpleImputer, IterativeImputer

```

```

# imputer = IterativeImputer(max_iter=200, tol=1e-3, random_state=random_seed)
imputer = SimpleImputer(strategy='median')

X_train_imputed = imputer.fit_transform(X_train)
X_test_imputed = imputer.fit_transform(X_test)
X_calib_imputed = imputer.fit_transform(X_calib)

joblib.dump({
    'X_train_imputed': X_train_imputed,
    'y_train': y_train,
    'X_test_imputed': X_test_imputed,
    'y_test': y_test,
    'X_calib_imputed': X_calib_imputed,
    'y_calib': y_calib,
}, os.path.join(RESULT_DIR, 'imputed_data.pkl'))

# %% [markdown]
#

# %% [markdown]
# ### Balancing

# %%
# Define the balancing method
balancing_method = 'smote' # Change to 'ros' to use Random Over-Sampling

from imblearn.under_sampling import RandomUnderSampler
from imblearn.over_sampling import RandomOverSampler, SMOTE

rus = RandomUnderSampler(sampling_strategy={0: 200}) # Adjust the majority class to a smaller size
X_rus, y_rus = rus.fit_resample(X_train_imputed, y_train)

# Apply the selected balancing method
if balancing_method == 'ros':
    ros = RandomOverSampler(sampling_strategy='not majority') # This will oversample all classes
    except the majority
    X_balanced, y_balanced = ros.fit_resample(X_rus, y_rus)
elif balancing_method == 'smote':
    smote = SMOTE(random_state=random_seed)
    X_balanced, y_balanced = smote.fit_resample(X_rus, y_rus)

```

```

print('RUS:', X_rus.shape, y_rus.shape)
print('Balanced:', X_balanced.shape, y_balanced.shape)

joblib.dump({
    'X_balanced': X_balanced,
    'y_balanced': y_balanced,
    'X_rus': X_rus,
    'y_rus': y_rus,
    'X_test_selected': X_test_imputed,
    'y_test': y_test,
}, os.path.join(RESULT_DIR, f'resampled_data_{balancing_method}.pkl'))

joblib.dump({
    'X_train_selected': X_balanced,
    'y_train_selected': y_balanced,
    'X_test_selected': X_test_imputed,
    'y_test_selected': y_test,
    'X_calib_selected': X_calib_imputed,
    'y_calib_selected': y_calib,
}, os.path.join(RESULT_DIR, f'full_selected_data.pkl'))

# %% [markdown]
# ### Feature Selection

# %% [markdown]
# ### 1. RFE

# %%
from sklearn.feature_selection import RFE, RFECV
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import StratifiedKFold

estimators = {
    'rf': RandomForestClassifier(random_state=random_seed),
    # 'lr': LogisticRegression(random_state=random_seed, max_iter=1000)
}

for est_name, estimator in estimators.items():
    rfe = RFE(estimator, n_features_to_select=10)
    # rfe = RFECV(estimator=estimator, step=1, cv=StratifiedKFold(5), scoring='roc_auc')

```

```

X_train_rfe = rfe.fit_transform(X_balanced, y_balanced)
X_test_rfe = rfe.transform(X_test_imputed)
X_calib_rfe = rfe.transform(X_calib_imputed)

print(X_train_rfe.shape, X_test_rfe.shape, X_calib_rfe.shape, y_calib.shape, y_balanced.shape,
y_test.shape)

features = X_train.columns

selected_features_mask = rfe.support_

# Get rankings
feature_ranking = rfe.ranking_

fet_ranking = pd.DataFrame({
    'Feature': features,
    'Selected': selected_features_mask,
    'Ranking': feature_ranking
})

FET_EXTRACTION_DIR = os.path.join(RESULT_DIR, 'fet_extraction')
if not os.path.exists(FET_EXTRACTION_DIR):
    os.makedirs(FET_EXTRACTION_DIR)

fet_ranking.to_csv(os.path.join(FET_EXTRACTION_DIR, f'rfe_fet_ranking_{est_name}.csv'))

selected_10_features = pd.read_csv(os.path.join(FET_EXTRACTION_DIR,
f'rfe_fet_ranking_{est_name}.csv'))

selected_10_features = selected_10_features[selected_10_features['Selected'] == True]['Feature']
print(selected_10_features)
joblib.dump({
    'est_name': 'rf',
    'X_train_rfe': X_train_rfe,
    'y_train_selected': y_balanced,
    'X_test_rfe': X_test_rfe,
    'y_test_selected': y_test,
    'X_calib_rfe': X_calib_rfe,
    'y_calib_selected': y_calib,
}, os.path.join(RESULT_DIR, f'selected_fet_data_RFE_{est_name}.pkl'))

```

```

# %%
data = joblib.load(os.path.join(RESULT_DIR, 'full_selected_data.pkl'))
X_train_selected = data['X_train_selected']
y_train_selected = data['y_train_selected']
X_test_selected = data['X_test_selected']
y_test_selected = data['y_test_selected']
X_calib_selected = data['X_calib_selected']
y_calib_selected = data['y_calib_selected']

# %% [markdown]
# ### 2. Backward elimination

# %%
import joblib
import numpy as np
from sklearn.feature_selection import SequentialFeatureSelector
from sklearn.linear_model import LinearRegression

sffs = SequentialFeatureSelector(LinearRegression(),
                                n_features_to_select=10,
                                direction='forward',
                                cv=5)
sffs.fit(X_train_selected, y_train_selected)

print(sffs.n_features_to_select_)
print(sffs.support_)
selected_fet = []
unselected_fet = []
for idx, fet in enumerate(X_train.columns.to_list()):
    if sffs.support_[idx]:
        selected_fet.append(fet)
    else:
        unselected_fet.append(fet)

selected_fet

joblib.dump({
    'features': selected_fet

```

```

}, os.path.join(RESULT_DIR, f'selected_10fet_forward.pkl'))
X_train_transformed = sffs.transform(X_train_selected)
print(X_train_transformed.shape)

# X_test_transformed = sffs.transform(X_test_selected)
# print(X_test_transformed.shape)

from statsmodels.regression.linear_model import OLS
from statsmodels.tools.tools import add_constant
import joblib
import numpy as np

def backward_elimination(data, target, significance_level = 0.05):
    features = data.columns.tolist()
    removed_features = {}
    all_features_iteration = {}
    all_features = []
    print(f'Fet num: {len(features)}')
    while(len(features)>0):
        features_with_constant = add_constant(data[features].values)
        # print(features_with_constant.shape)
        p_values = OLS(target, features_with_constant).fit().pvalues[1:]

        max_p_value = p_values.max()
        excluded_feature_index = p_values.argmax()
        excluded_feature = features[excluded_feature_index]

        all_features_iteration = {features[i]: p_values[i] for i in range(len(features))}
        all_features.append(all_features_iteration)

        if(max_p_value >= significance_level):
            # print('feature excluded:', excluded_feature, max_p_value)
            removed_features[excluded_feature] = max_p_value
            features.remove(excluded_feature)
        else:
            break

    return features, removed_features, all_features

features, removed_features, all_features = backward_elimination(pd.DataFrame(X_train_selected,
columns=X_train.columns), y_train_selected)

```

```

# %%
pd.DataFrame.from_dict(removed_features, orient='index',
columns=['pvalues']).to_csv(os.path.join(RESULT_DIR, f'be_removed_features_pvalues.csv'))

# %% [markdown]
# ### Iteratively Remove Each Feature and Test AUC
#
# * For Assessing only
#

# %%
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import roc_auc_score
from sklearn.model_selection import cross_val_score

# Convert numpy arrays to pandas DataFrames
X_train_selected_df = pd.DataFrame(X_train_selected, columns=X_train.columns)
X_test_selected_df = pd.DataFrame(X_test_selected, columns=X_test.columns)
X_calib_selected_df = pd.DataFrame(X_calib_selected, columns=X_calib.columns)

# Function to compute AUC with a given set of features
def compute_auc(X, y, model):
    auc = cross_val_score(model, X, y, cv=5, scoring='roc_auc').mean()
    return auc

def compute_test_auc(model, X_train, y_train, X_test, y_test):
    model.fit(X_train, y_train)
    y_pred_prob = model.predict_proba(X_test)[: , 1]
    return roc_auc_score(y_test, y_pred_prob)

# Define the model
model = RandomForestClassifier(random_state=random_seed)

# Calculate the AUC with all features
initial_auc = compute_auc(X_train_selected_df, y_train_selected, model)
initial_test_auc = compute_test_auc(model, X_train_selected_df, y_train_selected, X_test_selected_df,
y_test_selected)
print(f'Initial Train AUC with all features: {initial_auc}')
print(f'Initial Test AUC with all features: {initial_test_auc}')

```

```

# Test AUC for each feature removed
feature_auc = {}
feature_test_auc = {}
for feature in X_train_selected_df.columns:
    X_temp = X_train_selected_df.drop(columns=[feature])
    auc = compute_auc(X_temp, y_train_selected, model)
    test_auc = compute_test_auc(model, X_temp, y_train_selected,
X_test_selected_df.drop(columns=[feature]), y_test_selected)
    feature_auc[feature] = auc
    feature_test_auc[feature] = test_auc
    print(f'AUC without {feature}: {auc}, Test AUC without {feature}: {test_auc}')

# Save the AUC results
feature_auc_df = pd.DataFrame.from_dict(feature_auc, orient='index', columns=['AUC'])
feature_test_auc_df = pd.DataFrame.from_dict(feature_test_auc, orient='index',
columns=['Test_AUC'])
feature_auc_df = feature_auc_df.join(feature_test_auc_df)
feature_auc_df.to_csv(os.path.join(RESULT_DIR, 'feature_auc_removal.csv'))

# Print features that reduce the AUC significantly when removed
print("\nFeatures that reduce AUC significantly when removed:")
print(feature_auc_df[(feature_auc_df['AUC'] < initial_auc) | (feature_auc_df['Test_AUC'] <
initial_test_auc)].sort_values(by='AUC', ascending=True))

# %% [markdown]
#### 3. Remove Highly correlated Features

# %%
# Set the correlation threshold
correlation_threshold = 0.9

# Compute the correlation matrix
correlation_matrix = X_train_selected_df.corr()

# Generate a heatmap
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Feature Correlation Heatmap")
plt.savefig(os.path.join(RESULT_DIR, 'feature_correlation_heatmap.png'))
plt.show()

```

```

# Save the correlation matrix
correlation_matrix.to_csv(os.path.join(RESULT_DIR, 'feature_correlation_matrix.csv'))

# Identify highly correlated features
upper_triangle = correlation_matrix.where(
    np.triu(np.ones(correlation_matrix.shape), k=1).astype(bool)
)
high_correlation_features = [
    column for column in upper_triangle.columns if any(upper_triangle[column].abs() >
correlation_threshold)
]

# Ensure 'FVC' is not removed
if 'FVC' in high_correlation_features:
    high_correlation_features.remove('FVC')

# Print out the features that were removed
print("Features removed due to high correlation:")
for feature in high_correlation_features:
    print(feature)

# Drop highly correlated features
X_train_reduced = X_train_selected_df.drop(columns=high_correlation_features)
X_test_reduced = X_test_selected_df.drop(columns=high_correlation_features)
X_calib_reduced = X_calib_selected_df.drop(columns=high_correlation_features)

# Save the reduced dataset
reduced_data = {
    'X_train_reduced': X_train_reduced,
    'y_train_selected': y_train_selected,
    'X_test_reduced': X_test_reduced,
    'y_test_selected': y_test_selected,
    'X_calib_reduced': X_calib_reduced,
    'y_calib_selected': y_calib_selected,
}
joblib.dump(reduced_data, os.path.join(RESULT_DIR, 'reduced_data.pkl'))

print("Highly correlated features removed and data saved.")

# %% [markdown]

```

```

#### 4. Shap Based Feature Selection
#
# * train the initial balanced sample to identify best features

# %%
# Assuming `X_train` is already defined and contains your original training data
# Extract column names from X_train
column_names = X_train.columns.tolist()

# Now you can use these column names when creating your DataFrames for the selected data
X_train_selected = pd.DataFrame(data[X_train_selected], columns=column_names)
y_train_selected = pd.Series(data[y_train_selected])
X_test_selected = pd.DataFrame(data[X_test_selected], columns=column_names)
y_test_selected = pd.Series(data[y_test_selected])
X_calib_selected = pd.DataFrame(data[X_calib_selected], columns=column_names)
y_calib_selected = pd.Series(data[y_calib_selected])

# Print to verify
print("Column names:", column_names)

# %%
import shap
import matplotlib.pyplot as plt
from sklearn.ensemble import RandomForestClassifier
import numpy as np

best_model_name = 'RF'

# Define the model based on the best model name
if best_model_name == 'RF':
    model = RandomForestClassifier(random_state=42)
    model.fit(X_balanced, y_balanced)

    explainer = shap.Explainer(model, X_balanced)
    shap_values = explainer(X_balanced, check_additivity=False)

# Convert X_train back to DataFrame
X_train_intial_shap_df = pd.DataFrame(X_balanced, columns=X.columns)

shap.summary_plot(shap_values[:, :, 1], X_train_intial_shap_df,
feature_names=X_train_intial_shap_df.columns, show=False)

```

```

plt.savefig(os.path.join(RESULT_DIR, f'shap_summary_{best_model_name}_initial.png'))
plt.close()

# Save SHAP values as DataFrame
shap_values_df = pd.DataFrame(shap_values[:, :, 1].values,
columns=X_train_initial_shap_df.columns)

shap_values_df['outcome'] = y_balanced.values
shap_values_df.to_csv(os.path.join(RESULT_DIR, f'shap_values_{best_model_name}_initial.csv'),
index=False)

# Get mean absolute SHAP values for feature importance
mean_shap_values = np.abs(shap_values[:, :, 1].values).mean(axis=0)
feature_importance = pd.DataFrame(list(zip(X_train_initial_shap_df.columns, mean_shap_values)),
columns=['Feature', 'Importance'])

feature_importance = feature_importance.sort_values(by='Importance', ascending=False)
feature_importance.to_csv(os.path.join(RESULT_DIR,
f'feature_importance_{best_model_name}_initial.csv'), index=False)

print("Tabular SHAP values and feature importance before RFE have been saved.")
else:
    print(f'Model {best_model_name} is not implemented for SHAP analysis.')

# %%
# Load the top SHAP features
shap_values_df = pd.read_csv(os.path.join(RESULT_DIR, f'shap_values_RF_initial.csv'))
shap_feature_importance = pd.read_csv(os.path.join(RESULT_DIR,
f'feature_importance_RF_initial.csv'))
top_shap_features = shap_feature_importance['Feature'].tolist()

# Load the RFE results
rfe_ranking_df = pd.read_csv(os.path.join(RESULT_DIR, 'fet_extraction/rfe_fet_ranking_rf.csv'))
top_rfe_features = rfe_ranking_df[rfe_ranking_df['Selected'] == True]['Feature'].tolist()

# Find the common features
common_features = set(top_shap_features) & set(top_rfe_features)

# Print the best SHAP features
print("Best SHAP Features:")
print(top_shap_features)

# Print the best RFE features

```

```

print("\nBest RFE Features:")
print(top_rfe_features)

# Print the common best features
print("\nCommon Best Features between SHAP and RFE:")
print(common_features)

# Identify the worst SHAP features
worst_shap_features = shap_feature_importance['Feature'].iloc[-10:].tolist() # Adjust the number to
include more or fewer features

# Identify the features not selected by RFE
non_rfe_features = rfe_ranking_df[rfe_ranking_df['Selected'] == False]['Feature'].tolist()

# Print the worst SHAP features
print("\nWorst SHAP Features:")
print(worst_shap_features)

# Print the features not selected by RFE
print("\nFeatures not selected by RFE:")
print(non_rfe_features)

# Find the common worst features
common_worst_features = set(worst_shap_features) & set(non_rfe_features)
print("\nCommon Worst Features between SHAP and RFE:")
print(common_worst_features)

# %%
import os
import pandas as pd
import joblib
import pickle

# Define N values
N_values = [ 12,14,16,18,20,22,24,26] # Example values

# Save N values to a pickle file
with open('N_values.pkl', 'wb') as f:
    pickle.dump(N_values, f)

for N in N_values:

```

```

# Select top N SHAP features
selected_shap_features = top_shap_features[:N]

print(f"\nProcessing top {N} SHAP features:")
print(f"Selected SHAP features: {selected_shap_features}")

# Filter the data to only include the top N SHAP features
X_train_shap = X_train_selected[selected_shap_features]
X_test_shap = X_test_selected[selected_shap_features]
X_calib_shap = X_calib_selected[selected_shap_features]

# Save the selected top N SHAP features to a CSV file
feature_csv_path = os.path.join(RESULT_DIR, f'selected_top_{N}_features.csv')
pd.DataFrame(selected_shap_features, columns=['Selected Features']).to_csv(feature_csv_path,
index=False)

print(f"Saved selected top {N} SHAP features to {feature_csv_path}")

# Save the SHAP-reduced datasets using joblib
shap_data = {
    'X_train_shap': X_train_shap,
    'y_train_selected': y_train_selected,
    'X_test_shap': X_test_shap,
    'y_test_selected': y_test_selected,
    'X_calib_shap': X_calib_shap,
    'y_calib_selected': y_calib_selected,
    'N': N
}

shap_data_path = os.path.join(RESULT_DIR, f'shap_reduced_data_{N}.pkl')
joblib.dump(shap_data, shap_data_path)

print(f"Saved SHAP-reduced data for top {N} features to {shap_data_path}")

```

3. Model Train & evaluation

```

# %%
import os
from imblearn.pipeline import Pipeline as ImbPipeline
from sklearn.preprocessing import MinMaxScaler
import xgboost as xgb
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from mlxtend.classifier import StackingCVCClassifier
import joblib
from sklearn.model_selection import cross_val_predict
from sklearn.metrics import accuracy_score, confusion_matrix, roc_auc_score
import pandas as pd

random_seed = 1234

KFOLD_SPLIT = 5

# --- X_smote + 10 fet using rf-rfe ---
# 1234, 2345 is split 3
# 2345, 42 is split 5
# 333 : split 10

# --- X_ros + 10 fet using rf-rfe ---
# 111, split 3
# 222, split 5
# 555, split 10

# --- X_smote + 48 fet ---
# 25, split 3
# 35, split 5
# 65, split 10

# --- X_ros + 48 fet ---
# 255, split 3
# 355, split 5
# 655, split 10

# --- X_ros + 14 fet using be ---
# 616: split 3

```

```

# 626: split 5
# 636: split 10

# --- X_smote + 14 fet using be ---
# 716: split 3
# 726: split 5
# 736: split 10

# --- 10 fet using be sfs backward ---
# 666: split 3

# --- 10 fet using be sfs forward ---
# 888: split 3

DATA_DIR = '/Users/malithidesilva/fyp/model_1/final'
RESULT_DIR = f'/Users/malithidesilva/fyp/model_1/final/model1_results_{random_seed}'

os.makedirs(RESULT_DIR, exist_ok=True)

# Define models
rf_model = RandomForestClassifier(random_state=random_seed)
xgb_model = xgb.XGBClassifier(use_label_encoder=False, eval_metric='logloss', random_state =
random_seed)
dt_model = DecisionTreeClassifier(random_state=random_seed)
gbm_model = GradientBoostingClassifier(n_estimators=100, random_state=random_seed)
svc_model = SVC(probability=True, random_state=random_seed)
knn_model = KNeighborsClassifier(n_neighbors=5)

single_pipelines = {
    'RF': ImbPipeline([
        ('scaling', MinMaxScaler()),
        ('classifier', rf_model)
    ]),
    'XGBoost': ImbPipeline([
        ('scaling', MinMaxScaler()),
        ('classifier', xgb_model)
    ]),
    'DT': ImbPipeline([

```

```

        ('scaling', MinMaxScaler()),
        ('classifier', dt_model)
    ]),
    'GBM': ImbPipeline([
        ('scaling', MinMaxScaler()),
        ('classifier', gbm_model)
    ]),
    'SVC': ImbPipeline([
        ('scaling', MinMaxScaler()),
        ('classifier', svc_model)
    ]),
    'KNN': ImbPipeline([
        ('scaling', MinMaxScaler()),
        ('classifier', knn_model)
    ]),
}

ensemble_pipelines = {
    'ALL_RF': ImbPipeline([
        ('scaling', MinMaxScaler()),
        ('classifier', StackingCVClassifier(
            classifiers=[rf_model, xgb_model, dt_model, gbm_model, svc_model, knn_model],
            meta_classifier=rf_model,
            cv=KFOLD_SPLIT,
        ))
    ]),
    'ALL_XGBoost': ImbPipeline([
        ('scaling', MinMaxScaler()),
        ('classifier', StackingCVClassifier(
            classifiers=[rf_model, xgb_model, dt_model, gbm_model, svc_model, knn_model],
            meta_classifier=xgb_model,
            cv=KFOLD_SPLIT,
        ))
    ]),
    'ALL_SVC': ImbPipeline([
        ('scaling', MinMaxScaler()),
        ('classifier', StackingCVClassifier(
            classifiers=[rf_model, xgb_model, dt_model, gbm_model, svc_model, knn_model],
            meta_classifier=svc_model,
            cv=KFOLD_SPLIT,

```

```

    ))
]),
'ALL_GBM': ImbPipeline([
    ('scaling', MinMaxScaler()),
    ('classifier', StackingCVClassifier(
        classifiers=[rf_model, xgb_model, dt_model, gbm_model, svc_model, knn_model],
        meta_classifier=gbm_model,
        cv=KFOLD_SPLIT,
    ))
])
}

# %%
def extended_class_report(cm):
    tn, fp, fn, tp = cm.ravel()

    # Calculate additional metrics
    ppv = tp / (tp + fp) if (tp + fp) != 0 else 0
    npv = tn / (tn + fn) if (tn + fn) != 0 else 0
    sensitivity = tp / (tp + fn) if (tp + fn) != 0 else 0
    specificity = tn / (tn + fp) if (tn + fp) != 0 else 0

    return ppv, npv, sensitivity, specificity

from sklearn.preprocessing import label_binarize
from sklearn.metrics import roc_curve, auc
import matplotlib.pyplot as plt
import numpy as np

def draw_auc(actual_y, prob_y, saved_dir):

    y = actual_y #label_binarize(actual_y, classes=np.unique(actual_y))
    n_classes = y.shape[1]

    plt.figure(figsize=(15,15))
    # Compute ROC AUC for each class
    for i in range(n_classes):
        # Ensure prob_y is a numpy array, handle if it's a dataframe
        if hasattr(prob_y, "to_numpy"):
            prob_y = prob_y.to_numpy()

```

```

fpr, tpr, threshold = roc_curve(y[:, i], prob_y[:, i])
roc_auc_results = auc(fpr, tpr)
plt.plot(fpr, tpr, lw=2, label=f'Class {i} (area = {roc_auc_results:.2f})')

plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title(f'Multiclass ROC Curve by One-vs-Rest, thres:{threshold}')
plt.legend(loc="lower right")
# plt.show()
plt.savefig(saved_dir)
plt.close()

# Function to convert numpy array to DataFrame
def ensure_dataframe(data):
    if isinstance(data, np.ndarray):
        return pd.DataFrame(data)
    return data

# %% [markdown]
#

# %%
def evaluate_models(pipelines, X_train_selected, y_train_selected, X_test_selected, y_test_selected,
result_suffix):
    overview = []
    for model_name, pipeline in pipelines.items():
        model_result_dir = os.path.join(RESULT_DIR, f'{model_name}_{result_suffix}')
        if not os.path.exists(model_result_dir):
            os.makedirs(model_result_dir)
        print(f'----- {model_name} -----')

        print('>>> Fitting training model')

        pipeline.fit(X_train_selected, y_train_selected)
        y_train_pred = cross_val_predict(pipeline, X_train_selected, y_train_selected, cv=KFOLD_SPLIT)
        y_train_pred_proba = cross_val_predict(pipeline, X_train_selected, y_train_selected,
cv=KFOLD_SPLIT, method='predict_proba')[:, 1]

```

```

train_accuracy = accuracy_score(y_train_selected, y_train_pred)
train_conf_matrix = confusion_matrix(y_train_selected, y_train_pred)
train_pv, train_npv, train_sensitivity, train_specificity = extended_class_report(train_conf_matrix)
train_roc_auc = roc_auc_score(y_train_selected, y_train_pred_proba)

print('>>> Validating test set')

y_test_pred = pipeline.predict(X_test_selected)
y_test_pred_proba = pipeline.predict_proba(X_test_selected)[:, 1]

test_accuracy = accuracy_score(y_test_selected, y_test_pred)
test_conf_matrix = confusion_matrix(y_test_selected, y_test_pred)
test_pv, test_npv, test_sensitivity, test_specificity = extended_class_report(test_conf_matrix)
test_roc_auc = roc_auc_score(y_test_selected, y_test_pred_proba)

print(f'Results: {train_roc_auc}, {test_roc_auc}')

pd.DataFrame(train_conf_matrix).to_csv(os.path.join(model_result_dir,
f'{model_name}_train_cm.csv'))

pd.DataFrame(test_conf_matrix).to_csv(os.path.join(model_result_dir,
f'{model_name}_test_cm.csv'))

overview.append([model_name, test_accuracy, test_roc_auc, test_pv, test_npv, test_sensitivity,
test_specificity, train_accuracy, train_roc_auc, train_pv, train_npv, train_sensitivity, train_specificity,
train_roc_auc-test_roc_auc])

pred_to_save = {
    'y_train_pred': y_train_pred,
    'y_train_pred_proba': y_train_pred_proba,
    'y_test_pred': y_test_pred,
    'y_test_pred_proba': y_test_pred_proba,
}

joblib.dump(pred_to_save, os.path.join(model_result_dir, f'{model_name}_pred_values.pkl'))
joblib.dump(pipeline, os.path.join(model_result_dir, f'{model_name}_full_model.pkl'))

pd.DataFrame(overview, columns=['model_name',
                                'test_accuracy',
                                'test_roc_auc',
                                'test_ppv',
                                'test_npv',

```

```

        'test_sensitivity',
        'test_specificity',
        'train_accuracy',
        'train_roc_auc',
        'train_ppv',
        'train_npv',
        'train_sensitivity',
        'train_specificity',
        'auc_diff'
    ]).to_csv(os.path.join(RESULT_DIR, f'results_overview_{result_suffix}.csv'))

# %%
from sklearn.model_selection import cross_val_score

def evaluate_ensemble_models(pipelines, X_train_selected, y_train_selected, X_test_selected,
y_test_selected, result_suffix):
    overview = []
    for model_name, pipeline in pipelines.items():
        model_result_dir = os.path.join(RESULT_DIR, f'{model_name}_{result_suffix}')
        if not os.path.exists(model_result_dir):
            os.makedirs(model_result_dir)
        print(f'----- {model_name} -----')

        print('>>> Fitting training model')

        pipeline.fit(X_train_selected, y_train_selected)

        # Use cross_val_predict only for classification tasks where classes_ attribute exists
        if hasattr(pipeline, 'classes_'):
            y_train_pred = cross_val_predict(pipeline, X_train_selected, y_train_selected,
cv=KFOLD_SPLIT)
            y_train_pred_proba = cross_val_predict(pipeline, X_train_selected, y_train_selected,
cv=KFOLD_SPLIT, method='predict_proba')[:, 1]
        else:
            # Directly predict on training data
            y_train_pred = pipeline.predict(X_train_selected)
            y_train_pred_proba = pipeline.predict_proba(X_train_selected)[:, 1]

        train_accuracy = accuracy_score(y_train_selected, y_train_pred)
        train_conf_matrix = confusion_matrix(y_train_selected, y_train_pred)

```

```

train_pv, train_npv, train_sensitivity, train_specificity = extended_class_report(train_conf_matrix)
train_roc_auc = roc_auc_score(y_train_selected, y_train_pred_proba)

print('>>> Validating test set')

y_test_pred = pipeline.predict(X_test_selected)
y_test_pred_proba = pipeline.predict_proba(X_test_selected)[:, 1]

test_accuracy = accuracy_score(y_test_selected, y_test_pred)
test_conf_matrix = confusion_matrix(y_test_selected, y_test_pred)
test_pv, test_npv, test_sensitivity, test_specificity = extended_class_report(test_conf_matrix)
test_roc_auc = roc_auc_score(y_test_selected, y_test_pred_proba)

print(f'Results: {train_roc_auc}, {test_roc_auc}')

pd.DataFrame(train_conf_matrix).to_csv(os.path.join(model_result_dir,
f'{model_name}_train_cm.csv'))

pd.DataFrame(test_conf_matrix).to_csv(os.path.join(model_result_dir,
f'{model_name}_test_cm.csv'))

overview.append([model_name, test_accuracy, test_roc_auc, test_pv, test_npv, test_sensitivity,
test_specificity, train_accuracy, train_roc_auc, train_pv, train_npv, train_sensitivity, train_specificity,
train_roc_auc-test_roc_auc])

pred_to_save = {
    'y_train_pred': y_train_pred,
    'y_train_pred_proba': y_train_pred_proba,
    'y_test_pred': y_test_pred,
    'y_test_pred_proba': y_test_pred_proba,
}

joblib.dump(pred_to_save, os.path.join(model_result_dir, f'{model_name}_pred_values.pkl'))
joblib.dump(pipeline, os.path.join(model_result_dir, f'{model_name}_full_model.pkl'))

pd.DataFrame(overview, columns=['model_name',
                                'test_accuracy',
                                'test_roc_auc',
                                'test_ppv',
                                'test_npv',
                                'test_sensitivity',
                                'test_specificity',

```

```

        'train_accuracy',
        'train_roc_auc',
        'train_ppv',
        'train_npv',
        'train_sensitivity',
        'train_specificity',
        'auc_diff'
    ]).to_csv(os.path.join(RESULT_DIR, f'results_overview_{result_suffix}.csv'))

# %%

def evaluate_shap_models(pipelines, X_train_selected, y_train_selected, X_test_selected,
y_test_selected, result_suffix, is_shap=False):

    overview = []
    for model_name, pipeline in pipelines.items():
        model_result_dir = os.path.join(RESULT_DIR, f'{model_name}_{result_suffix}')
        if not os.path.exists(model_result_dir):
            os.makedirs(model_result_dir)
        print(f'----- {model_name} evaluation {result_suffix} -----')

        print('>>> Fitting training model')

        pipeline.fit(X_train_selected, y_train_selected)
        y_train_pred = cross_val_predict(pipeline, X_train_selected, y_train_selected, cv=KFOLD_SPLIT)
        y_train_pred_proba = cross_val_predict(pipeline, X_train_selected, y_train_selected,
cv=KFOLD_SPLIT, method='predict_proba')[:, 1]

        train_accuracy = accuracy_score(y_train_selected, y_train_pred)
        train_conf_matrix = confusion_matrix(y_train_selected, y_train_pred)
        train_pv, train_npv, train_sensitivity, train_specificity = extended_class_report(train_conf_matrix)
        train_roc_auc = roc_auc_score(y_train_selected, y_train_pred_proba)

        print('>>> Validating test set')

        y_test_pred = pipeline.predict(X_test_selected)
        y_test_pred_proba = pipeline.predict_proba(X_test_selected)[:, 1]

        test_accuracy = accuracy_score(y_test_selected, y_test_pred)
        test_conf_matrix = confusion_matrix(y_test_selected, y_test_pred)
        test_pv, test_npv, test_sensitivity, test_specificity = extended_class_report(test_conf_matrix)
        test_roc_auc = roc_auc_score(y_test_selected, y_test_pred_proba)

```

```

print(f'Results: Train ROC AUC = {train_roc_auc}, Test ROC AUC = {test_roc_auc}')

pd.DataFrame(train_conf_matrix).to_csv(os.path.join(model_result_dir,
f'{model_name}_train_cm.csv'))

pd.DataFrame(test_conf_matrix).to_csv(os.path.join(model_result_dir,
f'{model_name}_test_cm.csv'))

overview.append([model_name, test_accuracy, test_roc_auc, test_pv, test_npv, test_sensitivity,
test_specificity,
                  train_accuracy, train_roc_auc, train_pv, train_npv, train_sensitivity, train_specificity,
                  train_roc_auc-test_roc_auc])

pred_to_save = {
    'y_train_pred': y_train_pred,
    'y_train_pred_proba': y_train_pred_proba,
    'y_test_pred': y_test_pred,
    'y_test_pred_proba': y_test_pred_proba,
}

joblib.dump(pred_to_save, os.path.join(model_result_dir, f'{model_name}_pred_values.pkl'))
joblib.dump(pipeline, os.path.join(model_result_dir, f'{model_name}_full_model.pkl'))

pd.DataFrame(overview, columns=['model_name', 'test_accuracy', 'test_roc_auc', 'test_ppv',
'test_npv', 'test_sensitivity',
                              'test_specificity', 'train_accuracy', 'train_roc_auc', 'train_ppv', 'train_npv',
                              'train_sensitivity', 'train_specificity', 'auc_diff']).to_csv(os.path.join(RESULT_DIR,
f'results_overview_{result_suffix}.csv'))

# %%

def evaluate_shap_ensemble_models(pipelines, X_train_selected, y_train_selected, X_test_selected,
y_test_selected, result_suffix, is_shap=False):
    overview = []
    for model_name, pipeline in pipelines.items():
        model_result_dir = os.path.join(RESULT_DIR, f'{model_name}_{result_suffix}')
        if not os.path.exists(model_result_dir):
            os.makedirs(model_result_dir)

```

```

print(f'----- {model_name} evaluation {result_suffix} -----')

print('>>> Fitting training model')

pipeline.fit(X_train_selected, y_train_selected)

# Use cross_val_predict only for classification tasks where classes_ attribute exists
if hasattr(pipeline, 'classes_'):
    y_train_pred = cross_val_predict(pipeline, X_train_selected, y_train_selected,
cv=KFOLD_SPLIT)
    y_train_pred_proba = cross_val_predict(pipeline, X_train_selected, y_train_selected,
cv=KFOLD_SPLIT, method='predict_proba')[:, 1]
else:
    # Directly predict on training data
    y_train_pred = pipeline.predict(X_train_selected)
    y_train_pred_proba = pipeline.predict_proba(X_train_selected)[:, 1]

train_accuracy = accuracy_score(y_train_selected, y_train_pred)
train_conf_matrix = confusion_matrix(y_train_selected, y_train_pred)
train_pv, train_npv, train_sensitivity, train_specificity = extended_class_report(train_conf_matrix)
train_roc_auc = roc_auc_score(y_train_selected, y_train_pred_proba)

print('>>> Validating test set')

y_test_pred = pipeline.predict(X_test_selected)
y_test_pred_proba = pipeline.predict_proba(X_test_selected)[:, 1]

test_accuracy = accuracy_score(y_test_selected, y_test_pred)
test_conf_matrix = confusion_matrix(y_test_selected, y_test_pred)
test_pv, test_npv, test_sensitivity, test_specificity = extended_class_report(test_conf_matrix)
test_roc_auc = roc_auc_score(y_test_selected, y_test_pred_proba)

print(f'Results: Train ROC AUC = {train_roc_auc}, Test ROC AUC = {test_roc_auc}')

pd.DataFrame(train_conf_matrix).to_csv(os.path.join(model_result_dir,
f'{model_name}_train_cm.csv'))
pd.DataFrame(test_conf_matrix).to_csv(os.path.join(model_result_dir,
f'{model_name}_test_cm.csv'))

overview.append([model_name, test_accuracy, test_roc_auc, test_pv, test_npv, test_sensitivity,
test_specificity,

```

```

        train_accuracy, train_roc_auc, train_pv, train_npv, train_sensitivity, train_specificity,
        train_roc_auc-test_roc_auc])

pred_to_save = {
    'y_train_pred': y_train_pred,
    'y_train_pred_proba': y_train_pred_proba,
    'y_test_pred': y_test_pred,
    'y_test_pred_proba': y_test_pred_proba,
}

joblib.dump(pred_to_save, os.path.join(model_result_dir, f'{model_name}_pred_values.pkl'))
joblib.dump(pipeline, os.path.join(model_result_dir, f'{model_name}_full_model.pkl'))

pd.DataFrame(overview, columns=['model_name', 'test_accuracy', 'test_roc_auc', 'test_ppv',
                                'test_npv', 'test_sensitivity',
                                'test_specificity', 'train_accuracy', 'train_roc_auc', 'train_ppv', 'train_npv',
                                'train_sensitivity', 'train_specificity', 'auc_diff']).to_csv(os.path.join(RESULT_DIR,
f'results_overview_{result_suffix}.csv'))

# %%

import pickle

# Load N values from the file

with open('N_values.pkl', 'rb') as f:
    N_values = pickle.load(f)

# Define dataset files with patterns
dataset_files = {
    'resampled_data': 'resampled_data_smote.pkl',
    'rfe_data': 'selected_fet_data_RFE_rf.pkl',
    'be_data': 'selected_fet_data_be.pkl',
    'reduced_data': 'reduced_data.pkl',
    'shap_data': 'shap_reduced_data_{N}.pkl' # Pattern for SHAP data
}

# Main evaluation loop
for dataset_name, data_file in dataset_files.items():
    print(f"Loading and evaluating {dataset_name}...")

```

```

if dataset_name == 'shap_data':
    for N in N_values:
        file_path = os.path.join(DATA_DIR, data_file.format(N=N))
        try:
            data = joblib.load(file_path)
            X_train_selected = data['X_train_shap']
            y_train_selected = data['y_train_selected']
            X_test_selected = data['X_test_shap']
            y_test_selected = data['y_test_selected']

            # Evaluate single pipelines
            evaluate_shap_models(single_pipelines, X_train_selected, y_train_selected,
X_test_selected, y_test_selected, result_suffix=f'shap_data_N_{N}', is_shap=True)

            # Evaluate ensemble pipelines
            evaluate_shap_ensemble_models(ensemble_pipelines, X_train_selected, y_train_selected,
X_test_selected, y_test_selected, result_suffix=f'ensemble_shap_data_N_{N}', is_shap=True)

            print(f"Completed evaluation for SHAP data with N={N}.")
        except FileNotFoundError:
            print(f"File not found: {file_path}")
    else:
        file_path = os.path.join(DATA_DIR, data_file)
        try:
            data = joblib.load(file_path)
            if dataset_name == 'resampled_data':
                X_train_selected = data['X_balanced']
                y_train_selected = data['y_balanced']
                X_test_selected = data['X_test_selected']
                y_test_selected = data['y_test']
            elif dataset_name == 'rfe_data':
                X_train_selected = data['X_train_rfe']
                y_train_selected = data['y_train_selected']
                X_test_selected = data['X_test_rfe']
                y_test_selected = data['y_test_selected']
            elif dataset_name == 'be_data':
                X_train_selected = data['X_train_be']
                y_train_selected = data['y_train_selected']
                X_test_selected = data['X_test_be']
                y_test_selected = data['y_test_selected']
            elif dataset_name == 'reduced_data':

```

```

X_train_selected = data['X_train_reduced']
y_train_selected = data['y_train_selected']
X_test_selected = data['X_test_reduced']
y_test_selected = data['y_test_selected']

# Evaluate single pipelines
evaluate_models(single_pipelines, X_train_selected, y_train_selected, X_test_selected,
y_test_selected, result_suffix=dataset_name)

# Evaluate ensemble pipelines
evaluate_ensemble_models(ensemble_pipelines, X_train_selected, y_train_selected,
X_test_selected, y_test_selected, result_suffix=f'ensemble_{dataset_name}')

print(f"Completed evaluation for {dataset_name}.")
except FileNotFoundError:
    print(f"File not found: {file_path}")

print("Evaluation completed for all datasets.")

```

4. Model 1 Calibration & NRI

```

# %%

import os
import pandas as pd
import joblib

```

```

import numpy as np

from sklearn.metrics import brier_score_loss, log_loss, roc_auc_score, accuracy_score,
precision_score, recall_score, f1_score

from sklearn.utils import resample

from scipy.stats import ttest_ind


random_seed = 1234

DATA_DIR = '/Users/malithidesilva/fyp/model_1/calibration'
RESULT_DIR = '/Users/malithidesilva/fyp/model_1/calibration/result_calib'


# %% [markdown]
# #### Brief Score & Log Loss FOR NON CALLIBRATED


# %% [markdown]
# #### Non Callibrated Best Model VS Non Caliibrated FRS


# %%


# Function to calculate metrics
def calculate_metrics(pipeline, X_test, y_test):

    y_test_pred_proba = pipeline.predict_proba(X_test)[:, 1]
    y_test_pred = (y_test_pred_proba > 0.5).astype(int)

    brier_score = brier_score_loss(y_test, y_test_pred_proba)
    log_loss_value = log_loss(y_test, y_test_pred_proba)
    roc_auc = roc_auc_score(y_test, y_test_pred_proba)
    accuracy = accuracy_score(y_test, y_test_pred)
    precision = precision_score(y_test, y_test_pred)
    recall = recall_score(y_test, y_test_pred)
    f1 = f1_score(y_test, y_test_pred)

    return {
        'Brier Score': brier_score,
        'Log Loss': log_loss_value,
        'ROC AUC': roc_auc,
        'Accuracy': accuracy,
        'Precision': precision,
        'Recall': recall,
        'F1 Score': f1
    }

```

```

# Function to calculate p-values using bootstrapping
def calculate_p_values(probabilities1, probabilities2, y_test1, y_test2, n_bootstrap=1000):
    brier_scores_1 = []
    brier_scores_2 = []
    log_losses_1 = []
    log_losses_2 = []
    roc_aucs_1 = []
    roc_aucs_2 = []

    for _ in range(n_bootstrap):
        # Bootstrap samples
        y_test_sample_1, prob_sample_1 = resample(y_test1, probabilities1)
        y_test_sample_2, prob_sample_2 = resample(y_test2, probabilities2)

        # Brier Score
        brier_scores_1.append(brier_score_loss(y_test_sample_1, prob_sample_1))
        brier_scores_2.append(brier_score_loss(y_test_sample_2, prob_sample_2))

        # Log Loss
        log_losses_1.append(log_loss(y_test_sample_1, prob_sample_1))
        log_losses_2.append(log_loss(y_test_sample_2, prob_sample_2))

        # ROC AUC
        roc_aucs_1.append(roc_auc_score(y_test_sample_1, prob_sample_1))
        roc_aucs_2.append(roc_auc_score(y_test_sample_2, prob_sample_2))

    # Calculate p-values
    brier_score_p_value = ttest_ind(brier_scores_1, brier_scores_2).pvalue
    log_loss_p_value = ttest_ind(log_losses_1, log_losses_2).pvalue
    roc_auc_p_value = ttest_ind(roc_aucs_1, roc_aucs_2).pvalue

    return brier_score_p_value, log_loss_p_value, roc_auc_p_value

# Function to determine significance based on p-value
def significance_label(p_value, alpha=0.05):
    return 'Significant' if p_value < alpha else 'Not Significant'

# Load SHAP-reduced test data
shap_reduced_data = joblib.load(os.path.join(DATA_DIR, 'shap_reduced_data.pkl'))
X_test_shap = shap_reduced_data['X_test_shap']

```

```

y_test_selected = shap_reduced_data['y_test_shap']

# Load the SHAP model pipeline
model_result_dir = os.path.join(DATA_DIR)
pipeline = joblib.load(os.path.join(model_result_dir, 'RF_full_model.pkl'))

# Calculate metrics for the SHAP-reduced model
shap_metrics = calculate_metrics(pipeline, X_test_shap, y_test_selected)

# Load FRS data
frs_data = joblib.load(os.path.join(DATA_DIR, 'frs_data.pkl'))
X_test_frs_prob = frs_data['X_test_frs']['FRS_score_coeff3'] # FRS predicted probabilities

# Ensure binary outcome is 0/1
y_test_ = y_test_selected.copy()
y_test_[y_test_ > 0] = 1

# Calculate metrics for the non-calibrated FRS model
frs_metrics = {
    'Brier Score': brier_score_loss(y_test_, X_test_frs_prob),
    'Log Loss': log_loss(y_test_, X_test_frs_prob),
    'ROC AUC': roc_auc_score(y_test_, X_test_frs_prob),
    'Accuracy': accuracy_score(y_test_, (X_test_frs_prob > 0.5).astype(int)),
    'Precision': precision_score(y_test_, (X_test_frs_prob > 0.5).astype(int)),
    'Recall': recall_score(y_test_, (X_test_frs_prob > 0.5).astype(int)),
    'F1 Score': f1_score(y_test_, (X_test_frs_prob > 0.5).astype(int))
}

# --- Step 4: Calculate p-values and significance ---
# Calculate p-values and significance
brier_p_value, log_loss_p_value, roc_auc_p_value = calculate_p_values(
    pipeline.predict_proba(X_test_shap)[:, 1],
    X_test_frs_prob,
    y_test_selected,
    y_test_
)

brier_significance = significance_label(brier_p_value)
log_loss_significance = significance_label(log_loss_p_value)
roc_auc_significance = significance_label(roc_auc_p_value)

# Combine the results into a DataFrame

```

```

combined_results = pd.DataFrame({
    'Metric': list(shap_metrics.keys()),
    'Non-calibrated RF Model': list(shap_metrics.values()),
    'Non-calibrated FRS Model': list(frs_metrics.values()),
    'p-value': [brier_p_value, log_loss_p_value, roc_auc_p_value] + [""] * (len(shap_metrics) - 3),
    'Significance': [brier_significance, log_loss_significance, roc_auc_significance] + [""] *
(len(shap_metrics) - 3)
})

# Save to a CSV file
combined_csv_file_path = os.path.join(RESULT_DIR,
'non_calibrated_combined_metrics_with_p_values.csv')
combined_results.to_csv(combined_csv_file_path, index=False)
print(f"Non-calibrated combined metrics with p-values saved to {combined_csv_file_path}")

# %%

# %% [markdown]
# ### NRI for Non Callibrated Best Model and Non Callibrated FRS

# %%

# Function to calculate detailed NRI metrics
def calculate_detailed_nri(prob_old, prob_new, y_true, threshold=0.5):
    """
    Calculate detailed NRI metrics including upward and downward reclassification.

    Parameters:
    prob_old (array-like): Probabilities from the old model (e.g., FRS).
    prob_new (array-like): Probabilities from the new model (e.g., SHAP-based model).
    y_true (array-like): True binary outcomes (0 or 1).
    threshold (float): Decision threshold for classification. Default is 0.5.

    Returns:
    result (dict): Dictionary with detailed NRI metrics.
    """
    # Classify using the old and new model probabilities
    class_old = (prob_old > threshold).astype(int)

```

```

class_new = (prob_new > threshold).astype(int)

# Calculate upward and downward reclassification for events
events_upward = np.sum((class_new == 1) & (class_old == 0) & (y_true == 1))
events_downward = np.sum((class_new == 0) & (class_old == 1) & (y_true == 1))
total_events = np.sum(y_true == 1)

# Calculate upward and downward reclassification for non-events
non_events_upward = np.sum((class_new == 1) & (class_old == 0) & (y_true == 0))
non_events_downward = np.sum((class_new == 0) & (class_old == 1) & (y_true == 0))
total_non_events = np.sum(y_true == 0)

# Calculate NRI components
nri_events = (events_upward - events_downward) / total_events if total_events > 0 else 0
nri_non_events = (non_events_downward - non_events_upward) / total_non_events if
total_non_events > 0 else 0
nri_total = nri_events + nri_non_events

# Compile the results into a dictionary
result = {
    'threshold': threshold,
    'events_upward': events_upward,
    'events_downward': events_downward,
    'non_events_upward': non_events_upward,
    'non_events_downward': non_events_downward,
    'total_events': total_events,
    'total_non_events': total_non_events,
    'NRI': nri_total
}

return result

# Example usage

# Load SHAP-reduced test data and pipeline
shap_reduced_data = joblib.load(os.path.join(DATA_DIR, 'shap_reduced_data.pkl'))
X_test_shap = shap_reduced_data['X_test_shap']
y_test_selected = shap_reduced_data['y_test_shap']

model_result_dir = os.path.join(DATA_DIR) # Assuming you want to use the RF model
pipeline = joblib.load(os.path.join(model_result_dir, 'RF_full_model.pkl'))

```

```

# Calculate SHAP model predicted probabilities
shap_pred_prob = pipeline.predict_proba(X_test_shap)[: , 1] # SHAP model predicted probabilities

# Load FRS data
frs_data = joblib.load(os.path.join(DATA_DIR, 'frs_data.pkl'))
X_test_frs_prob = frs_data[X_test_frs]['FRS_score_coeff3'] # FRS predicted probabilities

# Ensure binary outcome is 0/1
y_test_ = y_test_selected.copy()
y_test_[y_test_ > 0] = 1

# Define the thresholds
thresholds = [0.01, 0.05, 0.1, 0.2, 0.3, 0.5]

# Calculate NRI for each threshold
nri_results = []
for threshold in thresholds:
    nri_result = calculate_detailed_nri(X_test_frs_prob, shap_pred_prob, y_test_, threshold=threshold)
    nri_results.append(nri_result)

# Convert results to a DataFrame and save as CSV
nri_df = pd.DataFrame(nri_results)
nri_csv_file_path = os.path.join(RESULT_DIR, 'nri_detailed_results_multiple_thresholds.csv')
nri_df.to_csv(nri_csv_file_path, index=False)
print(f"NRI detailed metrics saved to {nri_csv_file_path}")

# %%
print(RESULT_DIR)

# %% [markdown]
# #### Plot Calibration Curves - just to assess before calibration

# %%
import matplotlib.pyplot as plt
from sklearn.calibration import calibration_curve

# Calibration curve for SHAP model (Non-calibrated)
prob_true_shap, prob_pred_shap = calibration_curve(y_test_selected, shap_pred_prob, n_bins=10)

plt.figure(figsize=(8, 6))

```

```

plt.plot(prob_pred_shap, prob_true_shap, marker='o', label='Non-calibrated RF', color='blue')
plt.plot([0, 1], [0, 1], linestyle='--', color='gray')
plt.xlabel('Mean Predicted Probability')
plt.ylabel('Fraction of Positives')
plt.title('Calibration Curve: Non-calibrated RF')
plt.legend()
plt.show()

# Calibration curve for FRS model (Non-calibrated)
prob_true_frs, prob_pred_frs = calibration_curve(y_test_, X_test_frs_prob, n_bins=10)

plt.figure(figsize=(8, 6))
plt.plot(prob_pred_frs, prob_true_frs, marker='o', label='Non-calibrated FRS', color='orange')
plt.plot([0, 1], [0, 1], linestyle='--', color='gray')
plt.xlabel('Mean Predicted Probability')
plt.ylabel('Fraction of Positives')
plt.title('Calibration Curve: Non-calibrated FRS')
plt.legend()
plt.show()

# Combined Histogram of predicted probabilities for both models
plt.figure(figsize=(8, 6))
plt.hist(shap_pred_prob, bins=10, alpha=0.5, label='Non-calibrated RF', color='blue')
plt.hist(X_test_frs_prob, bins=10, alpha=0.5, label='Non-calibrated FRS', color='orange')
plt.xlabel('Predicted Probability')
plt.ylabel('Frequency')
plt.title('Histogram of Predicted Probabilities')
plt.legend()
plt.show()

# %% [markdown]
# ### Brief Score & Log Loss FOR CALLIBRATED

# %% [markdown]
# ### Callibrated BEST Model VS Callibrated FRS

# %%
# Load the original imputed data for FRS
splitted_data = joblib.load(os.path.join(DATA_DIR, 'train_test_data.pkl'))

X_train = splitted_data['X_train']

```

```

y_train = splitted_data['y_train']

X_calib = splitted_data['X_calib']
y_calib = splitted_data['y_calib']

X_test = splitted_data['X_test']
y_test = splitted_data['y_test']

# Ensure y_train, y_calib, and y_test are binary (0/1)
y_train = (y_train > 0).astype(int)
y_calib = (y_calib > 0).astype(int)
y_test = (y_test > 0).astype(int)

# %% [markdown]
#

# %%
# Load SHAP-reduced data
shap_reduced_data = joblib.load(os.path.join(DATA_DIR, 'shap_reduced_data.pkl'))

X_train_shap = shap_reduced_data['X_train_shap']
y_train_shap = shap_reduced_data['y_train_shap']
X_calib_shap = shap_reduced_data['X_calib_shap']
y_calib_selected = shap_reduced_data['y_calib_shap']
X_test_shap = shap_reduced_data['X_test_shap']
y_test_selected = shap_reduced_data['y_test_shap']

# Ensure y_train_shap, y_calib_selected, and y_test_selected are binary (0/1)
y_train_shap = (y_train_shap > 0).astype(int)
y_calib_selected = (y_calib_selected > 0).astype(int)
y_test_selected = (y_test_selected > 0).astype(int)

# Confirm the shapes are consistent with the respective datasets
assert X_train_shap.shape[0] == len(y_train_shap), "Mismatch in X_train_shap and y_train_shap"
assert X_calib_shap.shape[0] == len(y_calib_selected), "Mismatch in X_calib_shap and
y_calib_selected"
assert X_test_shap.shape[0] == len(y_test_selected), "Mismatch in X_test_shap and y_test_selected"

# %%
from sklearn.isotonic import IsotonicRegression

```

```

import joblib
import os

# Load FRS probabilities for training, calibration, and test sets
frs_data = joblib.load(os.path.join(DATA_DIR, 'frs_data.pkl'))

X_train_frs_prob = frs_data['X_train_frs']['FRS_score_coeff3'] # FRS probabilities for training
X_calib_frs_prob = frs_data['X_calib_frs']['FRS_score_coeff3'] # FRS probabilities for calibration
X_test_frs_prob = frs_data['X_test_frs']['FRS_score_coeff3'] # FRS probabilities for test

# Confirm the shapes are consistent with the respective datasets
assert X_train.shape[0] == len(X_train_frs_prob), "Mismatch in X_train and X_train_frs_prob"
assert X_calib.shape[0] == len(X_calib_frs_prob), "Mismatch in X_calib and X_calib_frs_prob"
assert X_test.shape[0] == len(X_test_frs_prob), "Mismatch in X_test and X_test_frs_prob"

# --- Step 1: Calibrate FRS Model using Isotonic Regression ---
# Calibrate the FRS probabilities using Isotonic Regression
frs_isotonic = IsotonicRegression(out_of_bounds='clip')
frs_isotonic.fit(X_train_frs_prob, y_train) # Fit on training data

# Apply the calibration to the test set
frs_calib_prob = frs_isotonic.transform(X_test_frs_prob)
from sklearn.calibration import CalibratedClassifierCV

# --- Step 2: Calibrate SHAP Model using Isotonic Regression ---
# Load the SHAP model pipeline
model_result_dir = os.path.join(DATA_DIR)
pipeline = joblib.load(os.path.join(model_result_dir, 'RF_full_model.pkl'))

# Calibrate the SHAP model using Isotonic Regression
shap_isotonic = CalibratedClassifierCV(pipeline, method='isotonic', cv='prefit')
shap_isotonic.fit(X_calib_shap, y_calib_selected) # Fit on calibration data

# Apply the calibration to the test set
shap_calib_prob = shap_isotonic.predict_proba(X_test_shap)[: , 1]

from sklearn.metrics import brier_score_loss, log_loss, roc_auc_score, accuracy_score,
precision_score, recall_score, f1_score
import pandas as pd
import os

```

```

# Function to calculate metrics
def calculate_metrics(probabilities, y_test):
    y_test_pred = (probabilities > 0.5).astype(int)

    brier_score = brier_score_loss(y_test, probabilities)
    log_loss_value = log_loss(y_test, probabilities)
    roc_auc = roc_auc_score(y_test, probabilities)
    accuracy = accuracy_score(y_test, y_test_pred)
    precision = precision_score(y_test, y_test_pred)
    recall = recall_score(y_test, y_test_pred)
    f1 = f1_score(y_test, y_test_pred)

    return {
        'Brier Score': brier_score,
        'Log Loss': log_loss_value,
        'ROC AUC': roc_auc,
        'Accuracy': accuracy,
        'Precision': precision,
        'Recall': recall,
        'F1 Score': f1
    }

# %%
from sklearn.utils import resample
from scipy.stats import ttest_ind

# Function to calculate p-values using bootstrapping
def calculate_p_values(probabilities1, probabilities2, y_test1, y_test2, n_bootstrap=1000):
    brier_scores_1 = []
    brier_scores_2 = []
    log_losses_1 = []
    log_losses_2 = []
    roc_aucs_1 = []
    roc_aucs_2 = []

    for _ in range(n_bootstrap):
        # Bootstrap samples
        y_test_sample_1, prob_sample_1 = resample(y_test1, probabilities1)
        y_test_sample_2, prob_sample_2 = resample(y_test2, probabilities2)

```

```

# Brier Score
brier_scores_1.append(brier_score_loss(y_test_sample_1, prob_sample_1))
brier_scores_2.append(brier_score_loss(y_test_sample_2, prob_sample_2))

# Log Loss
log_losses_1.append(log_loss(y_test_sample_1, prob_sample_1))
log_losses_2.append(log_loss(y_test_sample_2, prob_sample_2))

# ROC AUC
roc_aucs_1.append(roc_auc_score(y_test_sample_1, prob_sample_1))
roc_aucs_2.append(roc_auc_score(y_test_sample_2, prob_sample_2))

# Calculate p-values
brier_score_p_value = ttest_ind(brier_scores_1, brier_scores_2).pvalue
log_loss_p_value = ttest_ind(log_losses_1, log_losses_2).pvalue
roc_auc_p_value = ttest_ind(roc_aucs_1, roc_aucs_2).pvalue

return brier_score_p_value, log_loss_p_value, roc_auc_p_value

# Function to determine significance based on p-value
def significance_label(p_value, alpha=0.05):
    return 'Significant' if p_value < alpha else 'Not Significant'

# --- Step 3: Calculate Metrics ---
# Calculate metrics for the calibrated RF model
shap_calib_metrics = calculate_metrics(shap_calib_prob, y_test_selected)

# Calculate metrics for the calibrated FRS model
frs_calib_metrics = calculate_metrics(frs_calib_prob, y_test)

# --- Step 4: Calculate p-values and significance ---
# Calculate p-values and significance
brier_p_value, log_loss_p_value, roc_auc_p_value = calculate_p_values(shap_calib_prob,
frs_calib_prob, y_test_selected, y_test)
brier_significance = significance_label(brier_p_value)
log_loss_significance = significance_label(log_loss_p_value)
roc_auc_significance = significance_label(roc_auc_p_value)

# Create DataFrame with all metrics and significance
combined_calib_results = pd.DataFrame({
    'Metric': list(shap_calib_metrics.keys()),

```

```

'Calibrated RF Model': list(shap_calib_metrics.values()),
'Calibrated FRS Model': list(frs_calib_metrics.values()),
'p-value': [brier_p_value, log_loss_p_value, roc_auc_p_value] + [""] * (len(shap_calib_metrics) - 3),
'Significance': [brier_significance, log_loss_significance, roc_auc_significance] + [""] *
(len(shap_calib_metrics) - 3)
})

# Save to a CSV file
combined_calib_csv_file_path = os.path.join(RESULT_DIR,
'combined_calibrated_metrics_with_p_values_as_columns.csv')
combined_calib_results.to_csv(combined_calib_csv_file_path, index=False)
print(f"Combined calibrated metrics with p-values saved to {combined_calib_csv_file_path}")

# %%
import numpy as np

# Function to calculate detailed NRI metrics
def calculate_detailed_nri(prob_old, prob_new, y_true, threshold=0.5):
    class_old = (prob_old > threshold).astype(int)
    class_new = (prob_new > threshold).astype(int)

    events_upward = np.sum((class_new == 1) & (class_old == 0) & (y_true == 1))
    events_downward = np.sum((class_new == 0) & (class_old == 1) & (y_true == 1))
    total_events = np.sum(y_true == 1)

    non_events_upward = np.sum((class_new == 1) & (class_old == 0) & (y_true == 0))
    non_events_downward = np.sum((class_new == 0) & (class_old == 1) & (y_true == 0))
    total_non_events = np.sum(y_true == 0)

    nri_events = (events_upward - events_downward) / total_events if total_events > 0 else 0
    nri_non_events = (non_events_downward - non_events_upward) / total_non_events if
total_non_events > 0 else 0
    nri_total = nri_events + nri_non_events

    return {
        'threshold': threshold,
        'events_upward': events_upward,
        'events_downward': events_downward,
        'non_events_upward': non_events_upward,
        'non_events_downward': non_events_downward,
        'total_events': total_events,

```

```

        'total_non_events': total_non_events,
        'NRI': nri_total
    }

# --- Step 4: NRI Analysis ---
# Define the thresholds
thresholds = [0.005, 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09, 0.1, 0.2, 0.3, 0.5]

# Calculate NRI for the calibrated models across different thresholds
calibrated_nri_results = []
for threshold in thresholds:
    nri_result = calculate_detailed_nri(frs_calib_prob, shap_calib_prob, y_test_selected,
threshold=threshold)
    calibrated_nri_results.append(nri_result)

# Convert results to a DataFrame and save as CSV
calibrated_nri_df = pd.DataFrame(calibrated_nri_results)
calibrated_nri_csv_file_path = os.path.join(RESULT_DIR,
'calibrated_nri_detailed_results_multiple_thresholds.csv')
calibrated_nri_df.to_csv(calibrated_nri_csv_file_path, index=False)
print(f"Calibrated NRI detailed metrics saved to {calibrated_nri_csv_file_path}")

# %% [markdown]
# #### Plot Calibration Curves - just to assess before calibration

# %%
import matplotlib.pyplot as plt
from sklearn.calibration import calibration_curve

# Calibration curve for SHAP model
prob_true, prob_pred = calibration_curve(y_test_selected, shap_calib_prob, n_bins=10)
plt.plot(prob_pred, prob_true, marker='o', label='Calibrated RF')

# Calibration curve for FRS model
prob_true_frs, prob_pred_frs = calibration_curve(y_test, frs_calib_prob, n_bins=10)
plt.plot(prob_pred_frs, prob_true_frs, marker='o', label='Calibrated FRS')

plt.plot([0, 1], [0, 1], linestyle='--')
plt.xlabel('Mean Predicted Probability')
plt.ylabel('Fraction of Positives')
plt.title('Calibration Curves')

```

```

plt.legend()
plt.show()

# Calibration curve for SHAP model
prob_true_shap, prob_pred_shap = calibration_curve(y_test_selected, shap_calib_prob, n_bins=10)

plt.figure(figsize=(8, 6))
plt.plot(prob_pred_shap, prob_true_shap, marker='o', label='Calibrated RF', color='blue')
plt.plot([0, 1], [0, 1], linestyle='--', color='gray')

plt.xlabel('Mean Predicted Probability')
plt.ylabel('Fraction of Positives')
plt.title('Calibration Curve: Calibrated RF')
plt.legend()
plt.show()

import matplotlib.pyplot as plt
from sklearn.calibration import calibration_curve

# Calibration curve for FRS model
prob_true_frs, prob_pred_frs = calibration_curve(y_test, frs_calib_prob, n_bins=10)

plt.figure(figsize=(8, 6))
plt.plot(prob_pred_frs, prob_true_frs, marker='o', label='Calibrated FRS', color='orange')
plt.plot([0, 1], [0, 1], linestyle='--', color='gray')

plt.xlabel('Mean Predicted Probability')
plt.ylabel('Fraction of Positives')
plt.title('Calibration Curve: Calibrated FRS')
plt.legend()
plt.show()

# %%
import matplotlib.pyplot as plt
from sklearn.calibration import calibration_curve

# Create a figure with subplots
fig, ax = plt.subplots(2, 1, figsize=(8, 12), gridspec_kw={'height_ratios': [2, 1]})

# Reliability diagram (calibration curve) for SHAP model
prob_true_shap, prob_pred_shap = calibration_curve(y_test_selected, shap_calib_prob, n_bins=10)

```

```

ax[0].plot(prob_pred_shap, prob_true_shap, marker='o', label='Calibrated RF', color='blue')

# Reliability diagram (calibration curve) for FRS model
prob_true_frs, prob_pred_frs = calibration_curve(y_test, frs_calib_prob, n_bins=10)
ax[0].plot(prob_pred_frs, prob_true_frs, marker='o', label='Calibrated FRS', color='orange')

# Plot the perfect calibration line
ax[0].plot([0, 1], [0, 1], linestyle='--', color='gray')

# Formatting the plot
ax[0].set_xlabel('Mean Predicted Probability')
ax[0].set_ylabel('Fraction of Positives')
ax[0].set_title('Reliability Diagram')
ax[0].legend()

# Histogram of predicted probabilities
ax[1].hist(shap_calib_prob, bins=10, alpha=0.5, label='Calibrated RF', color='blue')
ax[1].hist(frs_calib_prob, bins=10, alpha=0.5, label='Calibrated FRS', color='orange')
ax[1].set_xlabel('Predicted Probability')
ax[1].set_ylabel('Frequency')
ax[1].set_title('Histogram of Predicted Probabilities')
ax[1].legend()

plt.tight_layout()
plt.show()

# %%
import joblib
import os
import pandas as pd
from sklearn.metrics import brier_score_loss, log_loss, roc_auc_score
from sklearn.utils import resample
from scipy.stats import ttest_ind

# Assuming the non-calibrated best model's probabilities are already calculated as `shap_pred_prob`
# Assuming the calibrated best model's probabilities are already calculated as `shap_calib_prob`

# Step 1: Calculate metrics for the non-calibrated best model (already done previously)
non_calib_metrics = calculate_metrics(shap_pred_prob, y_test_selected)

# Step 2: Calculate metrics for the calibrated best model (already done previously)

```

```

calib_metrics = calculate_metrics(shap_calib_prob, y_test_selected)

# Step 3: Calculate p-values and significance between non-calibrated and calibrated models
brier_p_value, log_loss_p_value, roc_auc_p_value = calculate_p_values(shap_pred_prob,
shap_calib_prob, y_test_selected, y_test_selected)
brier_significance = significance_label(brier_p_value)
log_loss_significance = significance_label(log_loss_p_value)
roc_auc_significance = significance_label(roc_auc_p_value)

# Step 4: Combine the results into a DataFrame
comparison_results = pd.DataFrame({
    'Metric': ['Brier Score', 'Log Loss', 'ROC AUC'],
    'Non-calibrated Best Model': [non_calib_metrics['Brier Score'], non_calib_metrics['Log Loss'],
non_calib_metrics['ROC AUC']],
    'Calibrated Best Model': [calib_metrics['Brier Score'], calib_metrics['Log Loss'], calib_metrics['ROC
AUC']],
    'p-value': [brier_p_value, log_loss_p_value, roc_auc_p_value],
    'Significance': [brier_significance, log_loss_significance, roc_auc_significance]
})

# Step 5: Save the results to a CSV file
comparison_csv_file_path = os.path.join(RESULT_DIR,
'non_calibrated_vs_calibrated_best_model_metrics_with_p_values.csv')
comparison_results.to_csv(comparison_csv_file_path, index=False)
print(f"Non-calibrated vs Calibrated best model metrics with p-values saved to
{comparison_csv_file_path}")

# %%
import matplotlib.pyplot as plt
from sklearn.calibration import calibration_curve

# Calibration curve for the non-calibrated best model
prob_true_non_calib, prob_pred_non_calib = calibration_curve(y_test_selected, shap_pred_prob,
n_bins=10)

# Calibration curve for the calibrated best model
prob_true_calib, prob_pred_calib = calibration_curve(y_test_selected, shap_calib_prob, n_bins=10)

# Plot the calibration curves
plt.figure(figsize=(8, 6))

```

```

plt.plot(prob_pred_non_calib, prob_true_non_calib, marker='o', label='Non-calibrated Best Model',
color='blue')
plt.plot(prob_pred_calib, prob_true_calib, marker='o', label='Calibrated Best Model', color='orange')

# Plot the perfect calibration line
plt.plot([0, 1], [0, 1], linestyle='--', color='gray')

# Formatting the plot
plt.xlabel('Mean Predicted Probability')
plt.ylabel('Fraction of Positives')
plt.title('Calibration Curves: Non-calibrated vs Calibrated Best Model')
plt.legend()
plt.show()

# %% [markdown]
#

# %% [markdown]
#

# %%
import numpy as np
import pandas as pd
import os

# Function to calculate detailed NRI metrics
def calculate_detailed_nri(prob_old, prob_new, y_true, threshold=0.5):
    """
    Calculate detailed NRI metrics including upward and downward reclassification.

    Parameters:
    prob_old (array-like): Probabilities from the old model (e.g., non-calibrated model).
    prob_new (array-like): Probabilities from the new model (e.g., calibrated model).
    y_true (array-like): True binary outcomes (0 or 1).
    threshold (float): Decision threshold for classification. Default is 0.5.

    Returns:
    result (dict): Dictionary with detailed NRI metrics.
    """
    # Classify using the old and new model probabilities
    class_old = (prob_old > threshold).astype(int)

```

```

class_new = (prob_new > threshold).astype(int)

# Calculate upward and downward reclassification for events
events_upward = np.sum((class_new == 1) & (class_old == 0) & (y_true == 1))
events_downward = np.sum((class_new == 0) & (class_old == 1) & (y_true == 1))
total_events = np.sum(y_true == 1)

# Calculate upward and downward reclassification for non-events
non_events_upward = np.sum((class_new == 1) & (class_old == 0) & (y_true == 0))
non_events_downward = np.sum((class_new == 0) & (class_old == 1) & (y_true == 0))
total_non_events = np.sum(y_true == 0)

# Calculate NRI components
nri_events = (events_upward - events_downward) / total_events if total_events > 0 else 0
nri_non_events = (non_events_downward - non_events_upward) / total_non_events if
total_non_events > 0 else 0
nri_total = nri_events + nri_non_events

# Compile the results into a dictionary
result = {
    'threshold': threshold,
    'events_upward': events_upward,
    'events_downward': events_downward,
    'non_events_upward': non_events_upward,
    'non_events_downward': non_events_downward,
    'total_events': total_events,
    'total_non_events': total_non_events,
    'NRI': nri_total
}

return result

# Define the thresholds
thresholds = [0.01, 0.05, 0.1, 0.2, 0.3, 0.5]

# Calculate NRI for the non-calibrated vs calibrated models across different thresholds
nri_results = []
for threshold in thresholds:
    nri_result = calculate_detailed_nri(shap_pred_prob, shap_calib_prob, y_test_selected,
threshold=threshold)
    nri_results.append(nri_result)

```

```
# Convert results to a DataFrame and save as CSV
nri_df = pd.DataFrame(nri_results)
nri_csv_file_path = os.path.join(RESULT_DIR, 'nri_non_calibrated_vs_calibrated_results.csv')
nri_df.to_csv(nri_csv_file_path, index=False)
print(f"NRI results for non-calibrated vs calibrated models saved to {nri_csv_file_path}")
```